

INTRODUCTION TO WELL LOGGING AND INTERPRETATION

(PGG434)

**Full Course Slides – 2021/2022
Session**

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Thursdays: 8 – 10 am

Course Description (Student Handbook)

- Driller's log
- Sample logs
- Mud logging
- Electric logging
- Radioactivity logging
- Miscellaneous logging devices
- Wire line logs
- Well logging interpretation

Course Contents

Content	Hrs	Instructor
1. Introduction – History, Present and future of a science and technology of Well Logging. Fundamental problems and way forward	0.5	Olamigoke
2. Reservoir Rocks and Properties – Reservoir rock types, Porosity, Saturation, Permeability, Core Analysis, Wettability and Capillarity	1.5	Olamigoke
3. Overview of Well Logging – Principle of log measurement, Classification of Logging methods, The tool (probe) and the environment of measurement	2	Olamigoke
4. Well Logging Methods – Electrical and electromagnetic logs, Nuclear logs, Acousticlog, Sonic log, NMR measurements Imaging Techniques	2	Olamigoke
5. Introduction to Well Log Interpretation – Overview, Quick Look Methods, Quantitative Interpretation: Shale Content	6	Olamigoke
6. Some Applications in Water Exploration – Water well in an unconsolidated formation, Mineral water well in a fractured carbonate formation	1	Olamigoke
7. Examples and Exercises – Oil-bearing Sandstone, Shaly sand profile, Mixed lithology Carbonate rock	2	Olamigoke
8. Field Management & Data Acquisition Planning and Execution – Logging Programs, Operational Challenges, Value of Information, Miscellaneous	4	Olamigoke
9. Revision and Contingency	4	Olamigoke

Learning Outcomes

- By the end of this course students should be able to:
 - explain in basic terms how petrophysical data is obtained from well logs
 - interpret basic wireline logs
 - quantitatively evaluate basic petrophysical properties from well logs.

Required Background

- **PGG 322 – Formation Evaluation and Applied Geophysical Methods**
- PGG 313 – Basic Petroleum Reservoir Engineering
- PGG 308 – Drilling Methods
- PGG 310 – Petroleum Engineering Lab. II
- Knowledge from other courses might come in handy

References

- Prof. Dr. Jurgen Schon, *Basic Well Logging and Formation Evaluation*, 2015 bookboon.com
- Baker Atlas. *Introduction to Wireline Log Analysis*, 2002
- Toby Darling, *Well Logging and Formation Evaluation*, 2005 Elsevier Inc
- Krygowski DA, Texas A, *Guide to Petrophysical Interpretation*, 2003
- Ellis, D.V. and Singer, J.M. *Well Logging for Earth Scientists*, 2008. 2nd Edition

Grading System

<u>Components</u>	<u>Marks</u>
Mid Semester Test(s)	30
Attendance	10
Group Presentation / Term Paper	20
Tutorial Questions	20
Final Examination	<u>60</u>
Total = <u>100</u>	

Ground Rules

- Keep your phones on vibration or silent mode
- You are free to note typographical errors on the slides or and inconsistencies. **Please take note of the slide number and notify me after the class.**
- Some slides appear to be repetition of earlier slides or material from PGG322. **IT IS FOR EMPHASIS.**
- If you have a question, raise your hand. Note that all question can't be entertained in class.
- The number of attendees at each class would be taken. This would be used to confirm the day's attendance list.

Introduction to Well Logging – Lecture 1

History of Logging Tools & Interpretation

- **1920s:** First Electrical log recorded in 1927 by Schlumberger brothers – Marcel and Conrad in France using stationed resistivity method. The first commercial electrical resistivity tool in 1929 was used in Venezuela, USA and Indonesia - useful for correlation purposes & identification of HC zones
- **1930s:** Spontaneous Potential (SP) measurement was included with the resistivity curve on the electric log. Core Sample Taker. Schlumberger developed the first continuous recording in 1931 via the first pen recorder. In 1936 the photographic film recorder was introduced. SP, Short Normal, Long Normal and Long Lateral resistivity curve combined.
- **1940s:** 3-arm Dipmeter using SP with an associated photocalinometer in 1943, in 1946 the SP sensor were replaced with short resistivity devices. The first use of Gamma ray and Neutron tools. Induction tool for resistivity in Non-Aqueous mud

History of Logging Tools & Interpretation

- **1950s to 1960s:** The first continuously recording electrical dipmeter sonde with 3 microresistivity areas & a fluxgate compass. Microlog to delineate permeable zones, Laterolog for deep resistivity, Sonic log, Formation Tester, Formation Density tool. Introduction of the sidewall neutron porosity tool
- **1970s:** Dual Spacing Neutrons, Advanced Dipmeter, Computerized Computer Systems, Repeat Tester Tools, Electromagnetic Propagation tool for dielectric measurements in formations with freshwater or varying salinity
- **1980s:** Resistivity Imaging Tools, Advanced Sonic Tools
- **1990s:** Advanced Formation Testing Tools, Induction Imaging tools, Azimuthal Laterolog Tools, Ultrasonic Imaging tools, Epithermal Porosity tools, Magnetic Resonance Tools.

Well Logging: Background

- Well logging is an indirect method. Therefore, measured parameters (see Table 1) must be transformed into reservoir parameters (porosity, saturation) by a process called interpretation.
- A formation or reservoir characterization cannot be derived from one type of logs alone – it needs a combination of various physical parameters in order to derive a consistent model of the formation.
- The suite of logs which have been produced must be correlated, to ensure that the petrophysical measurements made in a particular formation by the sondes are all represented at the same depth.
- **Definition:** A well log is a continuous record of downhole data organized and interpreted by depth and represented on a graph plotted to scale.

Evolution of Borehole Geophysics Tools

Table 1: Measured Parameters and Reservoir Parameter of Interest

Log Type	Parameter of Interest
Resistivity Log	Water Saturation determination from clean rocks
Spontaneous Potential (SP) Log	Separation of sand (clean rock, reservoir) & shale
Acoustic Log	The “Porosity log” based on Wyllie’s equation
Natural Gamma Ray Log	Estimation of Shale Content
Density Log	More accurate determination of porosity and mineral composition calculation using sophisticated combined techniques (with the Acoustic Log)
Neutron Log	
Nuclear Magnetic Resonance (NMR)	Derive pore space properties with respect to a permeability estimate; realize fluid characterization

How are Well Logs measurements taken?

- Logs are taken in the Open Hole (while drilling or after the hole section has been drilled) or in the Cased Hole.
- Logs can be obtained while drilling by running tools downhole (MWD) with the drilling assembly (complimented by Surface Logging Systems).
- Well Logging tools run downhole after the hole section has been drilled are either (i) drillpipe-conveyed; (ii) wireline-conveyed (iii) Coiled tubing – conveyed (iv) via well tractors to aid wireline deployment in challenging well trajectories,

The wellbore environment for Logging

- The specific environment related to the development of well logging and interpretation tools is: The tool is surrounded by mud in an (open) well typically referred to as the “wireline technique”. Logs are also obtained in “Cased-hole” environments – for control of completion and quality (e.g. Cement Bond Logs) and monitoring fluid saturation (Cased hole saturation Logging)
- The modern techniques of Measurement While Drilling (MWD) and Logging While Drilling (LWD) reflect the dream to “see” in real-time what is penetrated with the bit; such a measurement is important for an optimized and safe drilling operation also in deviated and horizontal wells.

Integrated Wireline Logging

- The Platform Express integrated wireline logging tool uses integrated sensors & improved pad contact. Logging speed: 3,600 ft/h [1,097 m/h]
- The toolstring includes either the AIT array induction imager tool or High-Resolution Azimuthal Laterolog Sonde (HALS) for Induction or Laterolog resistivity, SP and mud resistivity
- The High-Resolution Mechanical Sonde (HRMS) houses the Three-Detector Lithology Density (TLD) tool and the Micro-Cylindrical Focused Log (MCFL) for Bulk density, photoelectric factor, borehole caliper, microresistivity
- The Highly Integrated Gamma Ray Neutron Sonde (HGNS) is for gamma ray, neutron and tool acceleration measurements

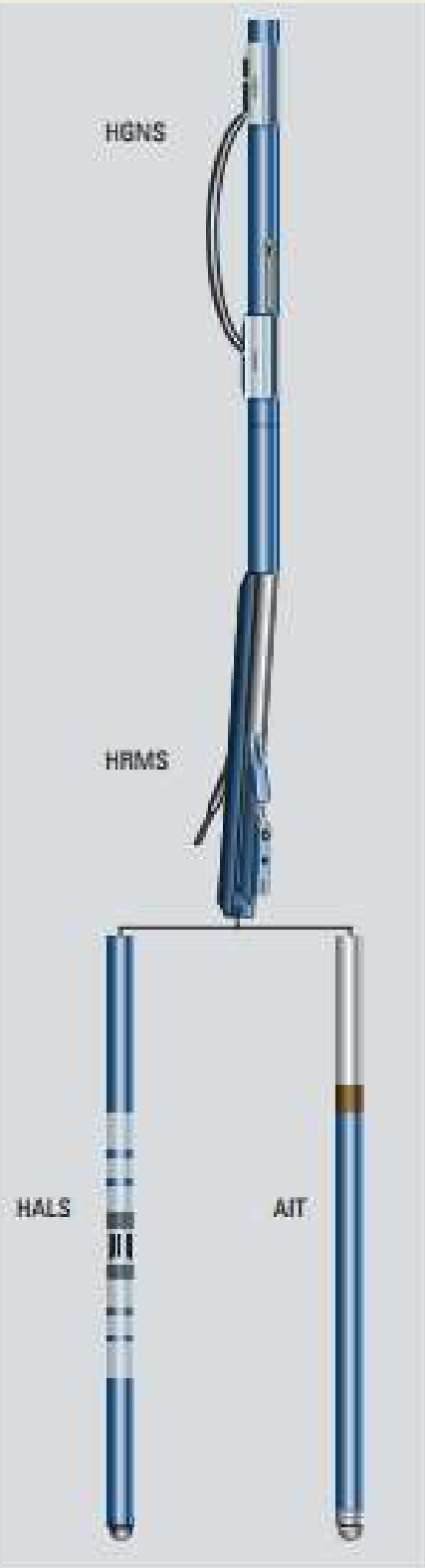


Figure 1: Schlumberger Platform Express Integrated Wireline Logging Tool

Objectives of Well Logging

- To provide data for evaluating petroleum reservoirs.
 - Qualitative: Assessment of Lithology with exact depth of formations / rock boundaries, fluid type, form log pattern, detect fractures
 - Quantitative: Estimation of effective porosity & permeability; Estimation of volume of clay fraction; Estimation of water and hydrocarbon saturation – initial and time lapse measurements; Determination of the depth and thickness of net pay.
- To aid in testing, completion, & workover of well.
 - Cement quality behind the casing
 - Mechanical rock properties for stability, sanding problems, frac operations
 - change in fluid content / saturation during production (monitoring, time lapse measurements).

Classification by Principle of Measurement

- Passive Tools: measure properties or parameters delivered by the formation or by interaction with the formation and the borehole fluid without any source.
 - Mechanical Caliper, Natural Gamma Measurements, Spontaneous Potential (SP), Temperature
- Active Tools: measure the response to a signal, pulse, radiation, current i.e. the result of an interaction with the formation in the vicinity of the tool (typically they have one source and one or more detectors)
 - Acoustic measurements
 $\Delta t_c, \Delta t_s, A_c, A_s$
 - Nuclear measurements
 $\rho_b, \phi_N, Pe, T_1, T_2$
 - Resistivity / Electromagnetic
 $R, t_{PL}, EATT$

See Figure 2 for more details

Measurement

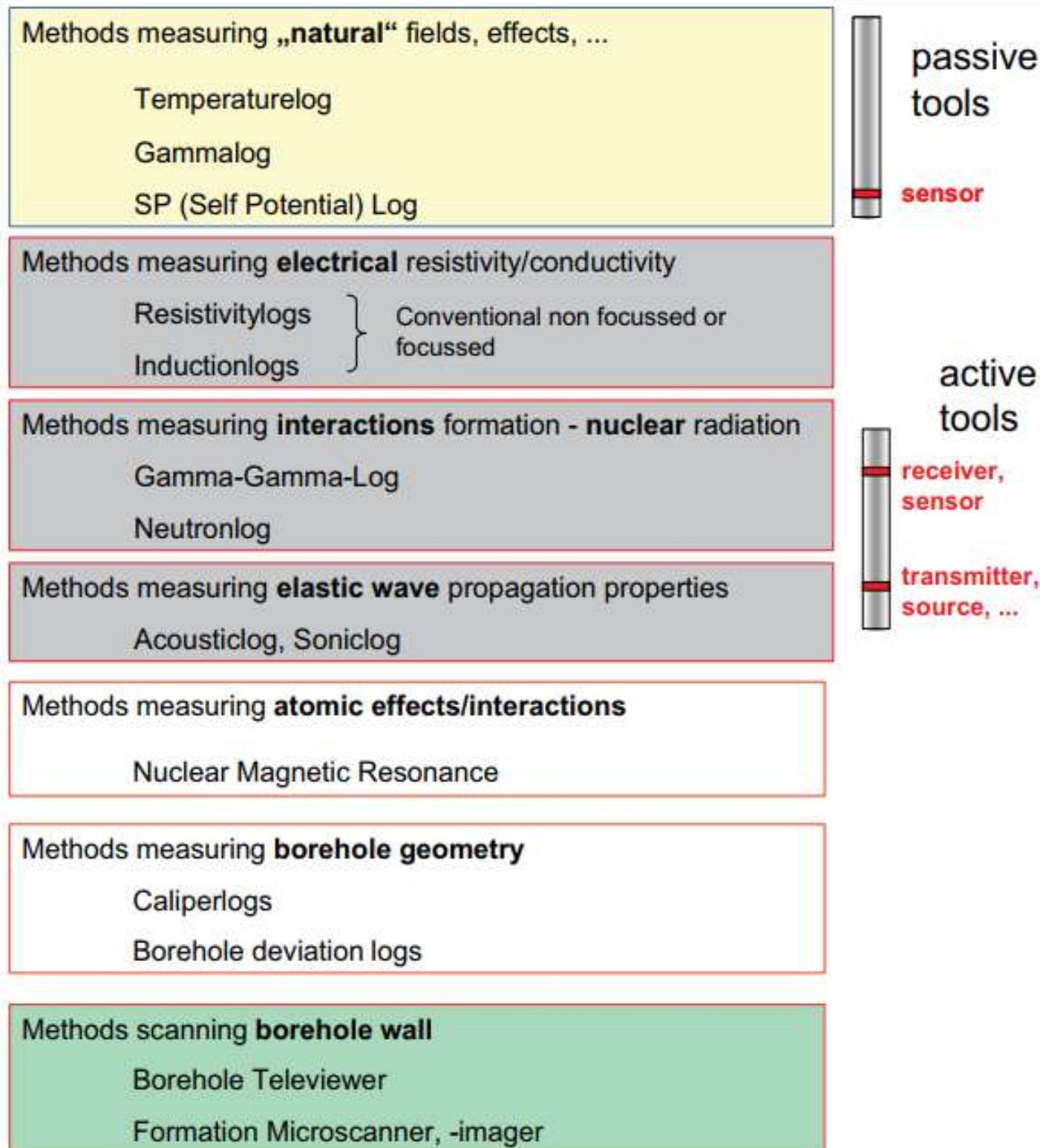
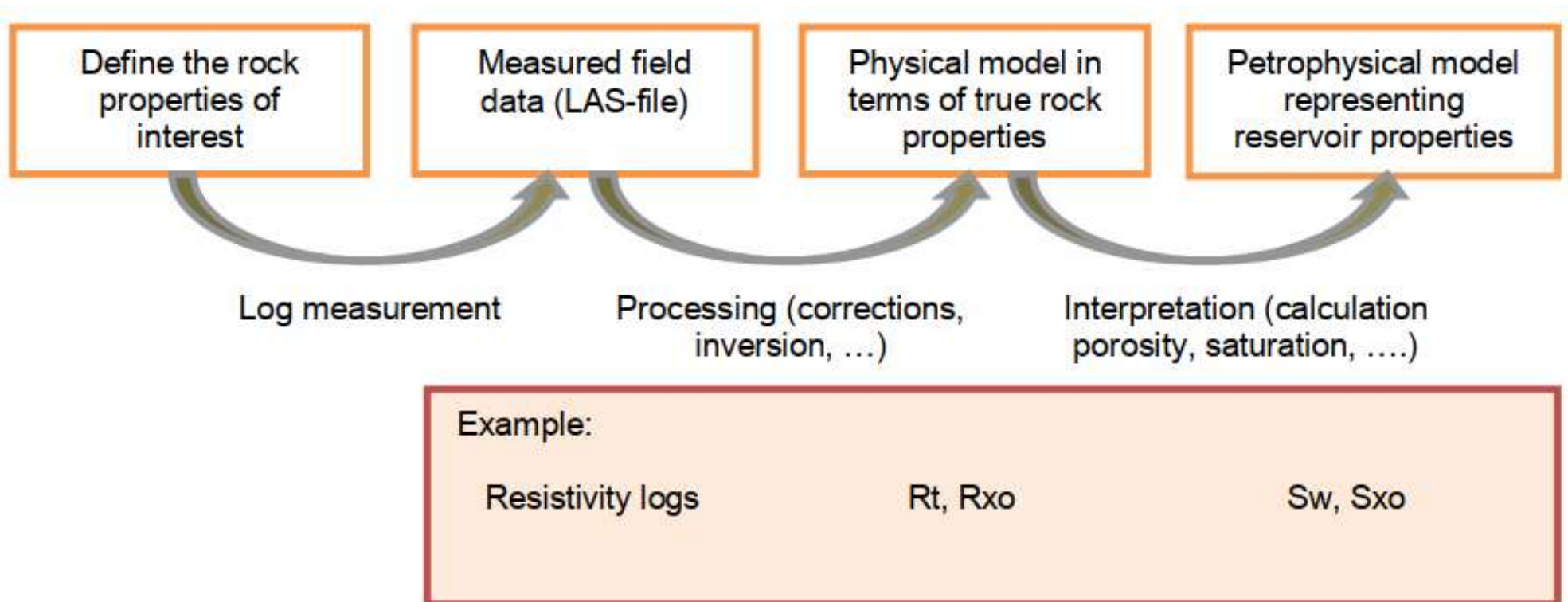


Figure 2:
Passive versus
Active tools

Transforming Property Measurement in the Well into Reservoir Properties

How can we derive equations & algorithms to transform measured properties to reservoir properties?

- Empirically using experiments.
- Theoretically using models.
- Combination of Theoretical and Empirical results



The Process of Borehole Measurements

- 1) Define the parameters of interest for the lithological type of the section. Design your logging program based on sensitivity of the individual tools.
- 2) Take the corresponding log measurements. The result is a dataset for each method i . With regard to the resistivity measurement, this could be Microlog, Laterolog-shallow, and Laterolog-deep
- 3) Derive a physical model in terms of the measured physical property p_i using processing algorithms. Regarding resistivity measurement, this results in R_{xo} and R_t . This step called “processing” has strong reference to the tool characteristics
- 4) Transform measured data into reservoir properties using interpretation methods.

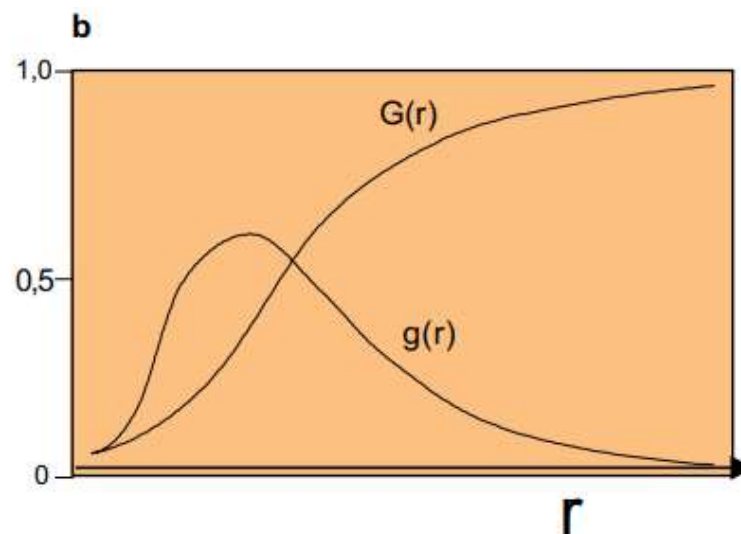
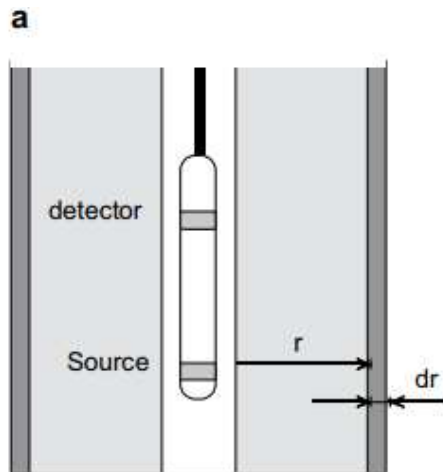
THE TOOL CHARACTERISTICS

Radius (Depth) of Investigation & Vertical Resolution

- The tool characteristic is the response of a tool controlled by the design of its elements. The tool characteristics are described by:
 - its response function and/or its vertical resolution, and
 - the radius (depth) of investigation.
- The specific **resolution** of the tool, describes the ability of the tool to detect and separate thin layers individually
- In permeable zones – depending on the radial characteristics of the tool and the depth of invasion – the device measures contributions from the invaded and non-invaded zone.
- See Figure 3 and Table 1 for typical values.

Radius (Depth) of Investigation & Vertical Resolution

- Illustrating this for a simple transmitter-receiver tool, the differential response $g(r)$ is the normalized contribution of a cylinder with radius r to the measured signal. Thus, the maximum can be considered as the radial distance with the strongest influence on the measured magnitude
- As cumulative presentation the integral response $G(r)$ describes the radial buildup of the total measured signal. For tool characterization the radius of $G(r) = 0.5$ (50%) is frequently used as “radius of investigation 50%” Radius of investigation 50% (r_{50}) implies that 50% of the total signal response is originated in a radial distance below r_{50} , and 50% originates from the space outside this cylinder.



Radial
characteristics and
response

Relationship between Mud Filtrate Invasion and Formation Porosity

- The specific resolution of the tool, describes the ability of the tool to detect and separate thin layers individually
- In permeable zones – depending on the radial characteristics of the tool and the depth of invasion – the device measures contributions from the invaded and non-invaded zone.

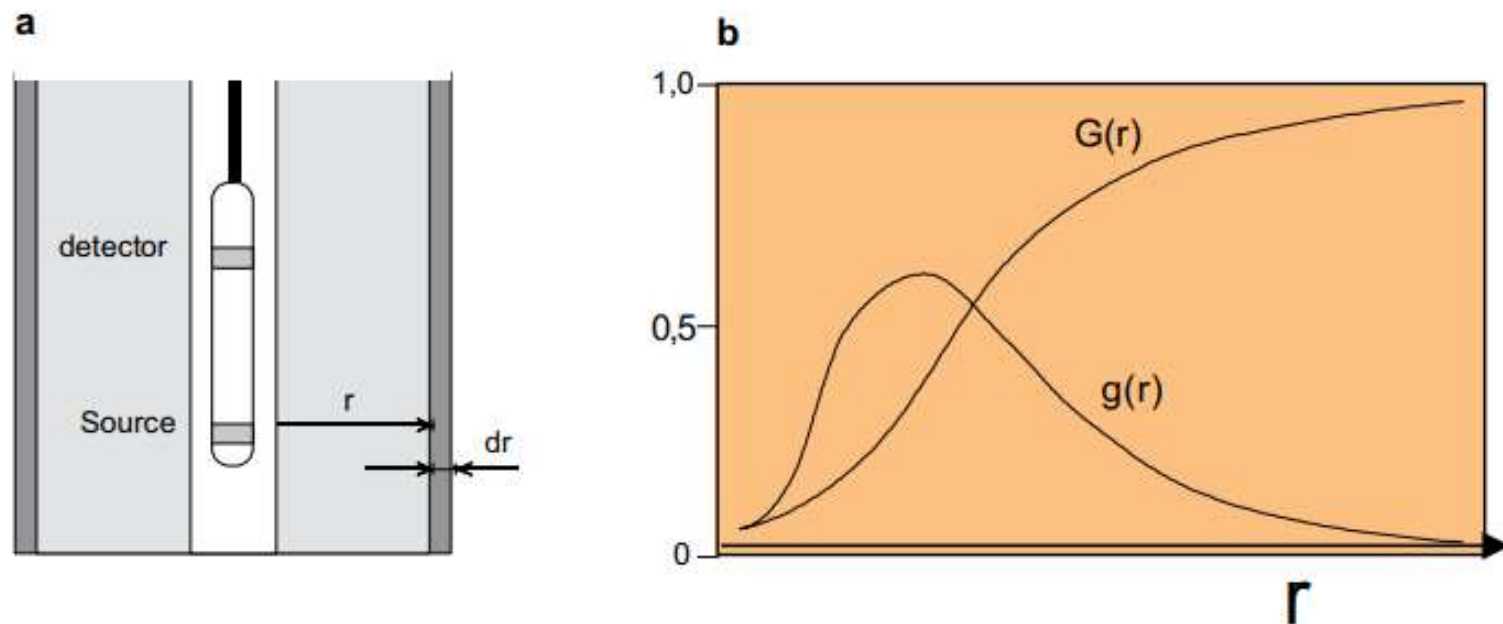


Figure 3-7: Radial characteristic and response, after Tittman (1986) ($g(r)$ Differential Response; $G(r)$ Integral Response).

Figure 3: Depth of Investigation vs. Resolution of Logging Tools

Logging Tools

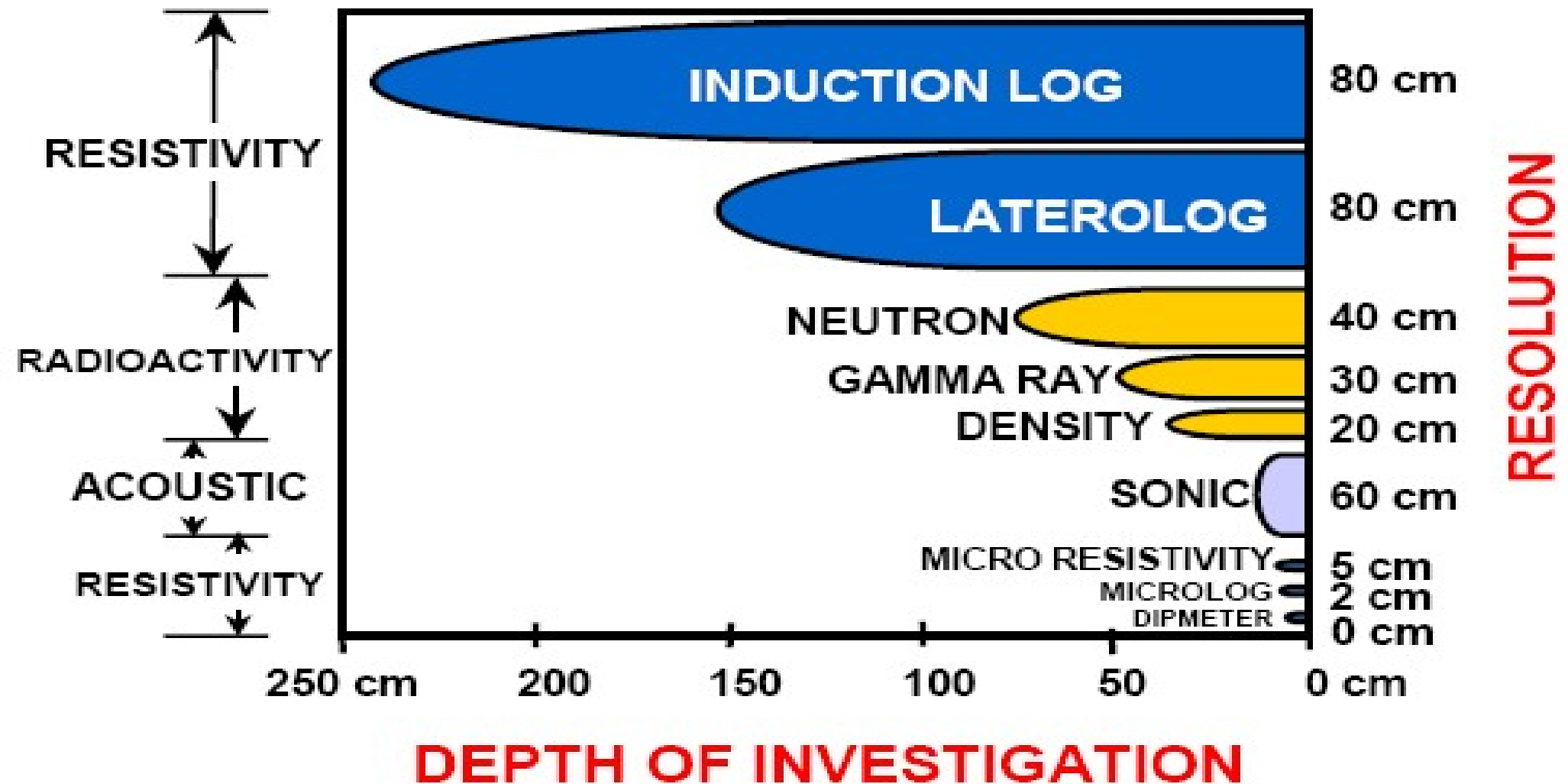


Table 1: Vertical Resolution & Radius of Investigation of Schlumberger Logging Tools

Tool	Vertical Resolution 90%	Radius of Investigation 50%
Natural Gamma	12 in. [30.48 cm]	24 in. [60.96 cm]
Density	15 in. [38.10 cm]	4 in. [10.16 cm]
Compensated Neutron	12 in. [30.48 cm]	9 in. [23 cm]
Acoustic/Sonic	2 ft. [0.61 m]	3 in. [7.62 cm]
Induction Log		
Deep	4 ft. [1.22 m]	30 – 90 in. [76.2 – 228.6 cm]
Medium	2 ft. [0.61 m]	20 – 60 in. [50.8 – 152.4 cm]
Shallow	1 ft. [0.20 m]	10 – 30 in. [25.4 – 76.2 cm]
Laterolog		
Deep	18 in. [45.72 cm]	50 in. [127.0 cm]
Medium	18 in. [45.72 cm]	32 in. [81.3 cm]
Shallow	8 in. [20.32 cm]	32 in. [81.3 cm]
Nuclear Magnetic Resonance	Std.: 18 in. HiRes.: 7.5 in	1.25 – 4 in (HiRes. – Max. Std)

An Overview of Well Logging Practice – Lecture 2

One takeaway is that for virtually every measurement and wireline tool, there is an equivalent LWD tool.

Notable exceptions were Fluid Sampling and Sidewall Coring****

WIRELINE LOGGING – FIELD OPERATION

Wireline Logging Operations Basics

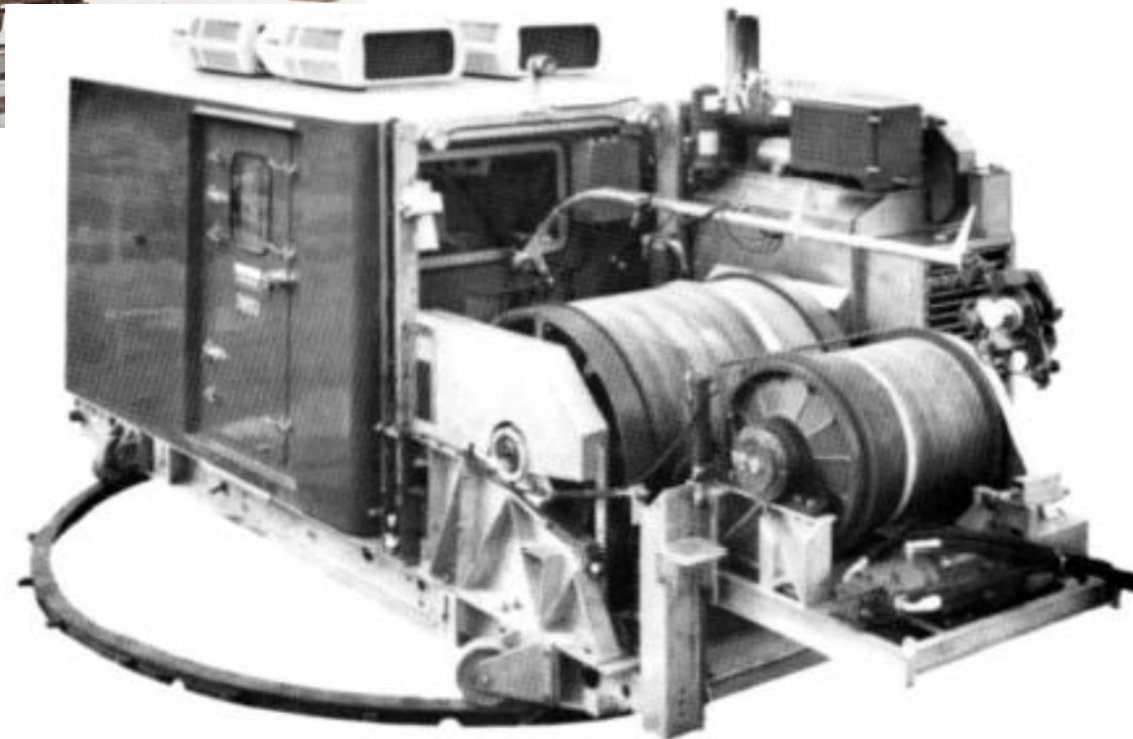
- To run wireline logs, the hole is circulated clean, stabilized and the drill string tripped out of hole.
- Wireline electrical logging is done using highly specialized equipment entirely separate from that used for drilling.
- Onshore, a motorized logging truck is used which brings its array of surface recorders, computers and a logging drum and cable to the drill site. Offshore, the same equipment is installed in a small cabin left permanently on the rig (See Figure 4).
- Both truck and cabin use a variety of interchangeable logging tools, which are lowered into the well on the logging cable

ing



(a) Truck-mounted Logging Unit

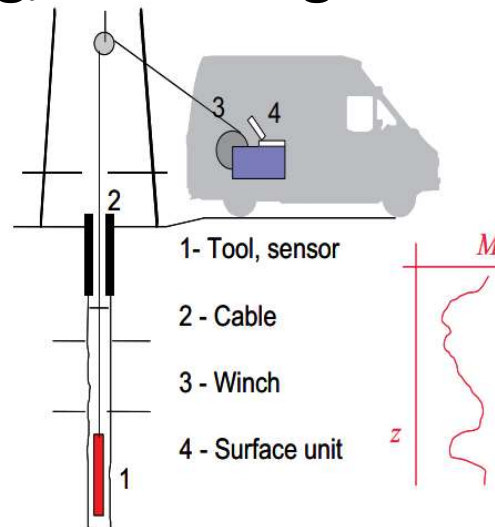
Figure 4



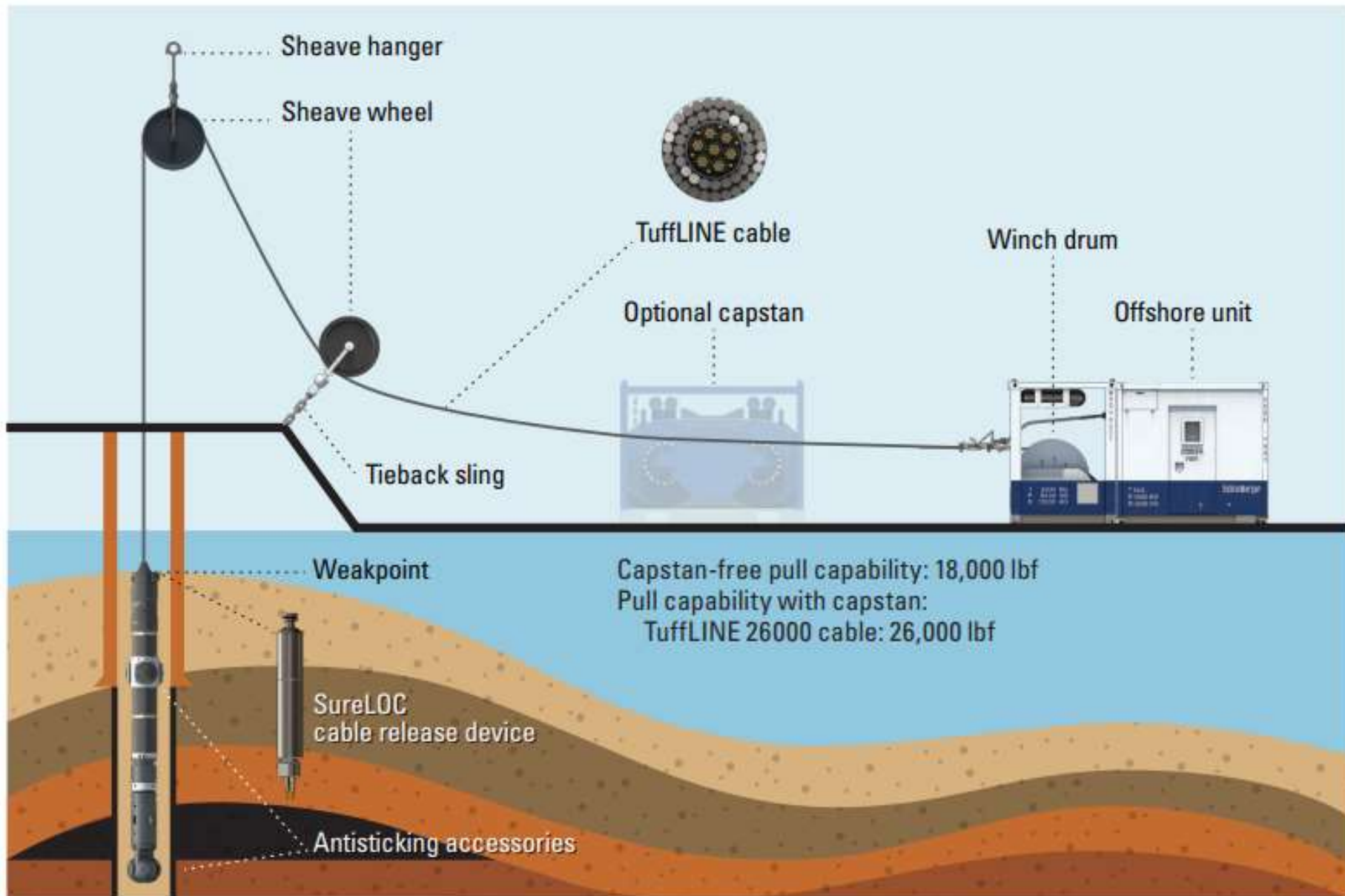
(b) Offshore Logging Unit

Wireline Logging Setup

- A Wireline Logging Setup has the following main components
 - Downhole measuring instruments (Tool or probe with sensors, transmitters, sources, the cartridge)
 - Electrical Cable connected with the probe by a cable-connector. The cable gives depth information about tool position, transmits the energy downward and the measured data upward
 - Winch with depth counter
 - Surface unit for powering downhole instruments, receive and process signals, visualizing, recording and storing measured data (Mobile Laboratory)



Wireline Logging Setup



The integrated wireline high-tension conveyance system provides efficient and reliable deployment.

Wireline Logging – The Sonde

- Of primary interest is the measurement device, or sonde which contains the sensors. There are over fifty different types of these logging tools developed in order to meet various information needs and functions.
- Superficially, sondes all resemble one another. They are generally cylindrical devices with an outside diameter on the order of 4 in. or less; this is to accommodate operation in boreholes as small as 6 in. in diameter. Their length varies depending on the sensor array used and the complexity of associated electronics required. It is possible to connect a number of devices concurrently, forming tool strings as long as 100 ft.
- The sonde housing may be constructed of steel and/or fiberglass

Wireline Logging – The Sonde

- Some sondes are designed to be operated in a centralized position in the borehole. This operation is achieved using bow-springs attached to the exterior, or by more sophisticated hydraulically actuated “arms.”
- Some measurements require that the sensor package (in this case called a pad) be in intimate contact with the formation. This is also achieved using a hydraulically actuated back-up arm.



Wireline Logging – Cartridge & Cable

- The other component of the downhole tool is the **cartridge** which contains the electronics that powers the sensors, process the resulting measurement signals, and transmits the signals up the cable to the truck. The cartridge may be a separate component screwed to the sonde to form the total tool, or it may be combined with the sensors into a single tool.
- Wireline refers to the armoured cable by which the measuring devices are lowered and retrieved from the well and, by a number of shielded insulated wires in the interior of the cable, provide for the electrical power of the device and a means for the transmission of data to the surface.
- Most cables contain seven insulated copper conductors. New cable developments include a fibre optics conductor in the centre of six conductors. The cable is wrapped with a steel armour to give it strength to support the tool weight and provide strength should the cable become stuck.

Wireline Logging – Depth Control

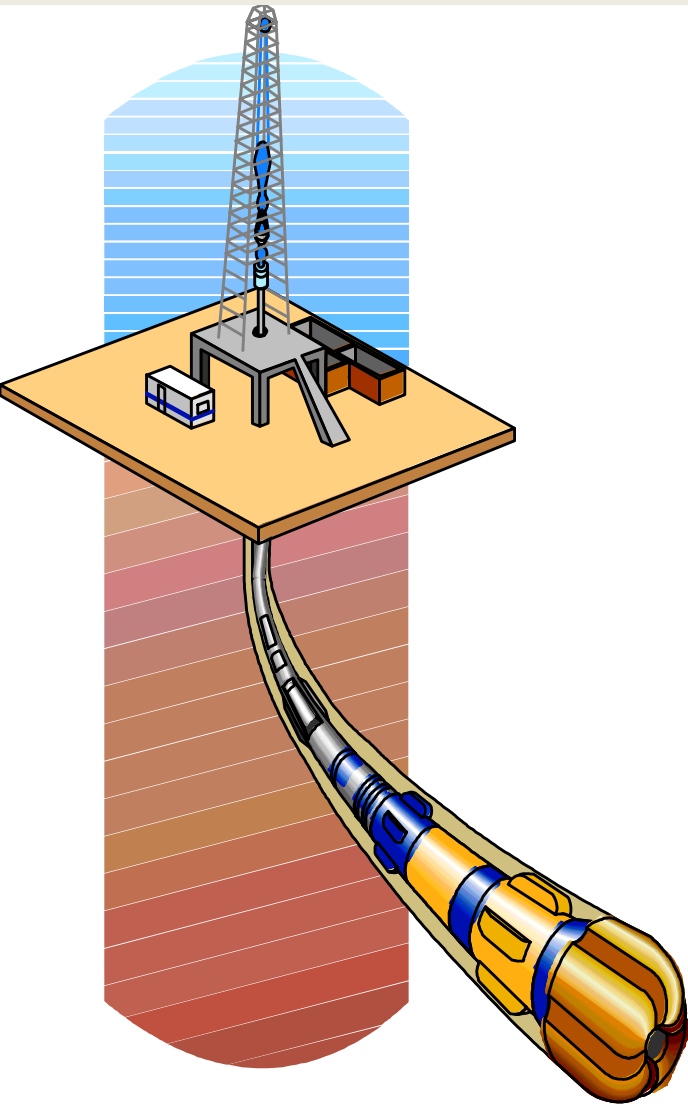
- A depth calibration wheel records the length of cable in the hole.
- Signal transmission over the cable is usually in digital form
- Survey is normally done from the bottom up. As the sonde is pulled up the hole, a continuous measurement signal is sent to the surface where the data is processed and recorded as a curve.
- The mobile laboratory provides the downhole power to the instrument package. It provides the cable and winch for the lowering and raising of the sonde, and is equipped with computers for data processing, interpretation of measurements, and permanent storage of the data.

LOGGING WHILE DRILLING (LWD)

Measurement While Drilling (MWD)

- To assist drillers in the complex task of a rotary drilling operation, a number of types of information like the downhole weight on bit and the downhole torque at bit are desirable in real time. To respond to this need, a type of service known as measurement while drilling (MWD) began to develop in the late 1970s.
- A typical MWD system consisted of a downhole sensor unit close to the drill bit, a power source, a telemetry system, and equipment on the surface to receive and display data. The telemetry system was often a mud pulse system that used coded mud pressure pulses to transmit at a few bits per second the measurements from the downhole subassembly. The power source was a combination of a generating turbine, deriving its power from the mud flow, and batteries.

Measurement While Drilling (MWD)



- MWD tools
 - Measure the direction & inclination of the wellbore
 - Allow drilling tools to be oriented
 - Provide mechanism for transmitting downhole data to the surface
- MWD comprises
 - Downhole telemetry and sensor system
 - Mud pulse transmission
 - Surface receiver decoding/display/ archive
- Depth
 - Drill line encoder
 - Measurements in time
 - Regular TIME sampling

To “convey” wireline logging tools through the complex geometry of the non-vertical well, another technology , referred to as LWD and provides, in addition to drilling-specific MWD measurements, a family of measurements entirely analogous to the measurements of traditional wireline logging.

Logging While Drilling (LWD)

- One of the major drawbacks of wireline information is that it is received several hours to several weeks after the borehole is drilled.
- During this time period, the formation can undergo significant alteration, especially in its fluid saturation, effective porosity, and relative permeability.
- LWD allows wireline-type information to be available as near as real-time as possible.
- Logging While Drilling (LWD) is a technique of conveying well logging tools into the well borehole downhole as part of the bottom hole assembly (BHA).
- The LWD tools are all built into heavy thick-walled drill collars – the special portion of the drill string used to counter buoyancy and provide stiffness to the lower segments of the drill string.

Logging While Drilling (LWD)

- These tools may transmit data as pulses in the mud column in real time while drilling to the surface or store the data in a downhole memory from which it may be downloaded when the assembly is brought back to the surface.
- In a typical operation, both modes will be used, with the memory data superseding the pulsed data once the tool is retrieved. The use of both modes is limited by:
 - Drilling mode: Data may be pulsed only if the drillstring is having mud pumped through it.
 - Battery life: Depending on the tools in the string, tools may work in memory mode only between 40 and 90 hours.
 - Memory size: Most LWD tools have a memory size limited to a few megabytes. Once the memory is full, the data will start to be overwritten. Depending on how many parameters are being recorded, the memory may become full within 20–120 hours.
 - Tool failure: It is not uncommon for a fault to develop in the tool such that the pulse data and/or memory data are not transmissible/recordable.

Logging While Drilling (LWD)

- Another important consideration with LWD tools is how close to the bit they may be placed in the drilling string. While the petrophysicist will obviously want the tools as close to the bit as possible, there may be limitations placed by drilling, whose ability to steer the well and achieve a high ROP is influenced by the placement of the LWD toolstring.
- As the drill collar is rotated, data can be acquired from multiple azimuths around the borehole, something not often achievable with a wireline.
- Unlike wireline tools that are generally of a standard diameter, many of the LWD tools come in families of sizes (e.g., 4, 6, and 8 in.). This is to accommodate popular drilling bit sizes and collar sizes since the LWD device must conform to the drilling string.

Logging While Drilling (LWD)

- However, LWD tools present a complication for drilling, as well as additional expense. However, their use may be justified when:
 - Real-time information is required for operational reasons, such as steering a well (e.g., a horizontal trajectory) in a particular formation or picking of formation tops, coring points, and/or casing setting depths
 - Acquiring data prior to the hole washing out or invasion occurring
 - Safeguarding information if there is a risk of losing the hole
 - The trajectory is such as to make wireline acquisition difficult (e.g., in horizontal wells)

Basic Data Differences between LWD & Wireline

LWD

- Almost always Recorded Down
- Time sampled
 - Resolution depends on ROP
 - Sensor spacing
- Real time data density depends on ROP and acquisition rate
- Memory – full dataset
- Downhole – storage/data limited
- Sensors fixed in drill collar
- Typically recorded with minimal invasion effects;
- Small borehole enlargement
- Affected by stick/slip
- Logging rate of slowest tool (e.g. SoundTrak™ – 15 sec acq.)
- Can log in any direction
- More capable in tough environment

Wireline

- Almost always Recorded Up
- Depth sampled
 - Surface controlled
 - High resolution possible
- All data telemetried while logging, but after drilling; Tools lowered in borehole on cable
- Future Memory options – slickline
- Storage limit is surface disk size
- Sensors positioned as required (cent/decent/ pads)
- Recorded after invasion (susceptible to hole condition)
- Significant to moderate borehole effects
- Affected by stick/pull
- Large combinations may affect logging speed

LWD versus Wireline



TOOL FAILURES AND STUCK PIPE

Tool Failures

- The standard procedure if a tool fails is to replace it with a backup and carry on as before. In general, if the program has specified that a tool is to be run over a specified interval (particularly a reservoir interval), then it is best to ensure that good-quality data are acquired, even if it means making additional runs. However, the following situations may arise:
- When logging with LWD and in the hole drilling near the end of a section, it may be far more cost-effective, if the data are not critical, to simply carry on drilling and re-acquire the data during a check trip. In some cases, the memory data may still be usable.
- If a wireline tool starts to act erratically, it may not justify re-running the tool, since the data may not be critical over the particular interval and can sometimes be corrected by post-processing the raw tool data at the surface.

Tool Failures

- If an advanced tool fails, it sometimes happens that no backup tool is available on the rig. The choices then are to either try and repair the tool, arrange a backup tool to be sent from another location, or replace the tool with an earlier version with less capability. Most commonly, the latter option is chosen.
- In the event of failures occurring with potential safety implications (e.g., explosive charges going off accidentally), it is normal for operations to be suspended until a full investigation has been carried out to establish the cause.

Stuck Tools

- Fairly regularly during logging operations, tools get stuck in the hole, either temporarily or permanently. There are often indications of bad hole conditions during drilling, and the logging program may be adjusted to take this into consideration. In general, it is found that the longer a hole is left open, the greater the likelihood of hole problems occurring. Three sorts of sticking are found to occur:
 - differential sticking,
 - key seating, and
 - holding up.
- Differential sticking occurs when either the cable or the toolstring becomes embedded in the borehole wall and gets held in place by the differential pressure between the mud and the formation. In such a situation, it is impossible to move the toolstring either up or down. The usual procedure is to alternately pull and slack off on the tool, pulling up to 90% of the weak point of the cable (that point at which the cablehead will shear off the top of the toolstring). This procedure is often successful, and a tool may become free after 30 minutes or so of cycling.

Stuck Tools

- Key seating occurs when a groove is cut into one side of the borehole, which allows the cable, but not the toolstring, to pass upward. The toolstring is effectively locked in place at a certain depth. Unfortunately, this often means that when the weak point is broken, the toolstring will drop to the bottom of the hole and may be hard to recover or may be damaged.
- Holding up occurs when a constriction, blockage, dogleg, or shelf occurs in the borehole such that the toolstring may not pass a certain depth, although it can be retrieved. The usual practice in such a situation is to pull out of the hole and reconfigure the tool in some way, making the toolstring either shorter or, in some cases, longer in an attempt to work past the holdup depth.
- Once it is no longer possible to recover a tool on wireline, there are two options: cutting-and-threading or breaking the weak point. In the cut-and-thread technique, the cable is cut at the surface. The drillpipe (with a special fishing head called an overshot) is run into the hole with the cable being laboriously threaded through each stand of pipe.

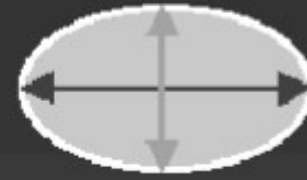
Stuck Tools

- At a certain depth it may be possible to install a Side-Entry Sub (SES), which means that when the toolstring is being recovered, log data may be acquired in a similar fashion to pipe-conveyed logging. If the weakpoint has been broken (either accidentally or on purpose), cutting-and-threading is not possible. Then the pipe is run into the hole until the toolstring is tagged, although often the tool will drop to the bottom of the hole before it can be engaged by the overshot.
- Most oil companies will specify that they do not wish to break the weak point on purpose, even if the cut-and-thread technique is much slower. This is particularly true where nuclear sources have been lost in the hole, and every effort should be made to ensure that they are recovered undamaged at the surface. In the event of nuclear sources being lost irretrievably in the hole, which occasionally happens (particularly with LWD tools), there are usually special procedures to be followed involving notification of government bodies and steps taken to minimize the risk of and monitor any potential nuclear contamination that may occur.

Passive Well Logging Measurements

– Lecture 3

Dual Axis (x,y)



CALIPER LOG

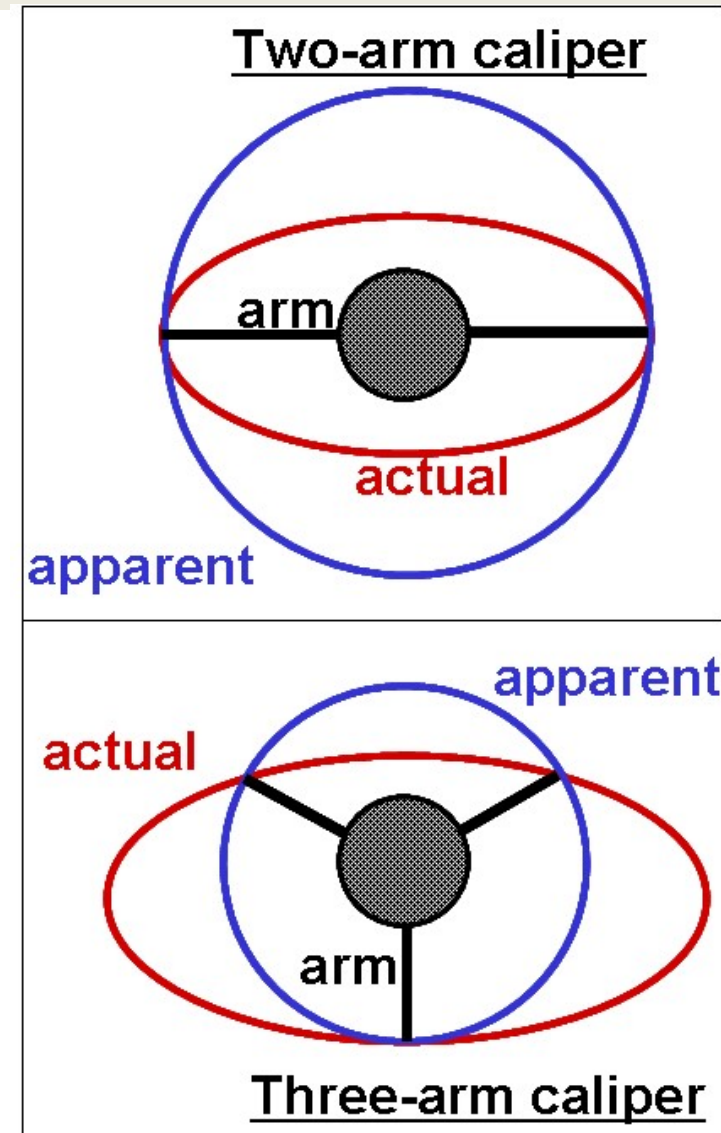
Calipers

- **The Caliper Log**

- is a continuous profile of the borehole wall showing variations in borehole diameter.

- **Uses**

- Indication of Hole volume and diameter: required for mud and cement volume in hole
- Environmental corrections: hole diameter from the caliper is used in charts for other logs and to indicate thickness of the mud cake for R_{xo} tool corrections.
- Qualitative indication of permeability: due to development of mudcake in association with gamma ray log.
- Correlation for lithological assessment
- Selection of consolidated formations for wireline pressure tests, fluid sampling, packer seating and for determining casing setting depth
- Log quality control: Indication that the hole is rough is a warning that the measurements are from tools pushed against the borehole wall such as Density, Neutron, Micro-resistivity curves, may not be reliable. Thus, it is an indicator of Hole Quality.



Calipers

- **Physics of Measurement**

- For wireline tools, the physical movement of arms on the tool (*Mechanical Systems*) is converted to a diameter measurement through electrical circuitry. The angle between the pad mechanism and the tool axis is a measure of pad distance from the tool.
- In some MWD tools the caliper curve is generated based on the response of the detectors as the tool rotates (gyro system). Others use ultrasonic sensors which measure the two-way travel time taken for an acoustic pulse from the sensor to the formation wall and back. Conversion into distance via mud wave velocity gives a detailed azimuthal caliper. Some tools come with built in calipers by design.

- **Properties**

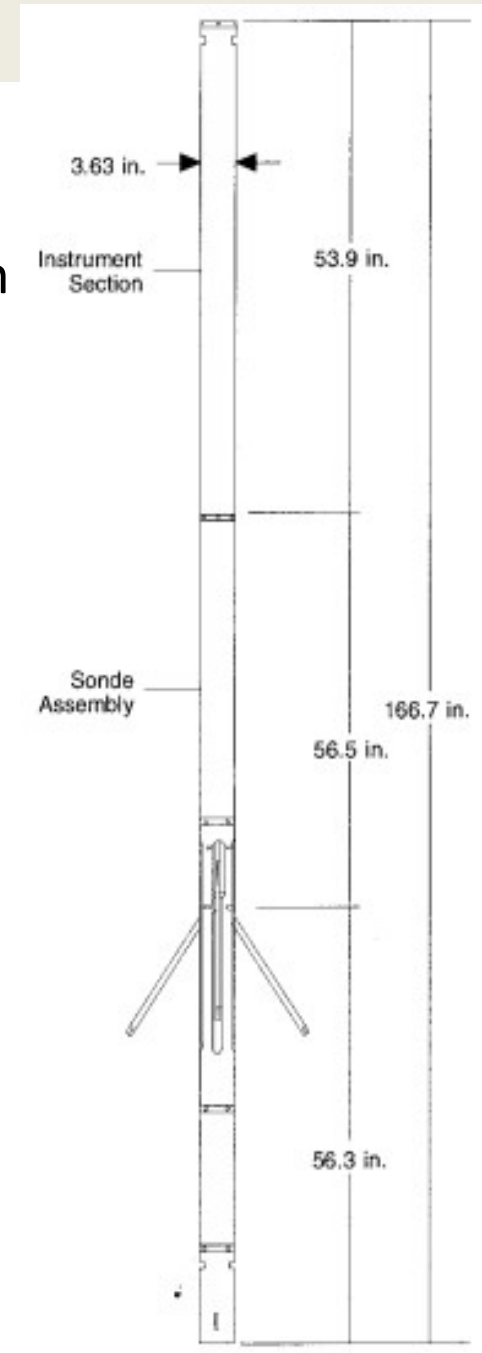
- two, three, four arms or more; linked or independent

- **Operational Constraints**

- In Open hole, the tool is run centered
- In Cased hole, the tool is run eccentric
- Can be run in air, water or oil-based mud

- **Secondary Effects**

- In highly deviated holes, the caliper mechanism may not be strong enough to support the weight of the logging tool thus may not indicate actual hole diameter
- Mud cake indications can be masked by a wash out



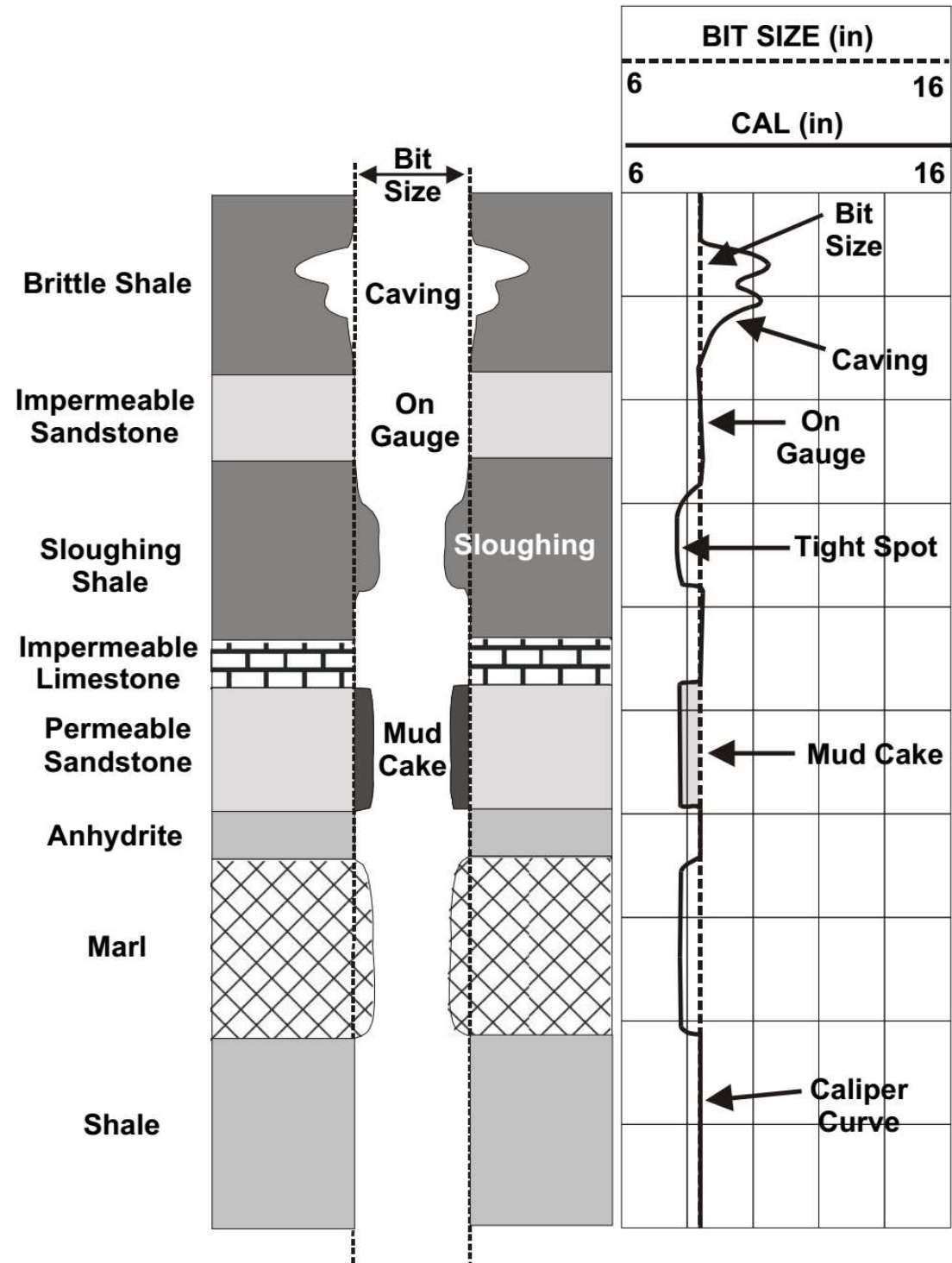
4-arm Caliper Tool

Caliper log

- **Log Presentation**

- The caliper logs are plotted in track 1 with the drilling bit size for comparison, or as a differential caliper reading,
- The scale is generally given in inches, which is standard for measuring bit sizes..

***N.B.:** Check that the caliper is reading correctly in casing and in non-permeable zones that are not washed out (when a hole is the same diameter as the bit-size it is called “on gauge”).*



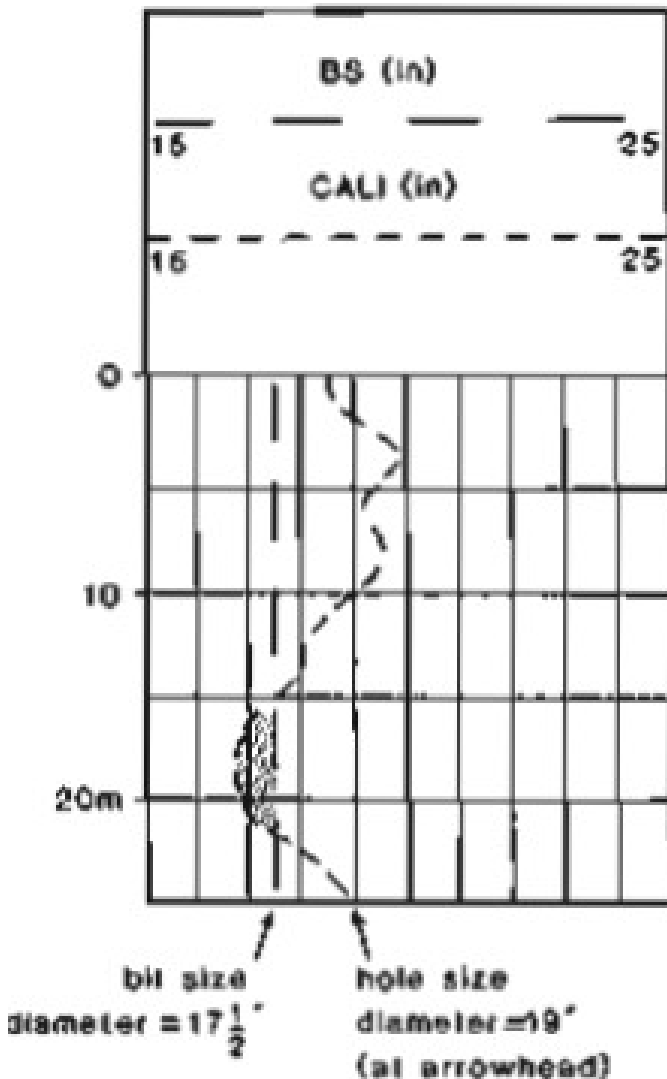
Typical caliper responses to various lithologies

Factors affecting Caliper tool responses

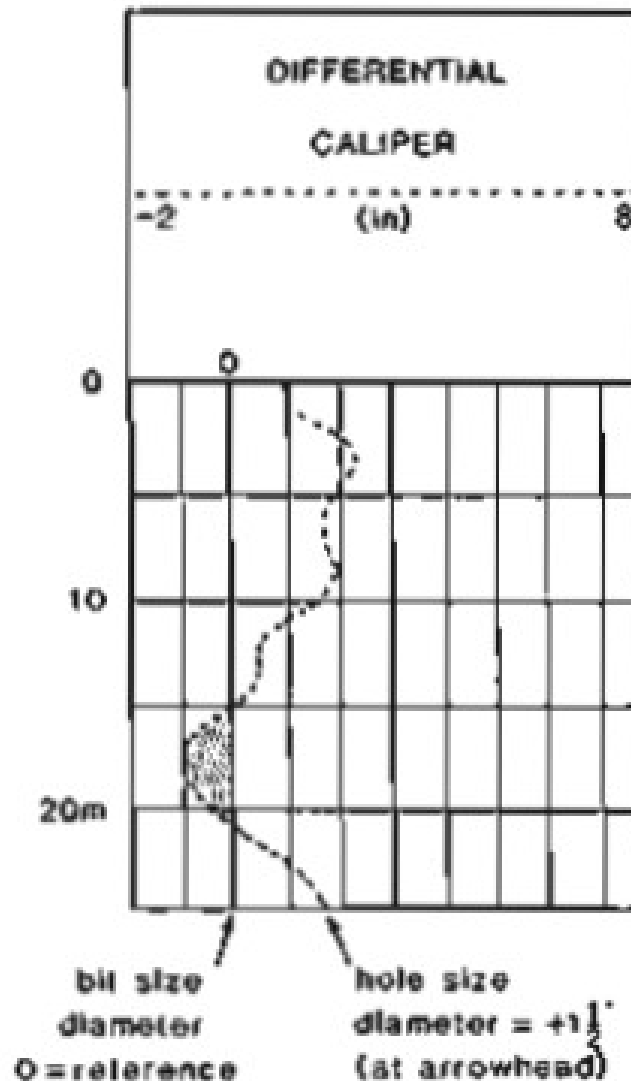
Hole Diameter	Cause	Possible Lithologies
On Gauge	Well consolidated formations Non-permeable formations.	Massive sandstones Calcareous shales Igneous rocks Metamorphic rocks
Larger than Bit Size	1. Formation soluble in drilling mud. 2. Formations weak and cave in.	1. Salt formations drilled with fresh water. 2. Unconsolidated sands, gravels, brittle shales.
Smaller than Bit Size	1. Formations swell and flow into borehole. 2. Development of mudcake for porous and permeable formations.	1. Swelling shales. 2. Porous, permeable sandstones.

Caliper log Presentations

1. CALIPER



2. DIFFERENTIAL CALIPER



The differential scale expresses the increase or decrease in hole diameter about a zero defined by the bit size.

Presentation of the caliper log;
Left – in ordinary format; Right – in differential format

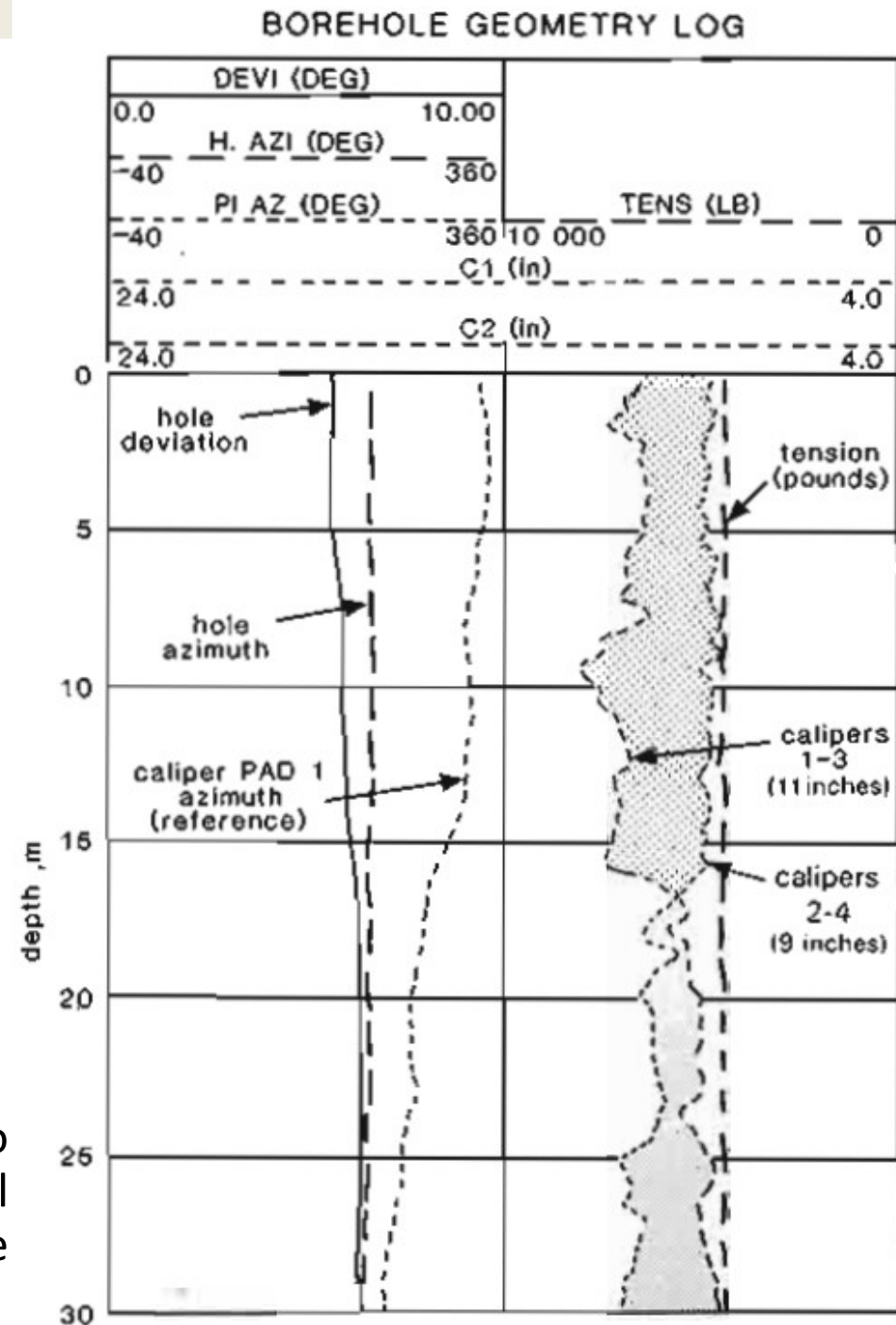
Borehole Geometry log Presentation

The geometrical data from a four-arm, dual-caliper tool is shown turning slowly as it moves upwards in a persistently oval hole with small and large diameters of 9" & 11" respectively.

The larger diameter is oriented nearly north to south as indicated by the pad 1 azimuth over the depth 0-15m (calipers 1-3 in larger diameter).

The rotation of the tool is indicated by the persistent change in pad 1 azimuth and explains the cross-over at depth 17m.

The two-hole diameters measured by the two calipers are combined with the directional elements of tool orientation (pad 1 azimuth), hole deviation and azimuth of the deviation.



THE SPONTANEOUS POTENTIAL (SP) LOG

Spontaneous Potential (SP)

- The SP is a passive measurement of very small direct current (DC) voltages resulting from electric currents in the borehole caused by the differences in the salinities (resistivities) of the formation connate water and the drilling mud filtrate (WBM), & by the presence of ion selective shale beds.
- The voltage changes are measured by a downhole electrode relative to fixed electrode located at the surface measured in **millivolts**.
- The SP is mostly measured simultaneously with the open hole resistivity log.
- Unlike other logging tools which are displayed on a specific scale with a specified reference value, the SP has no specified origin and values used for computation are referenced to deflection from a nearby shale baseline established by the interpreter.

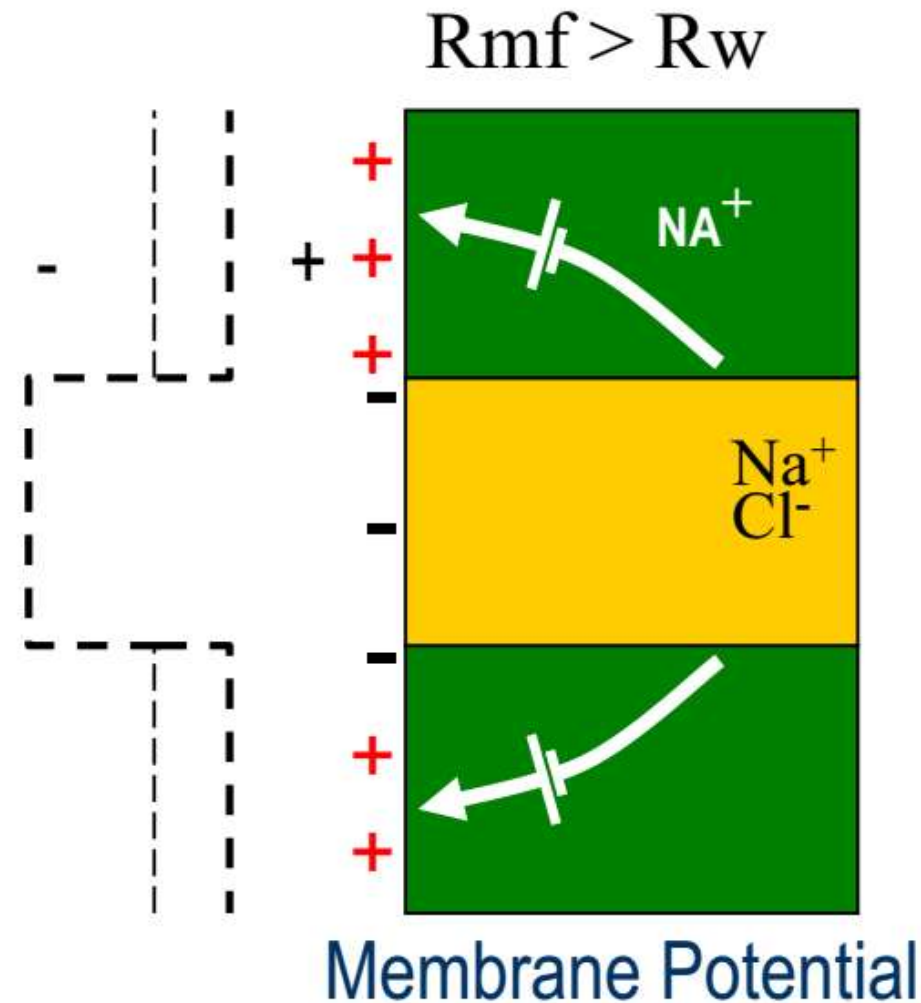
Spontaneous Potential (SP)

- SP is the charge exchange process driven by the difference of concentration (salinity) between formation water C_w and the mud filtrate C_{mf} . The concentration difference (in most cases $C_w > C_{mf}$) results in an ion motion controlled primarily by **electrochemical properties** of the:
 - Non-permeable shales creating a positive **membrane potential** E_m as a result of anion-blocking (ion-selectivity) of clay minerals i.e. selective to sodium ions (Na^+) but impervious to chloride ions (Cl^-).
 - **Liquid-junction potential** E_j : which arises due to net transfer of negative electric charges from the more concentrated sodium chloride solution to the less concentrated at the boundary between the invaded and un-invaded zones in the formation

Spontaneous Potential - Physics of Measurement

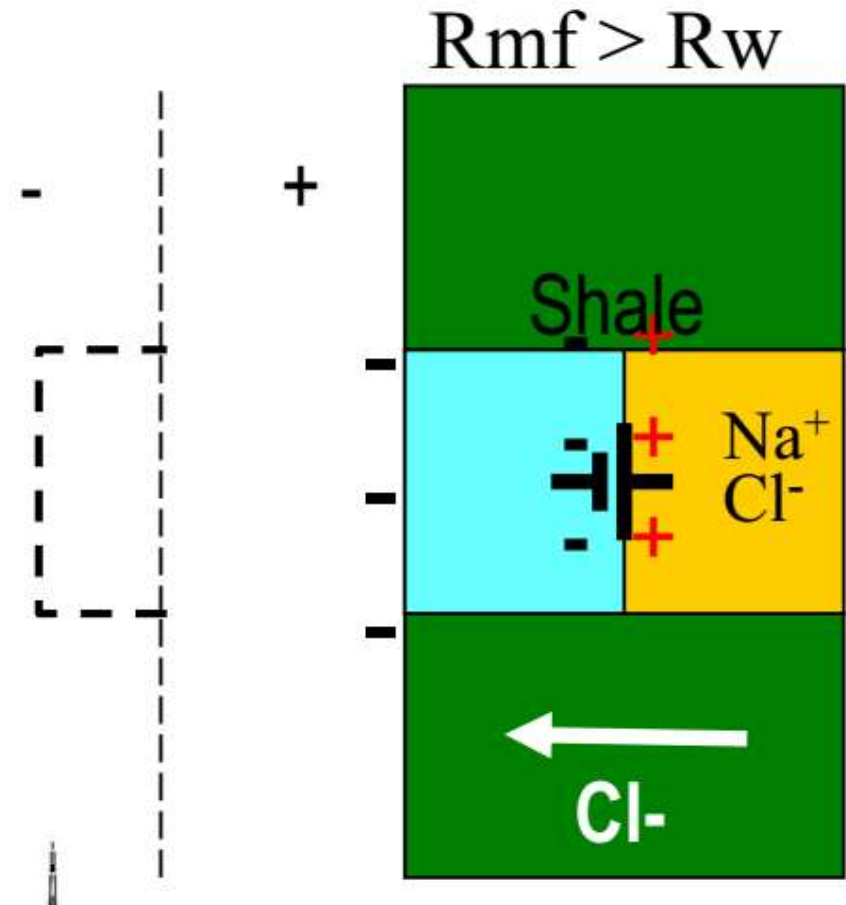
- **Shale membrane potential E_m :**

- Ion selective membranes allow ions with certain electrical charges (positive or negative) to pass through them. Shales act like ion selective membranes as they allow only cations (positively charged ions) to pass through. This flow of charged particles is an electric current.
- The Membrane Potential graphic illustrates the current flow for the case where $R_{mf} > R_w$ (fresh mud). The flow would be reversed if $R_{mf} < R_w$ (salt mud).
- Membrane potential is normally 5 times larger than the liquid junction potential



Spontaneous Potential - Physics of Measurement

- **Liquid-junction potential E_j :**
 - Ions will move between liquids of differing salinities. The ions will normally move from higher salinity to lower. The most abundant ions in a wellbore are sodium (Na^+) and chlorine (Cl^-).
 - Because chlorine is a smaller atom it moves easier and faster through the formation (permeable sandstone).
 - Current flow will be from the lower salinity to the higher salinity. In this case from the invaded zone to the true zone.



Spontaneous Potential

- **Operational Constraints**

- Only run in Open hole with the tool centered
- Only run in water-based mud. Because a conductive fluid is needed in the borehole for the SP log to operate, it **cannot be used** in non-conductive i.e OBM
- There is no separate SP tool; the SP band is usually run with induction logs and other electric logs

- **Environmental Effects**

- In highly deviated holes, the caliper mechanism may not be strong enough to support the weight of the logging tool thus may not indicate actual hole diameter
- Mud cake indications can be masked by a wash out

- **Caution as regards interpretation**

- Unlike other logging tools which are displayed on a specific scale with a specified reference value, the SP has no specified origin and values used for computation are referenced to deflection from nearby shale baseline established by the interpreter.
- A surface earth is required for the SP tool to work effectively. This is difficult offshore especially on floating rigs.

Spontaneous Potential - Uses

- Correlation of formations from well to well: Curves are scanned for similarities in shape and magnitude.
- Gross Lithology: Reservoir are shown as deflections (either positive or negative) from a relatively stable (and arbitrary) shale baseline. The direction of the deflection is determined by the relative salinities (resistivities) of the formation water (R_w) and the mud filtrate (R_{mf}), and is not directly related to formation porosity or permeability.
- As a rule of thumb the following relationships are true:
 - If $R_{mf} > R_w$, then SP deflection is negative
 - If $R_{mf} = R_w$, then SP deflection is zero
 - If $R_{mf} < R_w$, then SP deflection is positive
- Qualitative indication of permeability: The presence of an SP (positive or negative) opposite a bed indicates permeability. Only a minimal amount of permeability is required to develop an SP and therefore there is no technique to determine the magnitude of permeability from SP as permeability may be ionic not hydraulic.

Spontaneous Potential - Uses

- Identification of depositional environments: Depo. environment can be inferred from the shape of the SP. It however can be ambiguous; thus, it should be used in support of other data in an area of interest.
- Estimate of Clay Content: This method assumes a constant R_w for all zones. It also assumes that the response of SP to shaliness is linear.

Alternatively,

Shale/clay volume equation:

$$V_{clay} = V_{shale} = \left(\frac{SP_{clean} - SP_{log}}{SP_{clean} - SP_{shale}} \right)$$

$$V_{sh} = 1.0 - \frac{PSP}{SSP}$$

$V_{clay} = V_{shale}$ = Shale or clay volume.

SP_{log} = SP in the zone of interest (read from the log).

SP_{clean} = maximum SP deflection from a nearby clean wet zone in the same well.

SP_{shale} = SP value at the shale baseline (often considered to be zero).

Spontaneous Potential - Uses

- Estimate of formation water resistivity: Since the electrolyte's specific electrical resistivity and concentration are inverse with a good approximation, the ratio C_w/C_{mf} can be replaced with the ratio R_{mf}/R_w .
- The resistivity of the formation water R_w is a fundamental input for the calculation of water saturation. The equation offers the possibility of a determination from the SP-log at the formation temperature with a water sample.
- For R_w determination, R_{mf} – the mud resistivity (at formation temperature), SSP – the difference between the SP level in a clean zone and the shale baseline, and T – the formation temperature are used as input.
- A monogram is frequently used for R_w determination.
- A good estimate of R_w at formation temperature can be obtained from the following equation: (R_{mf} @ formation temp.)

$$SSP = k_{SP} \cdot \log\left(\frac{C_w}{C_{mf}}\right) = k_{SP} \cdot \log\left(\frac{R_{mf}}{R_w}\right)$$

$$R_w = 10^{(K \cdot \log(R_{mf}) + SP) / K}$$

Static spontaneous potential (SSP)

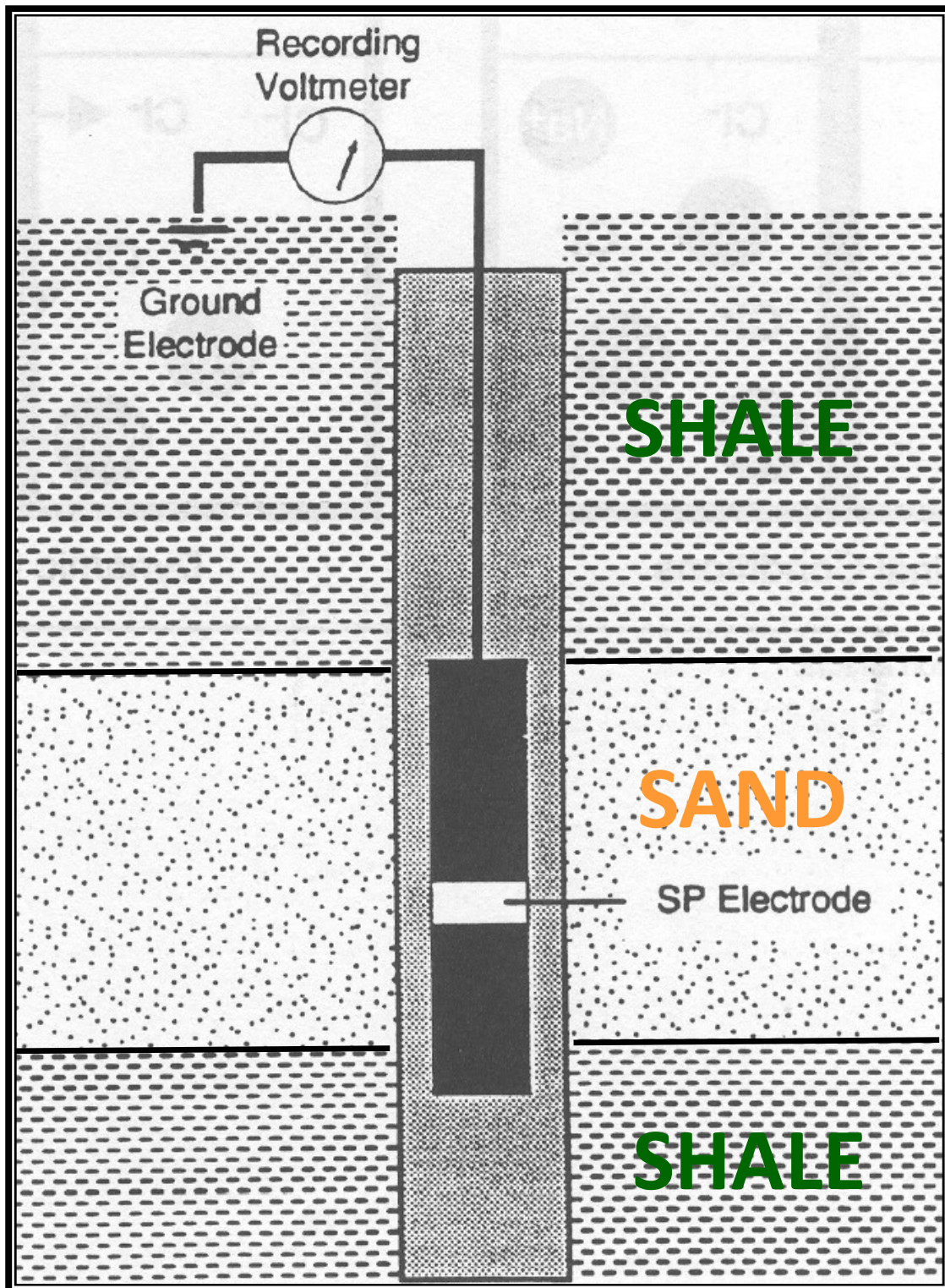
- The concept of **static spontaneous potential (SSP)** is important because SSP represents **the maximum SP** that a thick, shale-free, porous and permeable formation can have for a given ratio between R_{mf} / R_w .
- It is the maximum difference between the most negative and most positive level defined as:

$$SSP = |E_D| + |E_M| = k_{SP} \cdot \log\left(\frac{C_w}{C_{mf}}\right)$$

- SSP is determined by formula or chart and is a necessary element for determining accurate values of R_w and **volume of shale**.

Static spontaneous potential (SSP)

- The parameter k_{SP} is controlled by temperature, type of dissolved salts (electrolyte), and membrane activity of shale; it is about 71 mV at 25°C (or °F) for NaCl. For practical application, the following empirical equation can be used:
 - $k_{SSP}(\text{in } mV) = - [65 + 0.24 * T] (\text{in } ^\circ C)$
 - $k_{SSP}(\text{in } mV) = - [60 + 0.133 * T] (\text{in } ^\circ F)$
- The SP curve is influenced by several effects (correction charts eliminate most effects):
 - Bed not thick enough
 - The invasion is deep
 - There is shale in the formation
 - The presence of hydrocarbons
 - Adverse lithological effects (junction beds with high resistivity)
- Frequently a “shale baseline shift” can be observed as a result of change of clay mineralogy, water mineralization and/or temperature with depth. A baseline shift occurs whenever formation waters of different salinities are separated by a shale bed that is not a perfect cationic membrane.

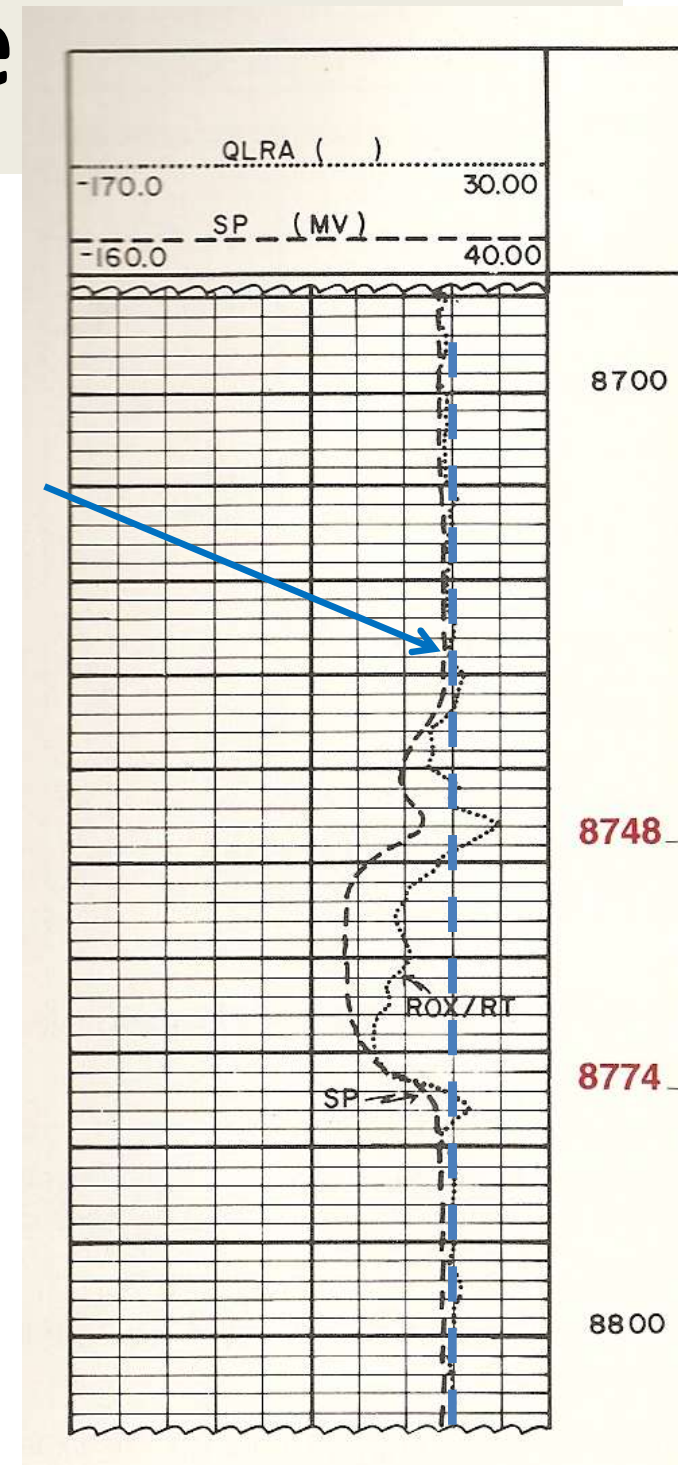


THE SP TOOL

- One movable electrode
- Insulators on either side
- Surface ground electrode – at a stable potential

Shale Base Line

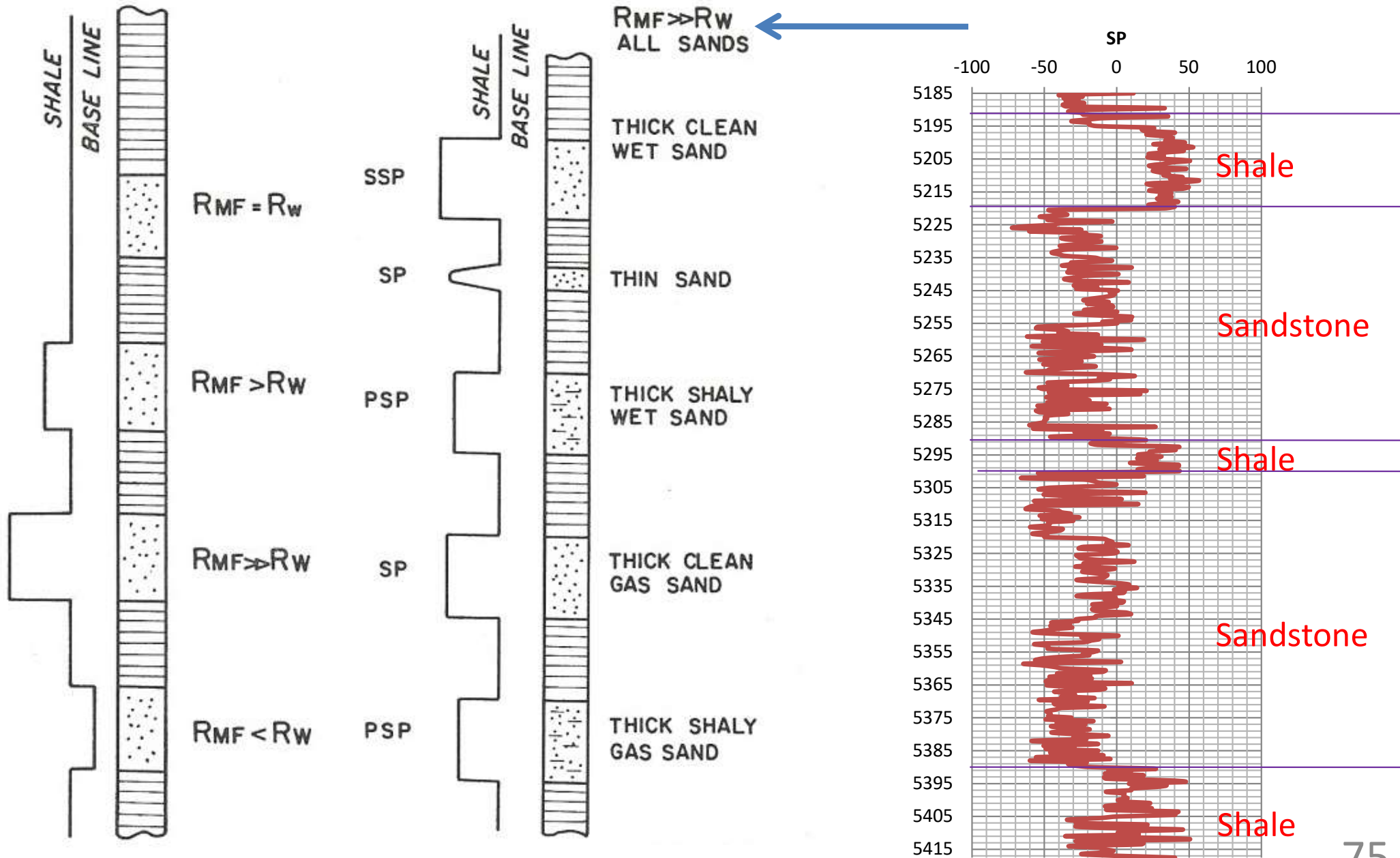
- The SP response of shales is relatively **constant** and follows a straight line called **a shale baseline**.
- SP curve **deflections** are measured from this shale baseline.
- *Permeable zones are indicated where there is **SP deflection** from **the shale baseline**.*



Log Characteristics

- Bed boundary definition: The sharpness of the boundary will depend on the shape and extent of the SP current patterns. Generally, when there is considerable difference between mud and formation resistivity, currents will be spread widely and the SP will deflect slowly. Conversely, when resistivity are similar, boundaries are sharper. The bed boundary should be placed at the point of maximum curve slope (inflexion point)
- SP bed resolution: It is poor. For a full SP deflection (SSP) and proper bed resolution, as a rule of thumb, a bed should be thicker than 20 times the borehole diameter. The exact minimum bed resolution will depend on depth of invasion and salinity differences

SP deflection



**Resistivity of Formation Water
(R_w) Calculates from the SP Curve**

Resistivity of Formation Water (R_w)

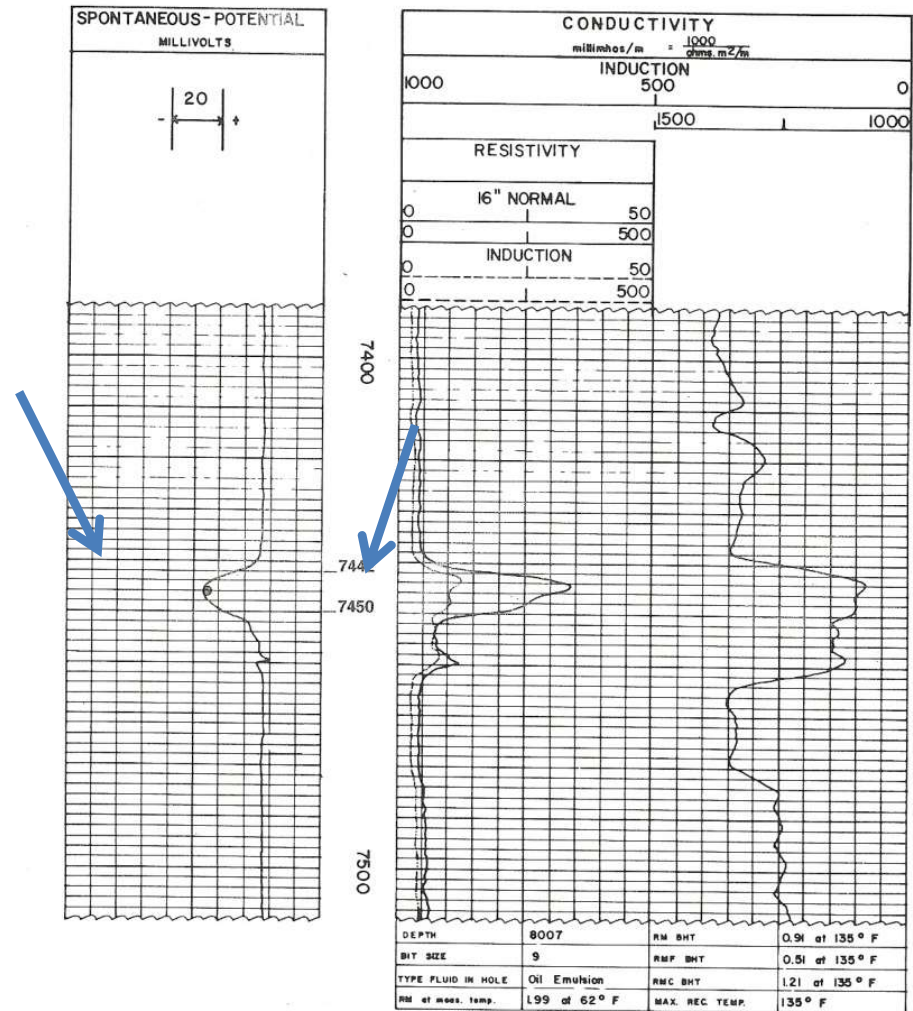
Calculated from the SP Curve

Step 0:

- Setup Shale-base-line
- read **SP log recording** and **16" Normal Resistivity log reading** every 2 ft from 7430 to 7460

Step 1:

- determine the formation temperature (T_f)



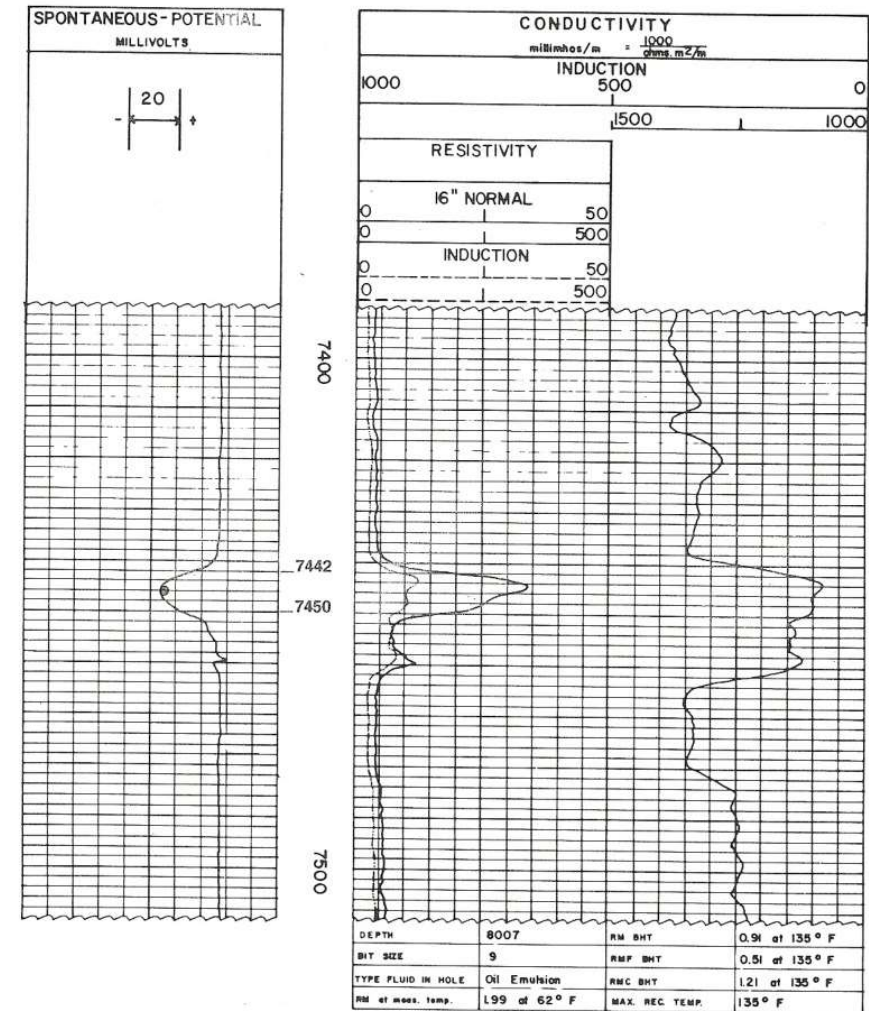
Resistivity of Formation Water (R_w) Calculated from the SP Curve (Cont.)

Step 2:

- correct the resistivities of the mud filtrate (R_{mf}) and drilling mud (R_m) to formation temperature (T_f)

Step 3:

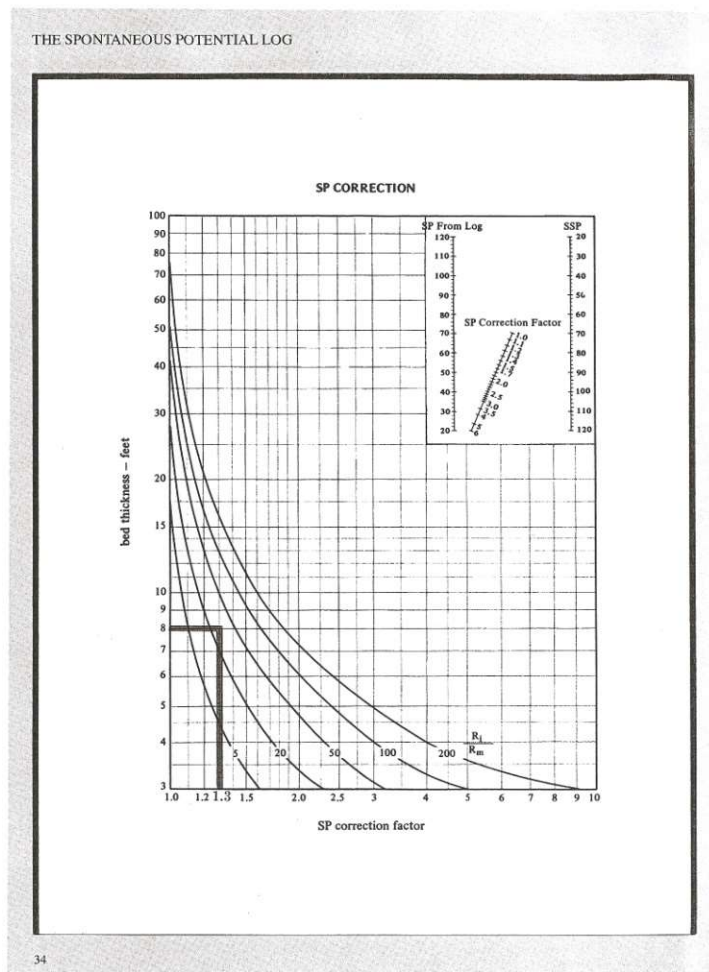
- Determine the SP reading (in water-bearing zone or wet-formation)



Resistivity of Formation Water (R_w) Calculated from the SP Curve (Cont.)

Step 4:

- Correct SP to SSP

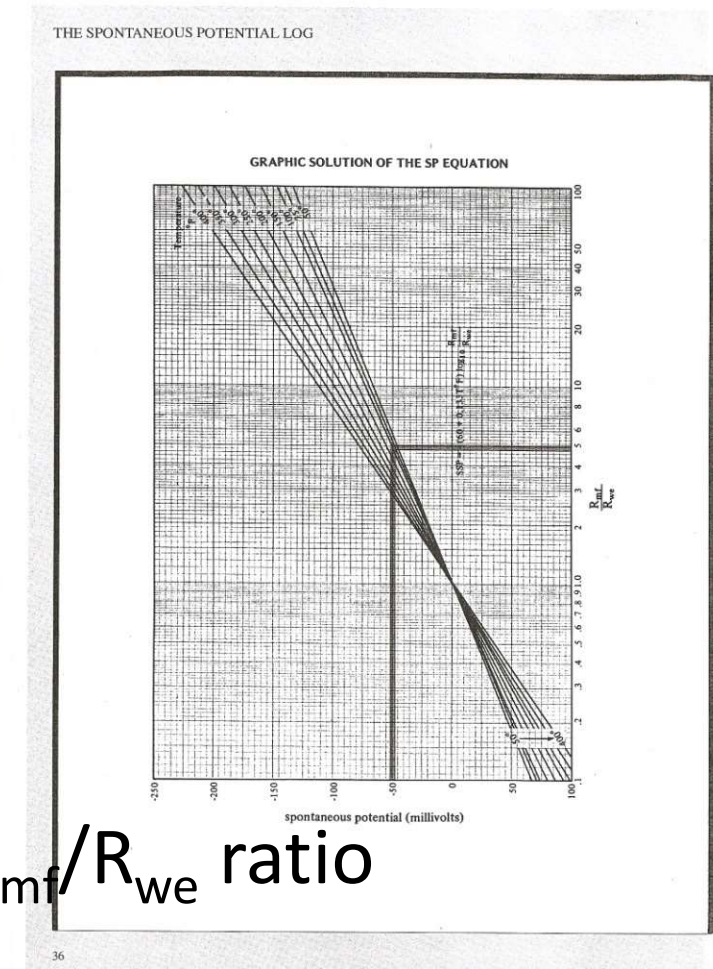


Step 5:

- Determine R_{mf}/R_{we} ratio

Step 6:

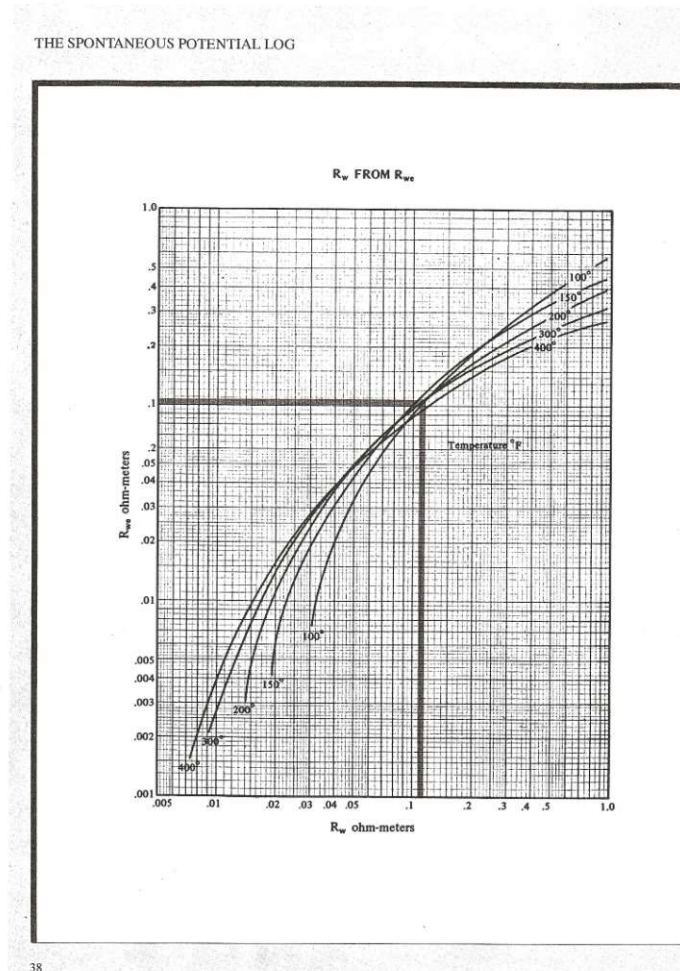
- Determine R_{we}



Resistivity of Formation Water (R_w) Calculated from the SP Curve (Cont.)

Step 7:

- Correct R_{we} to R_w



THE GAMMA RAY (GR) LOG

Natural Radioactivity

- Natural radioactivity is a spontaneous decay of a certain isotope into another isotope, characterized by emission of radiation. The three types of radiation (α , β , γ) have different ability to penetrate a material:
 - ✓ α , β : are particle radiation with a very shallow penetration or high absorption.
 - ✓ γ : is an electromagnetic wave with high penetration. Therefore, γ -radiation is preferred for practical application.
- Natural gamma activity of rocks and minerals is originated by:
 - ✓ Uranium-radium series (half-life time of 4.4×10^9 years)
 - ✓ Thorium series (half-life time of 1.4×10^9 years)
 - ✓ Potassium K^{40} (half-life time of 1.3×10^9 years) – mono-energetic radiation
- Occurrence and behaviour of these components in rocks are different. The abundance of these elements or isotopes in a formation controls the intensity of detectable natural radioactivity.
- In most cases the “API-unit” (American Petroleum Institute) is applied as measurement unit.

Gamma Ray Measurements

- The spectral character of decay results in two techniques of measurement:
 - ✓ **integral measurement**: measures all incoming counts above an energy threshold.
 - ✓ **spectral measurement**: measures in energy-windows and delivers count rates (and concentration) for the three components K (in %), U and Th (in ppm).
- With an integral gamma measurement the summarized contributions of all three origins are measured:
 - ✓ $I_{GR} = k * (a * K + U + b * Th)$
- The counts originated by each source-element are converted into equivalent U concentration by factors a and b is a factor characterizing the sensitivity of the sonde (determined by calibration).
- If a spectral gamma measurement is available, it is recommended to use a Gamma Ray Index calculated only from K and Th. The two components are dominated by the mineral composition, whereas U is controlled more by the environment of sedimentation and organic components. The spectral measurement helps us to detect k-feldspar, mica or glauconite as origin of radiation.
- N.B.: The spectral measurements are necessary for carbonate series as the integral gamma intensity is very often a poor clay indicator.

Gamma Ray Log

- **Types**

- Natural (Integral); -- Spectral (Thorium, Uranium, Potassium)

- **Physics of Measurement**

- The number of naturally occurring gamma rays from potassium, uranium, thorium and associated daughter products, is counted by the detector in both natural and spectral gamma ray tools.
- Spectral tools also measure the energy of each detected gamma ray. The range, or spectrum, of energy detected is divided into windows, or limited energy ranges, which indicate the elemental gamma ray source (i.e. specific isotopes of potassium, uranium or thorium)

- **Uses** (both Natural & Spectral Gamma Ray logs)

- Correlation: Curves are scanned for similarities in shape and magnitude. Lithological correlation from well to well if some wells are logged in a field
- Gross Lithology indicator; exploration for radioactive materials: In general, reservoir are less radioactive than shales. However some sandstones and dolomites can be radioactive.
- Evaluation of shale content: The magnitude of the gamma ray in the formation of interest (relative to that of nearby clean and shale zones) is related to the shale content of the formation
- Depth Adjustment: Discrepancies are caused for example by cable stretch. One GR log is chosen as a reference log for data quality. The other GR logs are matched to the reference using an automatic routine. Resulting depth shifts are applied to other logs on the tool string.

Shale Content from GR Log

- **Step 1:** Calculate Gamma Ray Shale Index

$$I_{GR} = \frac{GR - GR_{cn}}{GR_{sh} - GR_{cn}}$$

where

- GR = log reading at depth of interest
- GR_{cn} = Gamma Ray value in nearby clean zone
- GR_{sh} = Gamma Ray value in nearby shale

- **Step 2: Apply an empirical relationship between I_{GR} & Shaliness, V_{shale}**

- Linear relationship; $V_{sh} = I_{GR}$
- Non-Linear relationships

Steiber:

$$V_{shale} = \frac{I_{GR}}{3.0 - 2.0 \cdot I_{GR}}$$

Larionov (Tertiary rocks):

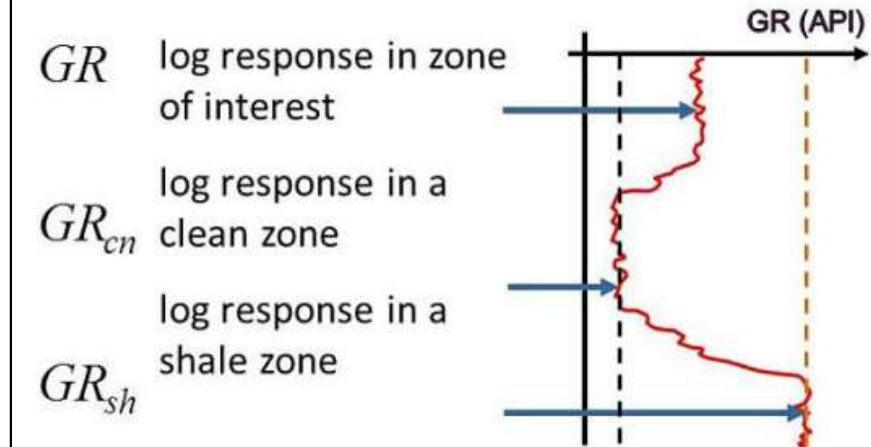
$$V_{shale} = 0.083 \cdot (2^{3.7 \cdot I_{GR}} - 1)$$

Clavier:

$$V_{shale} = 1.7 \cdot [3.38 \cdot (I_{GR} + 0.7)^2]$$

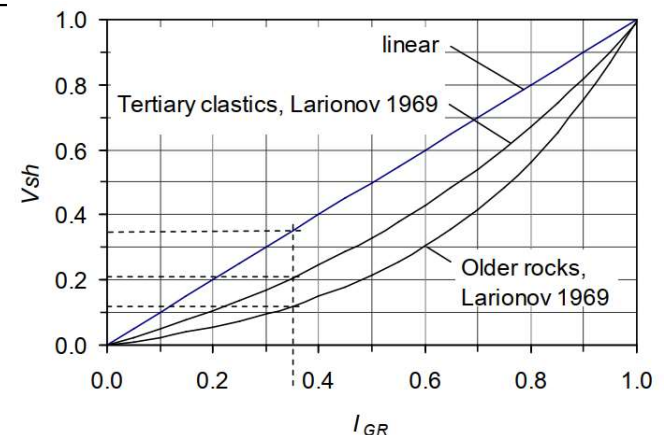
Larionov (older rocks):

$$V_{shale} = 0.33 \cdot [(2^{2 \cdot I_{GR}}) - 1.0]$$



This normalization of the logs results in $I_{GR} = 0$ for clean rocks (0% shale) & $I_{GR} = 1$ for pure shale (100% shale)

An empirical equation is recommended in most cases because the Gamma Ray Index is not identical with the volumetric determined shale content from core analysis



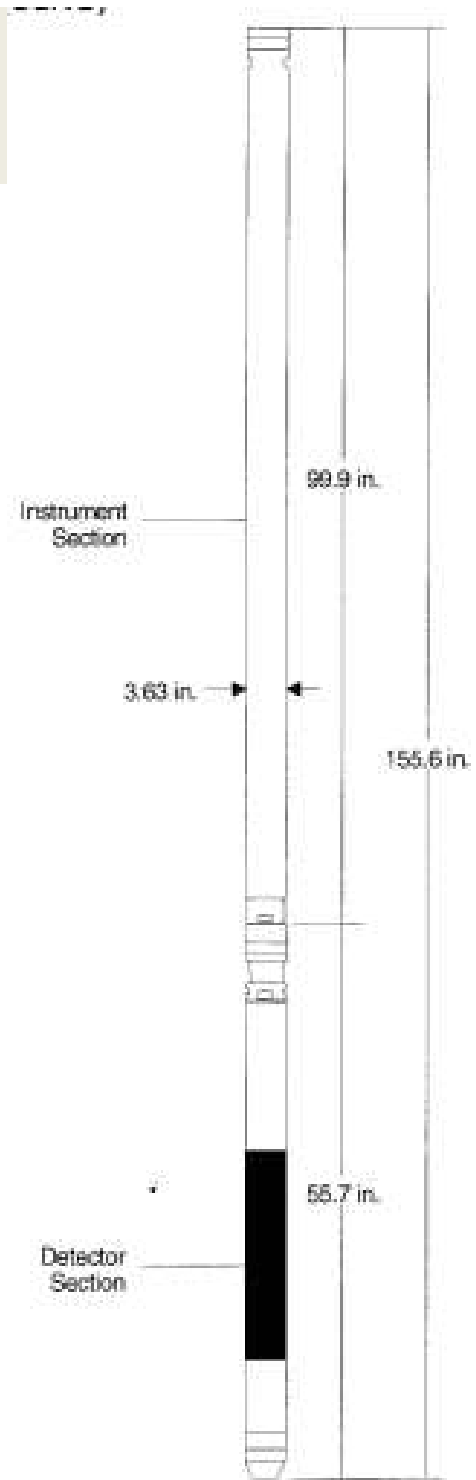
Gamma Ray Log

- **Operational Constraints**

- In Open hole, the tool is run centered
- In Cased hole, the tool is run eccentric
- Can be run in air, water or oil-based mud
- Logging speed: For standard gamma ray, it is constrained by other tools while for spectral gamma ray ≈ 10 feet/minute

- **Secondary Effects**

- Hole Size: increasing hole size decreases count rates
- Mud weight: increasing mud weight decreases count rates
- Mud type: KCl mud increases potassium count rates in spectral tools; barite tools affect all count rates
- Centering: centering the tool decreases count rates



Gamma Ray Log Interpretation Summary

- Clean sands, sandstones, limestones, and dolomites have **low radioactivity**. Anhydrite, salt, lignite and coal have also a low radioactivity.
- Ordinary shales have a much **higher radioactivity** than the rocks listed above. Shales are sufficiently high in radioactivity and can generally be easily distinguished from the other rocks on a gamma ray log.
- Sandstones and dolomites may be occasionally radioactive and respond as shales. A Density-Sonic crossplot may help distinguish between radioactive (“hot”) reservoirs from shales.

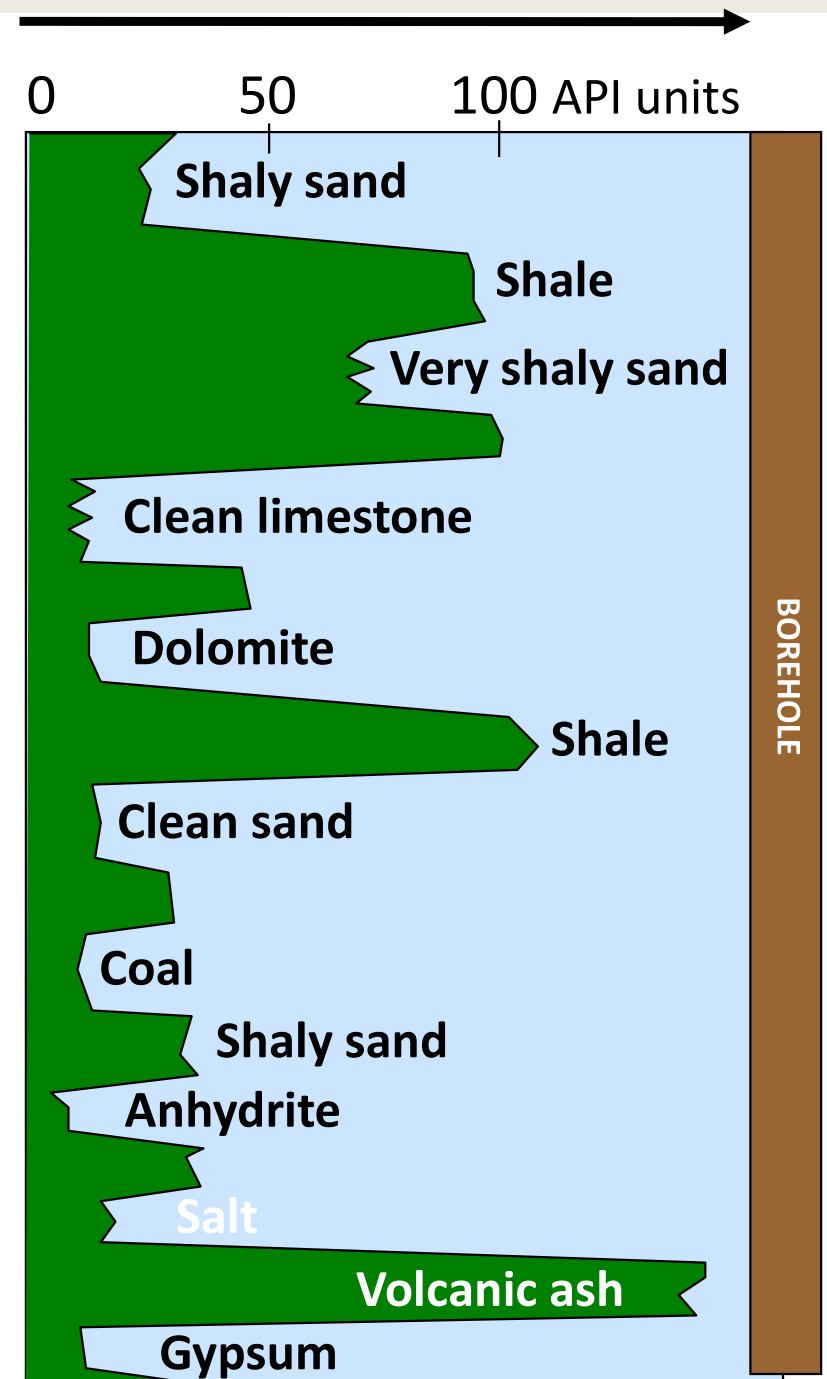
Curves Displayed

(Curves are listed by generic name, common mnemonics (if any) and measurement units.)

Curve Name	Mnemonics	Units of Measurement
Gamma Ray, Total Gamma Ray	GR	API Units
Uranium-Free Gamma Ray	GRS, SGR, KTH	API Units
Potassium	POTA, K	Percent
Uranium	URAN, U	ppm
Thorium	THOR, TH	ppm

GR Response In Common Formations

- Shales often radioactive
 - Clays
 - Trace and heavy minerals
- Sandstones may be radioactive
 - Non-clay minerals, e.g., mica, feldspar
 - Clays
- Units
 - GR calibrated to standard
 - Response in “mid-continent shale” equals 200 API units
 - Calibration pits

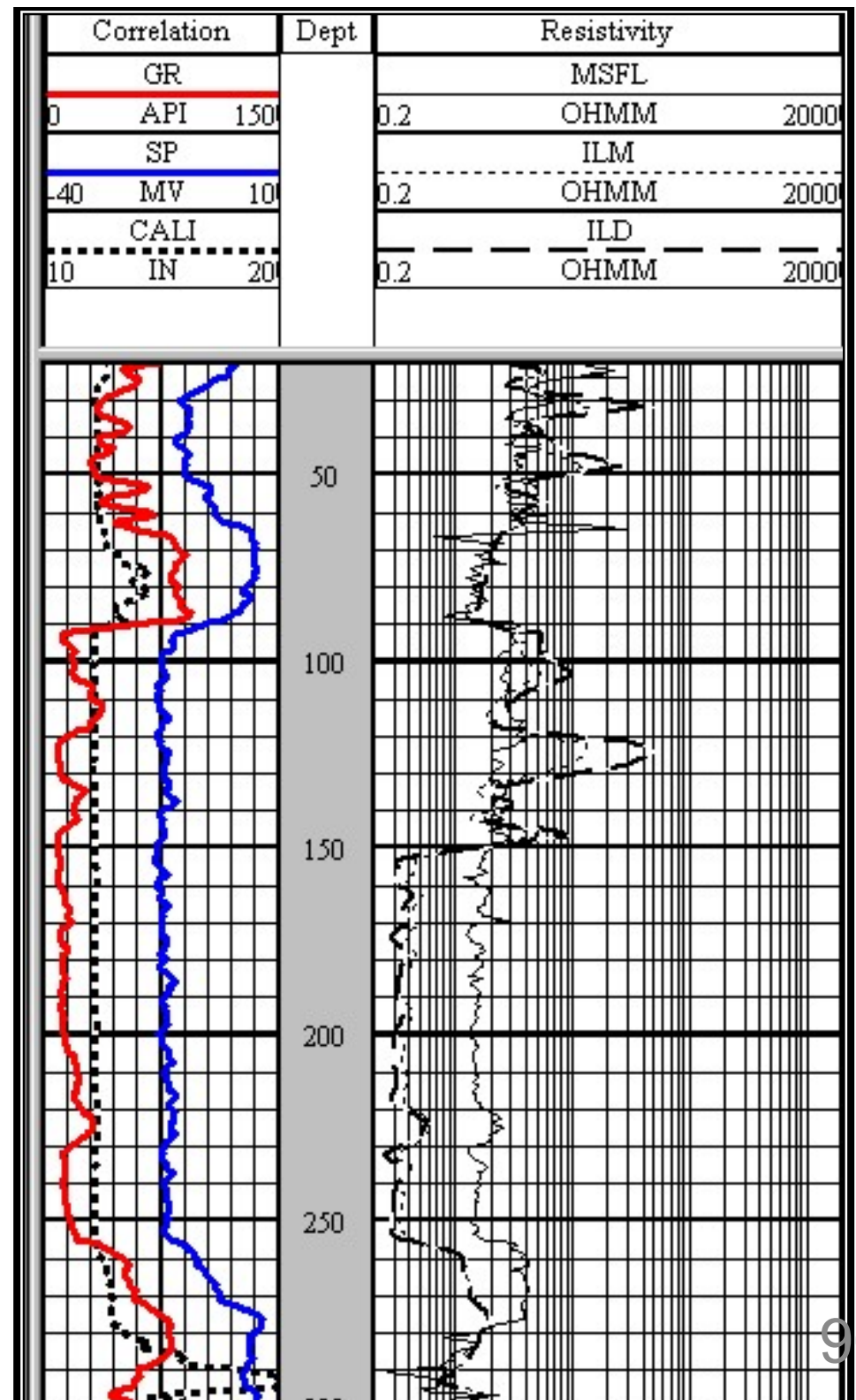


Gamma Ray Log – Quality Control

- The gamma ray should agree with other shale indicators except in radioactive beds.
- The uranium-free curve should always be less than or equal to the total gamma ray curve.
- The uranium curve should never be negative.
- Shale values should be similar to those of nearby wells.
- Check repeatability: curves should have the same values and character as those previous runs or repeat sections.
- Note that corrections are usually made for the borehole, mud weight and borehole salinity.

Passive Log Correlation

- GR, SP, and CAL
 - often correlate
 - different measurements
 - different reasons
- Correlation helps
 - GR instead of SP in OBM
 - Easier detection of shales
 - Facilitates “zonation”



Gamma Ray Log – Facies and Depositional Environment Analysis

- We have seen that the gamma ray log is often used to measure the shaliness of a formation. In reality the shaliness often does not change suddenly, but occurs gradually with depth.
- Such gradual changes are indicative of the litho-facies and the depositional environment of the rock, and are associated with changes in grain size and sorting that are controlled by facies and depositional environment as well as being associated with the shaliness of the rock.
- Figure X analyses the shape of gamma ray log responses for various depositional environments.

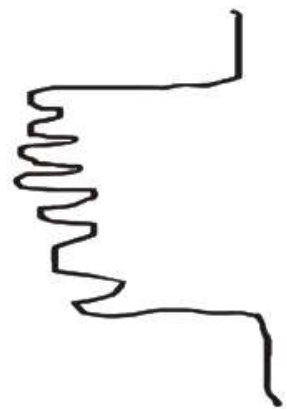
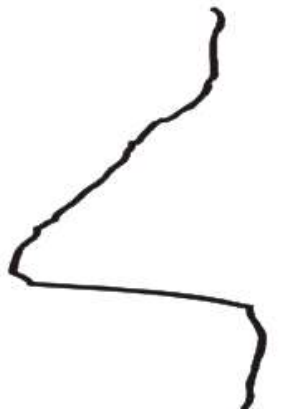
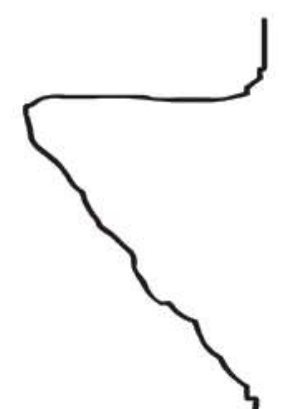
Shape	Smooth	Environments	Serrated	Environments
Cylinder Represents uniform deposition.		Aeolian dunes Tidal sands Fluvial Channels		Deltaic distributaries Turbidite channels Proximal deep-sea fans
Bell Shape Fining upwards sequences.		Tidal sands Alluvial sands Braided streams Fluvial channels Point bars		Lacustrine sands Deltaic distributaries Turbidite channels Proximal deep-sea fans
Funnel Shape Coarsening upward sequences.		Barrier bars Beaches Crevasse splays		Distributary mouth bars Delta marine fringe Distal deep-sea fans

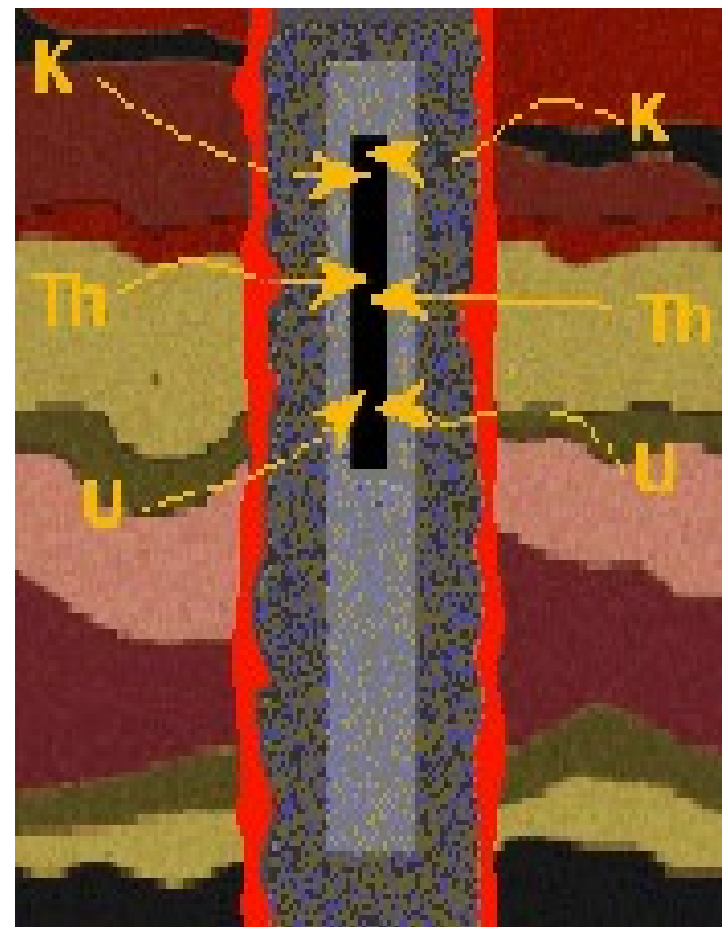
Figure X The gamma ray log and depositional environments.

Spectral Gamma Ray Log

- It is displayed as three curves from all sources of thorium, uranium and potassium.

Applications:

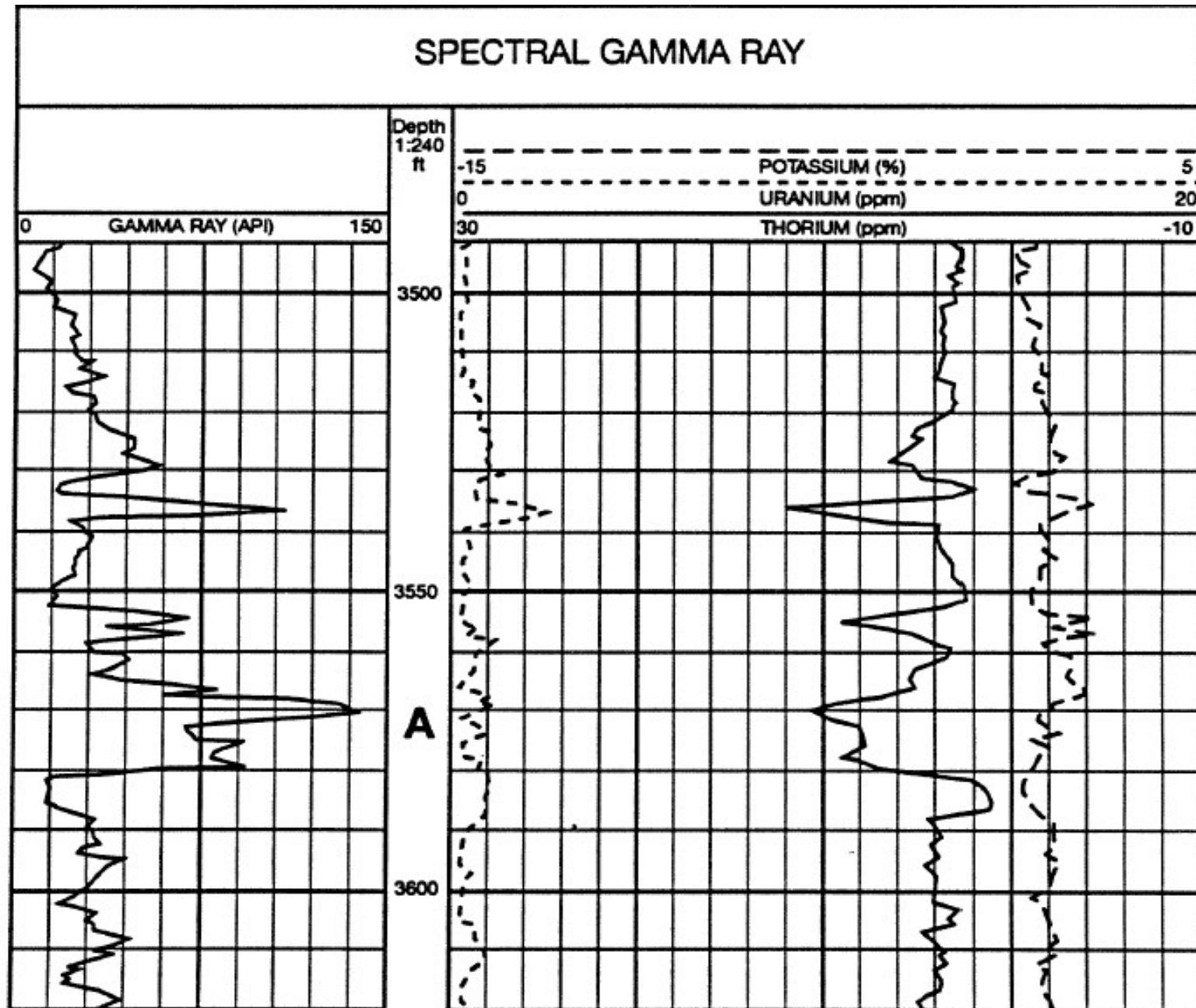
- Source Rock identification: Consistently high uranium readings in shales may indicate high source rock potential due to the uranium compounds associated with organic material. As a rule of thumb, the Th/U ratio is recommended as an indicator for environment.
- Clay typing: The uranium free curve is often a good indicator of shaliness
- Fracture detection: Spikes to higher values of uranium may indicate fractures due to deposition of soluble uranium compounds in fractures during reservoir fluid movement.



Indications for Carbonates from Spectral Gamma Ray Measurements

- Pure Carbonate (Chemical origin) which has a Th and K level near zero. If U is zero too, this carbonate was precipitated in an oxidizing environment.
- If there is variable U content, the carbonate could either have been deposited in a reducing environment, or it corresponds to a carbonate with stylolithes or to phosphate bearing layers .
- If Th and K are present along with with U, this indicates a certain clay content of the carbonate
- If K is present with or without U it can correspond to a carbonate of algal origin or carbonate with glauconite.
- A kind of clay typing is possible because clay minerals show different Th/K ratio i.e. clay minerals give different slopes on Th(ppm) vs K(%) plot
- The high U content of organic components is an indicator for source rocks and gives link to Total Organic Content (TOC). As a rule of thumb, the Th/U ratio is recommended as an indicator for environment.
 - $\text{Th/U} > 7$ continental, oxidizing
 - $\text{Th/U} < 7$ marine, grey...green shales
 - $\text{Th/U} < 2$ marine, black shales, phosphates

Spectral Gamma Ray



Active Well Logging Tools: Porosity Estimation – Lecture 4

Sources of Porosity Measurement

- Density Logs (ZDL) – Nuclear Logs
- Compensated Neutron Logs (CN) – Nuclear Logs
- Acoustic (Sonic) Logs (AC, XMAX).
- Nuclear Magnetic Resonance (NMR) MRIL.

These logs do not measure porosity directly. To accurately calculate porosity, the analyst must know:

- Formation lithology
- Fluid in pores of sampled reservoir volume

Nuclear Logs

- Are important group of log measurements using nuclear radiation. Measurements are possible in both open & cased hole.
- Basic types of nuclear methods are:
 - Measurement of natural gamma radiation of the formation (Gammalog)
 - Measurement of the radiation as a result of an interaction of the source-emitted radiation with the formation
 - Gamma-Gamma-Logs (uses a gamma source)
 - Neutronlogs (uses a chemical neutron source or neutron generator)
- Interactions of radiation with the target material (formation) are processes connected with the atomic structure. Radiation energy is given in Electronvolt (eV): 1 eV is the energy needed to move one electron with the charge of $1.6 \cdot 10^{-19}$ C (Coulomb) through a potential difference of 1 Volt
- Nuclear processes are statistically determined. The statistical variation obeys a Poisson distribution with the standard deviation σ equal to the square root of the total number of counts N .
$$\sigma = \sqrt{N}$$
- It follows that the precision of a nuclear measurement depends on the number of counts recorded – therefore statistical precision can be improved by logging slower and/or using sensors with high sensitivity.

Fundamentals of Gamma Ray Interactions

- Gamma-Gamma methods use the interaction of gamma radiation produced by a source (as part of the tool) with the formation. The main component of a Gamma-Gamma probe are:
 - ✓ Gamma source (Cs or Co),
 - ✓ One or more gamma detectors detecting the back scattered radiation,
 - ✓ A shield between the source and detectors to suppress direct radiation.
- There are three types of gamma ray interactions in earth formations that are of interest:
 - ✓ the Photoelectric Effect,
 - ✓ Compton scattering, and
 - ✓ Pair Production.
- The probability of a specific gamma ray interaction occurring will depend on the atomic number of the material and the energy of the gamma ray.
- The ordering of these three interactions in the following discussion reflects the change of the dominant process as the gamma ray energy increases.

Fundamentals of Gamma Ray Interactions

- The **photoelectric effect** results from interaction of a gamma ray (GR) with an atom in the material. In this process the incident GR disappears and transfers its energy to a bound electron.
- If the incident GR energy is large enough, the electron is ejected from the atom and begins interacting with the adjacent material. Normally the ejected electron is replaced by another less tightly bound electron with the accompanying emission of a characteristic fluorescence x-ray with an energy which is dependent on the atomic number of the material. Where E_γ is gamma ray energy
- For most earth formations, the PE effect becomes the dominant process for GR energies below about 100 keV.
- The photoelectric effect is the mechanism by which one type of well logging tool is made sensitive to the lithology of the formation. This tool measures the so-called photoelectric absorption factor (or Index), Pe , which is proportional to the photoelectric cross section per electron.

$$\sigma_{pe} \propto \frac{Z^{4.6}}{E_\gamma^{3.15}}$$

$$\frac{\sigma_{pe}}{Z}$$

Fundamentals of Gamma Ray Interactions – Photoelectric Effect

- Since PE is very sensitive to the average atomic number of the medium (Z), it can be used to obtain a direct measurement of lithology or rock type. This is because the principal rock matrices (sandstone, limestone, and dolomite) have different average atomic numbers and thus, considerably different photoelectric absorption characteristics, and that the pore fluids play only a minor role because of their low average atomic numbers.
- In the case of several minerals in a formation, the Pe index is not a linear function of the Pe -values of the mineral components, a second parameter U (volumetric photoelectric adsorption index) was introduced:

$$U = PE * \rho$$

- In a multi-mineral lithology this index is the volume weighted index of the individual contributions.

Fundamentals of Gamma Ray Interactions

- Moving up the gamma ray energy scale, the next dominant process is **Compton scattering (CS)** which involves interactions of gamma rays and individual electrons.
- It is a process in which only part of the gamma ray energy is imparted to the electron; the remaining gamma ray is of reduced energy. Unlike the *PE* effect, the cross section for CS changes relatively slowly with energy.
- The attenuation of gamma rays due to Compton scattering will be a function of the bulk density ρ_b and the ratio Z/A . The fact that Z/A is constant ($\approx 1/2$) for most elements of interest is the basis for the determination of bulk density from gamma ray scattering devices.
- The Compton effect is a scatter reaction with the orbiting electrons of the target material. Therefore, the probability of an interaction is controlled by the number of electrons, which is equal to Z . For the most common earth materials, the ratio Z / A is constant.

Fundamentals of Gamma Ray Interactions – Compton Scattering

- Since rock bulk density ρ_b is dominated by Atomic Mass A, the following relationship connects the “electron density” ρ_e as seen by the Compton effect via this ratio: $\rho_b = 2 * (Z / A) * \rho_e \approx \rho_e$

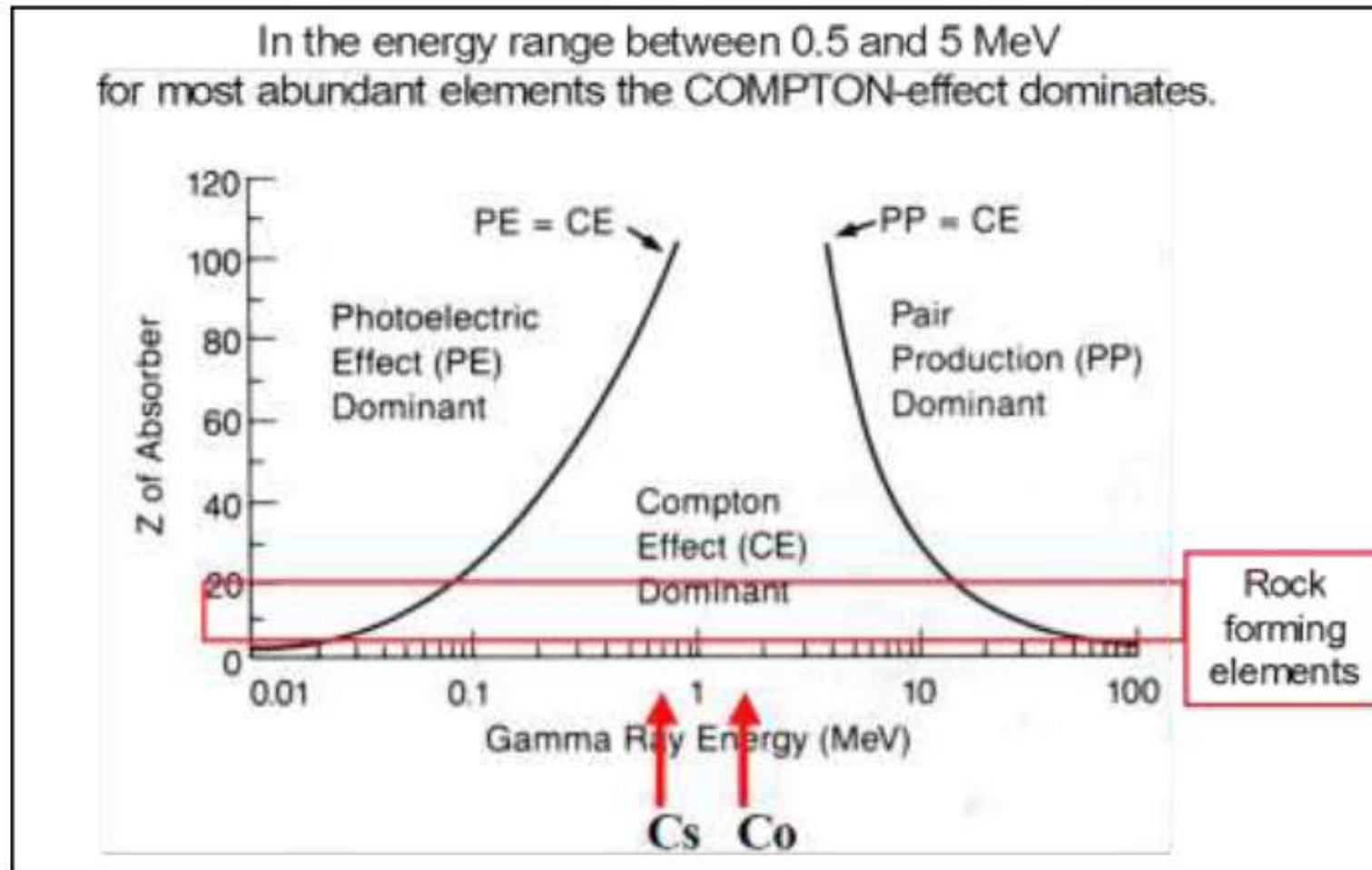
Mineral, fluid	ρ	$(Z/A)_{\text{eff}}$	ρ_{GG}	Pe	U
Quartz	2,654	0,499	2,650	1,81	4,79
Calcite	2,710	0,500	2,708	5,08	13,77
Dolomite	2,850	0,499	2,864	3,14	9,00
Anhydrite	2,960	0,499	2,957	5,05	14,95
Gypsum	2,320	0,511	2,372	3,42	8,11
Halite	2,165	0,479	2,074	4,65	9,65
Montmorillonite	2,12	0,500	2,12	2,04	7,28
Illite	2,65	0,496	2,63	3,55	10,97
Kaolinite	2,44	0,500	2,44	1,84	6,14
Barite	4,50	0,446	4,01	267	1070
Water (fresh)	1,000	0,555	1,110	0,36	0,40
Water (120000 ppm)	1,086	0,545	1,185	0,81	0,96
Oil (mean)	0,850	0,558	0,948	0,12	0,11

Mean Values of density in 10^3 kg/m^3 , electron density, or gamma-gamma density, effective ratio Z / A , PE index in barn/electron & volumetric PE absorption index U in barn/cm³

Fundamentals of Gamma Ray Interactions

- The third and final gamma ray interaction is **pair production**. Like the *PE* process, pair production is one of absorption rather than scattering.
- In this case the gamma ray interacts with the electric field of a nucleus, and if the gamma ray energy is above the threshold value of 1.022 MeV, it disappears and an electron-positron pair is formed. The onset of this interaction corresponds to the rest mass energy of the electron and positron.
- The subsequent annihilation of the positron (positively charged electron) results in the emission of two gamma rays of 511 keV each. The nuclear cross section of this process is zero below the threshold energy of 1.022 MeV and rises quite rapidly with increasing energy. It is also dependent on the charge of the nucleus, varying approximately as Z^2 .

Fundamentals of Gamma Ray Interactions



- Regions of Dominance of the three principal gamma ray scattering / absorption mechanisms as a function of energy and atomic number of the scattering material.

DENSITY LOG

DENSITY LOGS – Introduction

- Another important property of an earth formation is bulk density. Its use in seismic interpretation is well known, but more important, bulk density is linearly dependent on formation porosity, a key ingredient for the interpretation of electrical measurements.
- Gamma rays are typically used to measure bulk density since their scattering and transmission are strongly affected by this material property.
- The average electron density in the volume of formation probed by the tool controls the scattered gamma ray count rates at the detectors. The average electron density, in turn, correlates strongly (but not perfectly) with bulk density. Formation Bulk density, ρ_b , is dependent upon the (i) Lithology (ii) Porosity (iii) Density and saturation of fluids in pores.

Density Log

- **Uses**

- Porosity determination: from bulk density
- Lithology Identification: Lithology is determined by comparison of bulk density with Sonic and Neutron data in cross plots, in Matrix identification (MID) plots and in M-N (A-K) plots. The charts may vary by Neutron tool type, Sonic equation response type, and by service company. The Photoelectric effect (PEF) can be used alone to determine single lithology, or in combination with bulk density, or bulk density and Neutron curves to determine mixed lithologies.
- Gas Indication: with the Neutron.
- Synthetic Seismograms: Sonic compressional and Density data are used to determine acoustic impedance of the formations along the borehole, and reflection coefficients at the bed boundaries.
- Formations mechanical properties: with the Sonic
- Clay content (shaliness): with the Neutron .
- Borehole size: from an attached caliper.

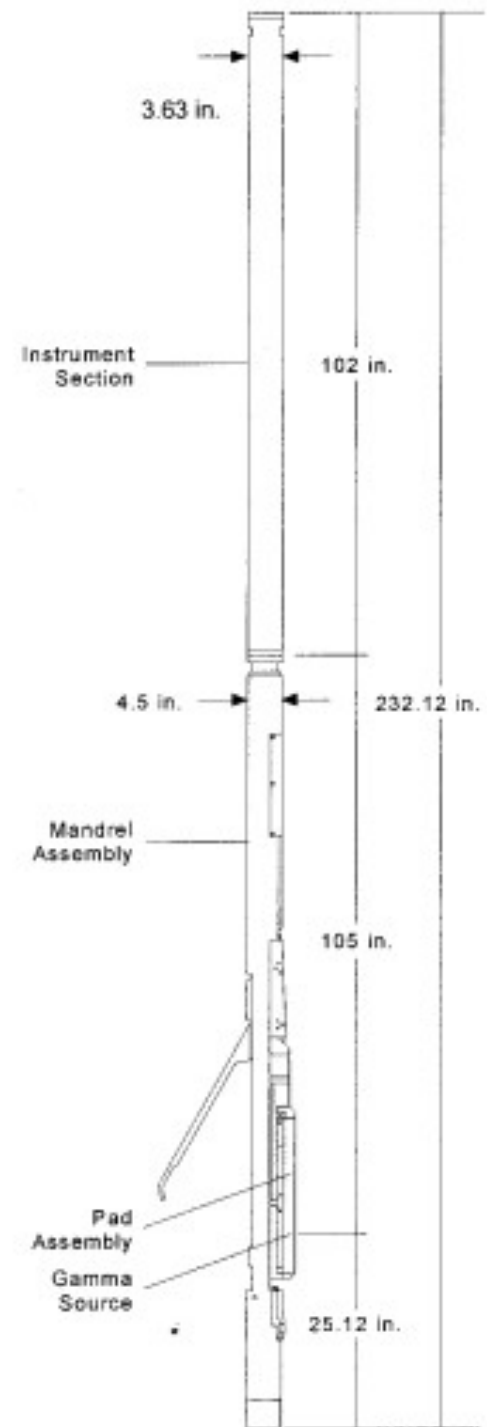
Density Logging Tool

- **Physics of Measurement**

- High energy gamma rays are emitted from a chemical source (usually Cesium-137) of 0.66MeV and interact with the electrons of the elements in the formation. Two detectors (7in & 16in from the source) in the tool count the number of returning (back-scattered) gamma rays which are related to the formation bulk density. For most earth materials of interest, the electron density is related to the formation bulk density through a constant.
- The number of returning gamma rays are measured at two energy ranges. The higher energy gamma rays (from Compton Scattering) determine bulk density, and therefore porosity, while the lower energy gamma rays (due to photoelectric effect) are used to determine formation lithology. The lower energy gamma rays are related to the lithology of the formation and show little dependence on porosity or fluid type. The average atomic number (which correlates with rock type) sets the amount of PE absorption that a formation exhibits.

- **Operational Constraints**

- In Open hole or Cased hole (in special conditions), the tool is run eccentric
- Can be run in air, water or oil-based mud



Spectral Density Tool

DENSITY LOGS – Tool Physics

- The source and both detectors are heavily shielded (collimated) to ensure that the radiation only goes into the mudcake and formation, and that detected gamma rays only come from the mudcake or formation. The leading edge of the shield is fashioned into a plough which removes part of the mudcake as the tool is pulled up the well.
- The tool is pressed against one side of the borehole using a servo-operated arm with a force of 800 pounds force. Under this pressure and the pulling power of the wireline winch the plough can make a deep impression in the mudcake. The large eccentricing force also means that there is much wear of the surface of the tool that is pressed against the borehole wall. The heavy shielding also doubles as a skid and a wear plate that protects the source and detectors and can be replaced easily and cheaply when worn down (see Figure DL1).
- The density logging tool detects gamma rays that have been supplied to the formation by the source on the tool and have undergone scattering. The natural gamma rays from the formation however confuse the measurement. A background gamma ray count is therefore carried out so that the gamma rays coming from the formation can be removed from the measurement.

DENSITY LOG – Tool Physics

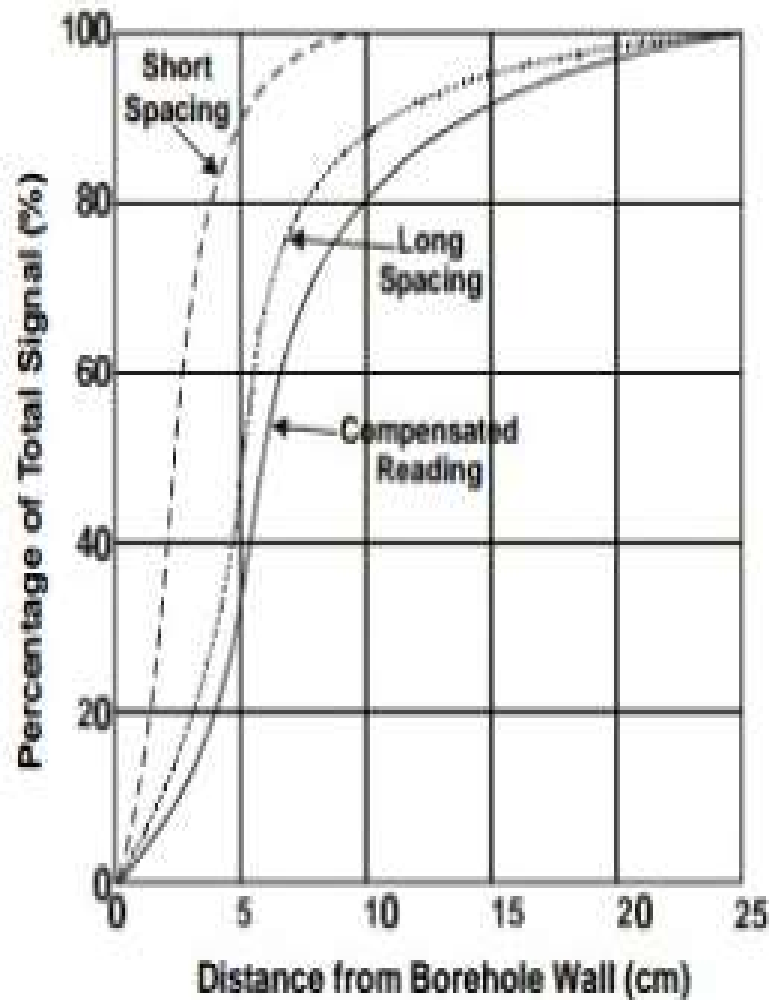


Figure DL2

A correction is needed when the contact between the skid and the formations is not perfect (due to mud cake or roughness of the borehole wall). Using two detectors, a correction can be made for unfavorable conditions.

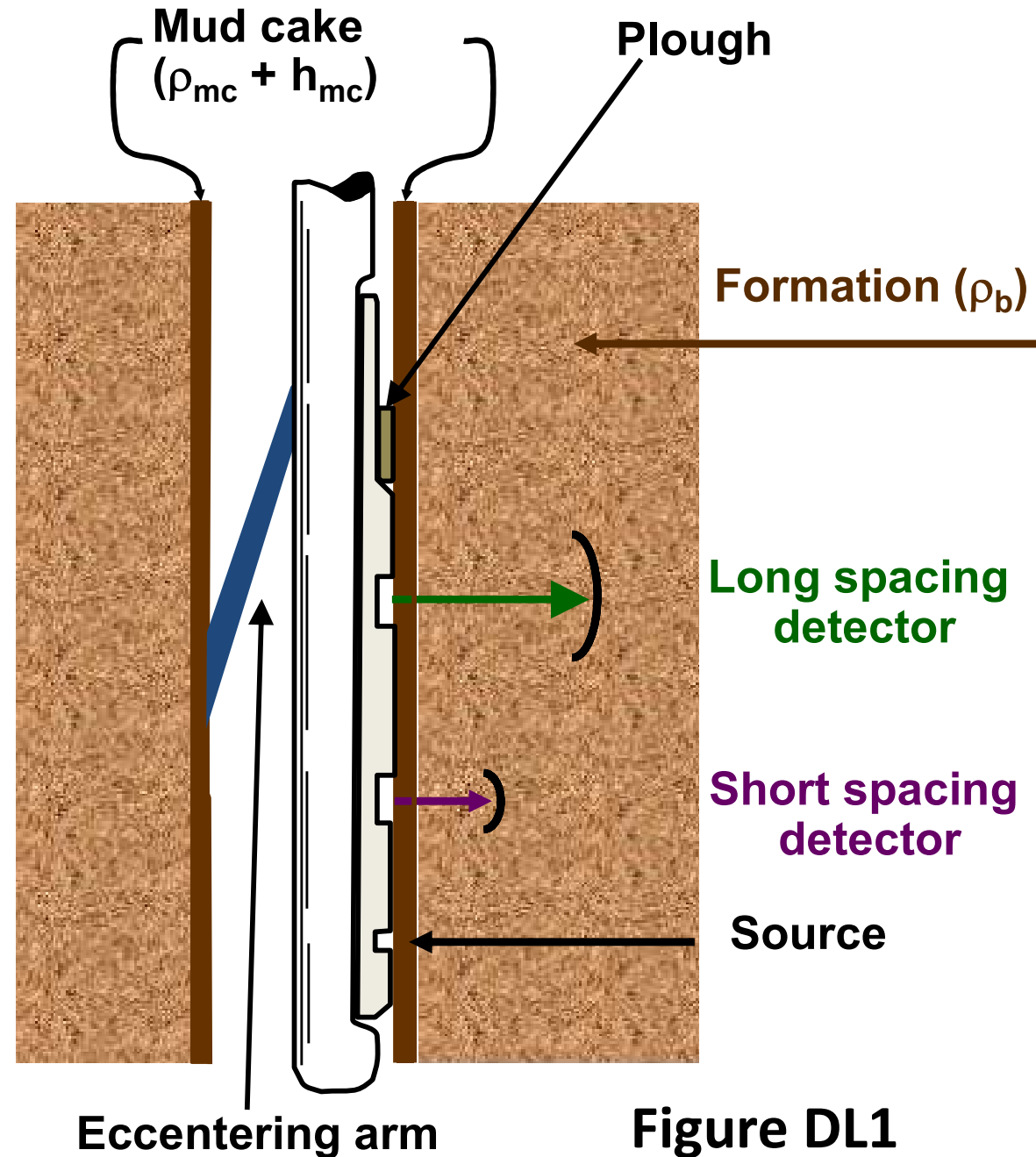


Figure DL1

DENSITY LOGS – Tool Physics

- Note in Figure DL2 that for the short spacing detector over 80% of its signal comes from within 5 cm of the borehole wall, which is commonly mainly mudcake. About 80% of the long spacing signal comes from within 10 cm of the borehole wall. Therefore, the tool has a shallow depth of investigation. When the mudcake compensation has been carried out, one can see from Figure DL2 that the depth of investigation has been improved, and less signal comes from the first 5 cm region. The depth of investigation of a density tool decreases with increasing density.
- The distance between the near and far detectors sets the vertical resolution, approximately 10 in. typically. Correction is by the spine-and-ribs technique. The spine is the normal calibrated relationship between the density measured by the near-spaced and far-spaced detectors in the absence of any gap between the tool and the borehole wall. Gaps cause departures from this spine and lead to density corrections that are applied to the density from the long-space detector.

DENSITY LOGS – Interpretation Effects

The Effect of Fluid Density:

- The porosity may also be in error if the fluid density is misjudged. The fluid existing in the zone of the rock measured by the formation density tool is usually mud filtrate. The density of these fresh and salt waters is approximately 1.0 & 1.1 g/cm³, respectively.
- However, these vary with temperature and composition, so accurate values for the actual reservoir formation water at the relevant reservoir temperatures should be used wherever possible. Such data can be obtained from samples of reservoir fluid from RFT analysis, or from the analysis of mud filtrate.
- Bearing in mind that the tool measures the invaded zone, the relevant fluid density to use in the porosity calculations is most often the mud filtrate density. Mud filtrate densities are now corrected automatically for temperature and pressure in most petrophysical software. However, the correction used to be carried out with the aid of a nomogram.

DENSITY LOGS – Interpretation Effects

The Effect of Gas:

- If gas is present in the formation, porosities can be overestimated. Gas density is very low ($\approx 0.0001 \text{ g/cm}^3$) compared to aqueous fluids. If the formation is gas bearing a significant amount of gas is always left in the invaded zone which will reduce the mean fluid density of the invaded zone, and cause porosity overestimation if a fluid density of $1.0 - 1.1 \text{ g/cm}^3$ is used.

The Effect of Oil:

- The density of oil ($\approx 0.7 \text{ g/cm}^3$) is less than that of aqueous fluids. However, the presence of oil-bearing formations rarely causes problems in porosity calculation because both the oil and water in the invaded zone, which the tool measures, is replaced by mud filtrate.

The Effect of Shale:

- The density of shales varies greatly, and if present as a proportion of a lithology (such as a shaly sandstone) can make the derivation of a reliable porosity inaccurate. If we have zones of clean lithology and zones of shale with a shaly sandstone in between (or a full fining-up or coarsening-up sequence), we can use the bulk densities in the shale and clean sandstone together with V_{sh} to obtain a corrected density for the shaly sandstone at any given depth.

DENSITY LOGS – Environmental Effects

Enlarged Borehole (> 9 inches):

- $RHOB < \text{formation bulk density}$ ($DPHI > PHI_{\text{actual}}$).

Rough Hole:

- $RHOB < \text{formation bulk density}$ ($DPHI > PHI_{\text{actual}}$). This is due to the sensor pad losing contact with the borehole wall. Other indications of a rough hole will be a variable Caliper curve, and a high-valued density correction (DRHO) curve. There are no environmental corrections than can be applied to correct for loss of pad contact.
- DRHO is a significant measure of log quality (QC):
 - Correction curve DRHO should be near zero in smooth holes,
 - $DRHO > 0.05$; Data quality may be questionable due to loss of pad contact,
 - $DRHO > 0.10$; indicates that the density value is not reliable

Barite Muds:

- $RHOB > \text{formation bulk density}$ ($DPHI < PHI_{\text{actual}}$), and $PEF > PEF_{\text{actual}}$

BULK DENSITY

$$\rho_b = \rho_{ma} (1 - \phi) + \rho_f \phi$$

Matrix

Fluids in
flushed zone

$$\phi = \frac{\rho_{ma} - \rho_b}{\rho_{ma} - \rho_f}$$

Porosity equation

We usually assume the fluid density (ρ_f) is between 1.0 and 1.1. If gas is present, the actual ρ_f will be < 1.0 and the calculated porosity will be too high.

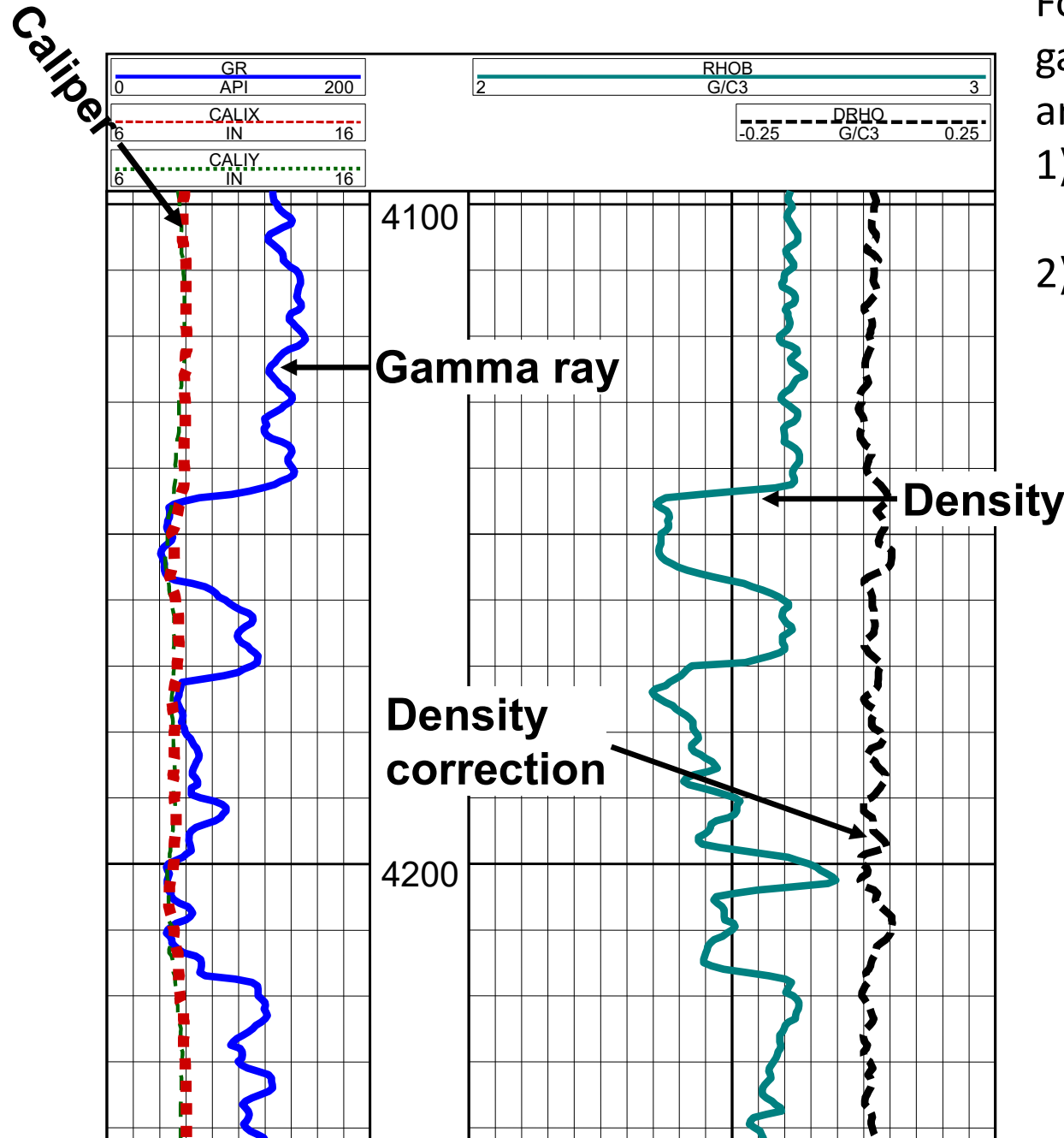
ρ_{mf} is the mud filtrate density, g/cc

ρ_h is the hydrocarbon density, g/cc

S_{xo} is the saturation of the flush/zone, decimal

- Measures electron density of a formation
- Strong function of formation bulk density
- Matrix bulk density varies with lithology
 - Sandstone 2.65 g/cc
 - Limestone 2.71 g/cc
 - Dolomite 2.87 g/cc

DENSITY LOG

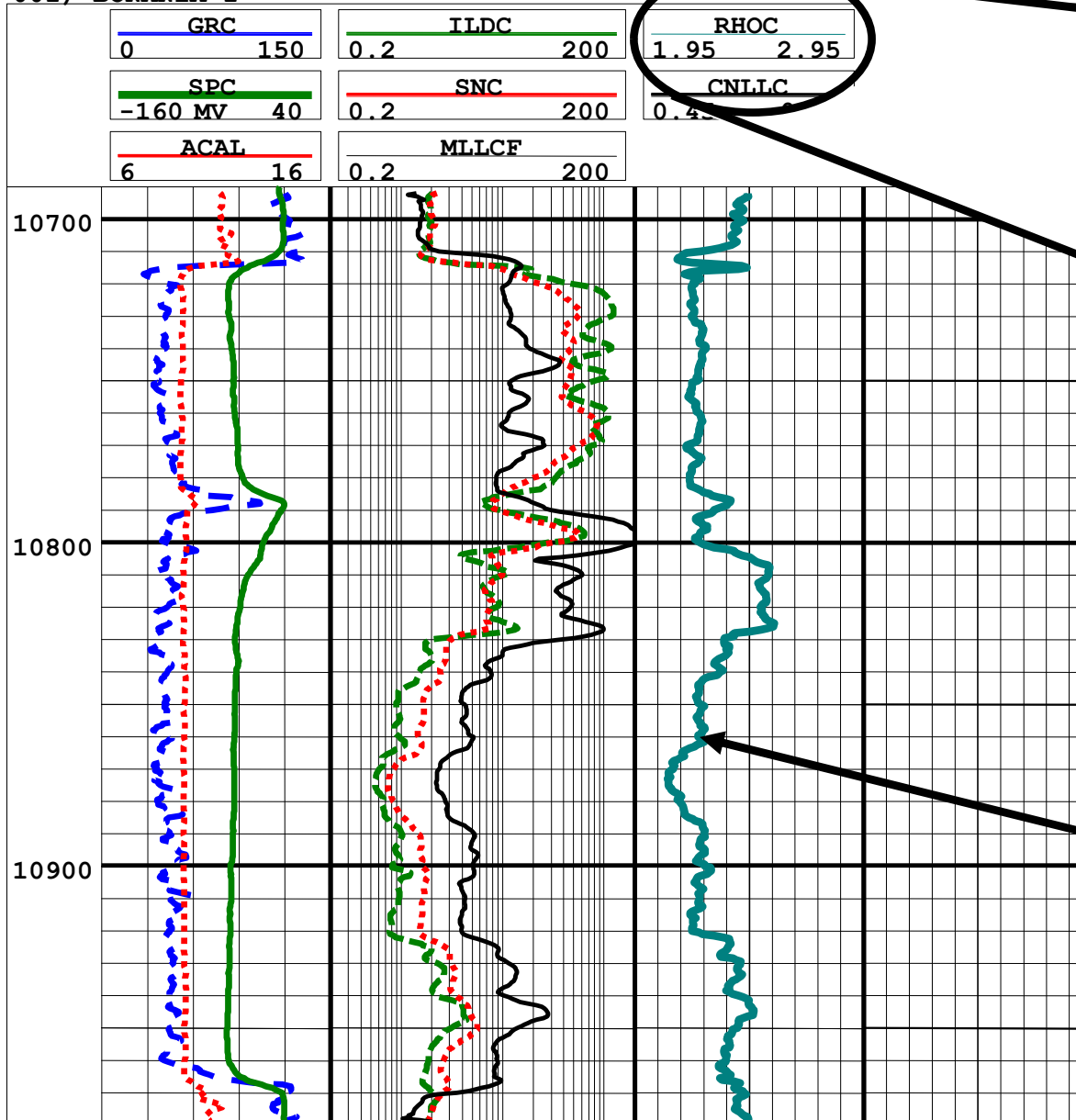


For porosity calculation from gamma-gamma-density two-input parameters are necessary:

- 1) Matrix density (or information about mineralogy).
- 2) Fluid density: The depth of investigation for gamma-gamma measurements is very shallow; thus, the measurement is controlled by the flushed zone in most cases and the density of mud filtrate can be used. Only in the case of shallow gas invasion density is lowered, for calculation an average fluid density of 0.7 g/cm^3 is recommended.

BULK DENSITY LOG

001) BONANZA 1



RHOC
1.95 2.95

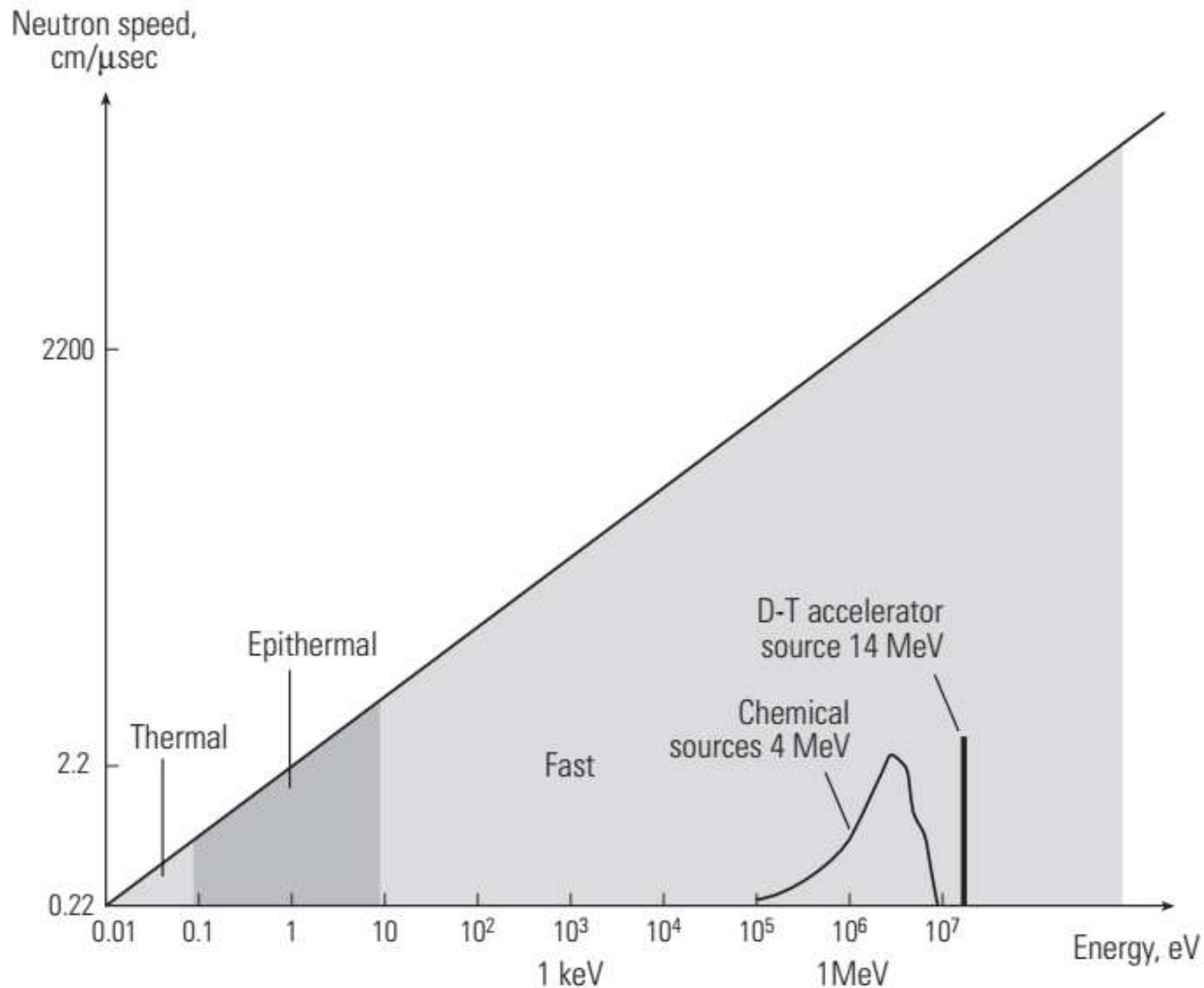
Bulk Density
Log

NEUTRON LOG

Fundamentals of Neutron methods

- Neutron methods use the interaction of neutron radiation from a source (as part of the tool) with the formation. The main component of the neutron probe are:
 - ✓ Neutron source (Chemical source or Neutron Generator). One or more detectors (neutron and gamma) detecting neutron radiation after slowing down processes and induced gamma radiation,
 - ✓ A shield between the source and detectors to suppress direct radiation.
- The neutron tool emits high energy (5-15 MeV) neutrons which move very fast. Their energy is related to their speed.
- Generated neutrons are classified by their energy:
 - ✓ Fast neutrons have energies greater than 10 eV;
 - ✓ epithermal neutrons have energies in the range 0.2 to 10 eV;
 - ✓ thermal neutrons distributed at about 0.025 eV at room temperature.
- There are different types of interactions:
 - ✓ Moderating interactions result in energy decrease (slowing down effect);
 - (a) Inelastic scattering (b) Elastic scattering
 - ✓ Capturing interactions;

Fundamentals of Neutron methods



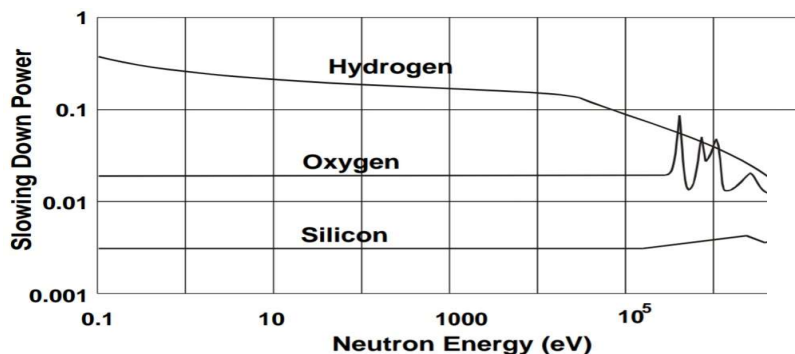
- The classification of neutrons according to broad energy ranges and their corresponding velocities

Neutron Scattering

- Inelastic Scattering: Neutrons collide with the nucleus and transfer a fraction of their kinetic energy to the nucleus which results in excitation of the nucleus followed by the emission of a characteristic gamma photon (de-excitation). The emitted gamma radiation are characteristic of the target material. An application is the Carbon/Oxygen log.
- Elastic Scattering: Neutrons collide with the nucleus and lose kinetic energy. The maximum energy loss results for a target material with the same atomic mass as the neutron; this is the case for hydrogen. The other rock forming elements have a lower slowing down effect and are different. This leads to the conclusion:
 - ✓ Elastic scattering is dominated by hydrogen in the formation; there a strong combination is expected from the fluids with a high H-concentration (water & oil) and somewhat a lower contribution from gas with lower H-concentration. In general elastic scattering is dominated by pore fluids
 - ✓ Other elements have a distinctly smaller effect; but this effect is different for each element. Therefore a different neutron response for different materials (quartz, calcite and dolomite) is expected

Neutron Elastic Scattering

- Neutron Elastic Scattering: The fast neutrons interact with the nuclei of atoms within the formation. The interaction is a form of elastic scattering involving the neutrally charged neutron and a stationary positively charged nucleus. At each interaction the neutron loses some energy and slows down, and the nucleus of the atom in the formation material gains energy. Such collisions occur with nuclei of all elements.
- However, the process of energy transfer is most efficient when the masses of the neutron and the nucleus are the same, and becomes much less efficient when the nuclei of the formation material are more massive than the neutron. The neutron has approximately the same mass as hydrogen nuclei (the lightest element). Hence the neutrons lose energy by elastic scattering most efficiently by interaction with hydrogen nuclei, and lose energy much less efficiently by interaction with more massive nuclei such as silicon or oxygen.



This figure shows the efficiency of several elements in slowing down fast neutrons for a clean sandstone

Neutron Capture

- Neutrons can be captured when they are thermalized. Thermal neutrons have small energies associated with the random kinetic motion at room temperatures. With this process a compound nucleus in an excited state is formed. When it returns to the ground state, it emits gamma radiation with an exactly defined energy. This capturing effect is used with the Thermal neutron diffusion / Decay measurements (TDT) log.

Element	Gamma energy in MeV - inelastic scattering -	Gamma energy in MeV - capturing -
H		2.2
O	6.1, 7.0	
C	4.43	
Si	1.78	3.5, 4.9
S	2.2	3.2, 4.9, 5.4
Ca	3.7	2.0, 4.4, 6.4
Al	2.2	7.7
Mg		3.9
Fe		6.0, 7.3, 7.6

Inelastic scattering and capturing – Some characteristic energies of emitted gamma radiation

Neutron Logging Processes

- Neutron Absorption: Thermal (and to some extent epithermal) neutrons can be absorbed by the nuclei of the formation atoms. The efficiency of neutron absorption varies from element to element. The only element which exhibit significant neutron absorbing behaviour and exist in reasonable quantities in rocks are hydrogen and chlorine. The process of neutron absorption by hydrogen and chlorine are:



- In neutron logging, some tools measure the epithermal neutrons, some the thermal neutrons and some the gamma rays emitted when a neutron is absorbed.

Neutron Log

- **Uses**

- Porosity determination: display directly on the log
- Lithology Identification: with the Sonic and/or Density.
- Gas Indication: with the Density.
- Clay content (shaliness): with the Density .
- Correlation: especially in cased holes.
- Characterization of mineralogical composition of the formation via Elemental Analysis based on Spectral Nuclear Measurements

	thermal	epithermal	neutron-gamma
Radius of investigation	6 in = 15 cm (50 %)		8 in = 20 cm (50 %)
Vertical resolution	36 in = 84 cm; 20 in = 51 cm enhanced resolution processing	30 ... 44 in = 76 ... 112 cm	20 in = 51 cm
Precision	0.4 p.u	1 p.u.	
Logging speed	< 60 feet/min = 18 m/min Lower for enhanced resolution processing		

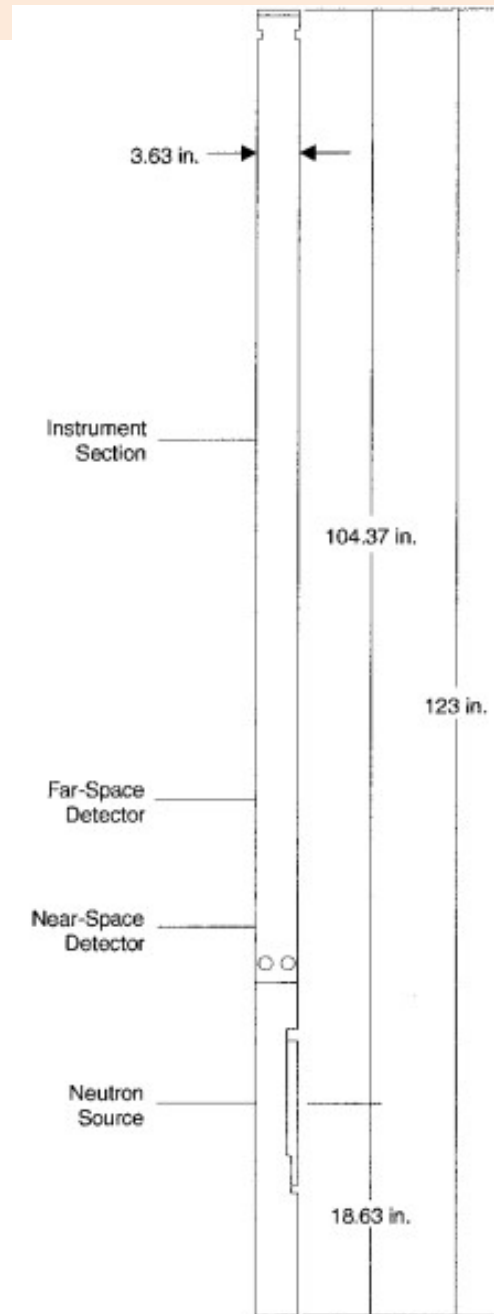
Neutron Logging Tool

- **Physics of Measurement**

- A chemical source (alpha particles from the radium, plutonium or americium interacting with beryllium-9) or a particle accelerator (which forces deuterium and tritium collisions at high energy levels) emit fast, high energy which are slowed down by the formation nuclei. Two detectors in the tool count the number of returning captured gamma rays or neutrons (depending on the tool type). The detector count rates are inversely proportional to the amount hydrogen in the formation (“Hydrogen index”).
$${}_4^9\text{Be} + {}_2^4\text{He} \Rightarrow {}_6^{12}\text{C} + {}_0^1\text{n} + \gamma$$
- “Gamma-ray-neutron” tools detect gamma rays and thermal neutrons; “sidewall” tools detect epithermal neutrons, “compensated” tools detect thermal neutrons.

- **Operational Constraints**

- In Open hole or Cased hole (in special conditions), the tool is run eccentric
- Can be run in air, water or oil-based mud



Neutron Tool

Neutron Tool – Hydrogen Index

- We can define a partial concentration of hydrogen per unit mass $(C_H)_{mass}$ of a material as the mass of hydrogen atoms in the material divided by the mass of all the atoms of all elements in the material.

$$(C_H)_{mass} = \frac{n_H A_H}{\sum_i n_i A_i + n_H A_H}$$

- A_H = the atomic mass of hydrogen atoms in the material
 - A_i = the atomic mass of non-hydrogen element “i”
 - n_H = the number of hydrogen atoms in a molecule of the material
 - n_i = the number of non-hydrogen atoms of element “i” in a molecule of the material
- Thus, for pure water (H_2O), where the atomic mass of hydrogen is 1.0 and the atomic mass of oxygen is 16.0, the partial concentration of hydrogen is

$$(C_H)_{mass} = \frac{2 \times 1.0}{(1 \times 16.0) + (2 \times 1.0)} = \frac{1}{9}$$

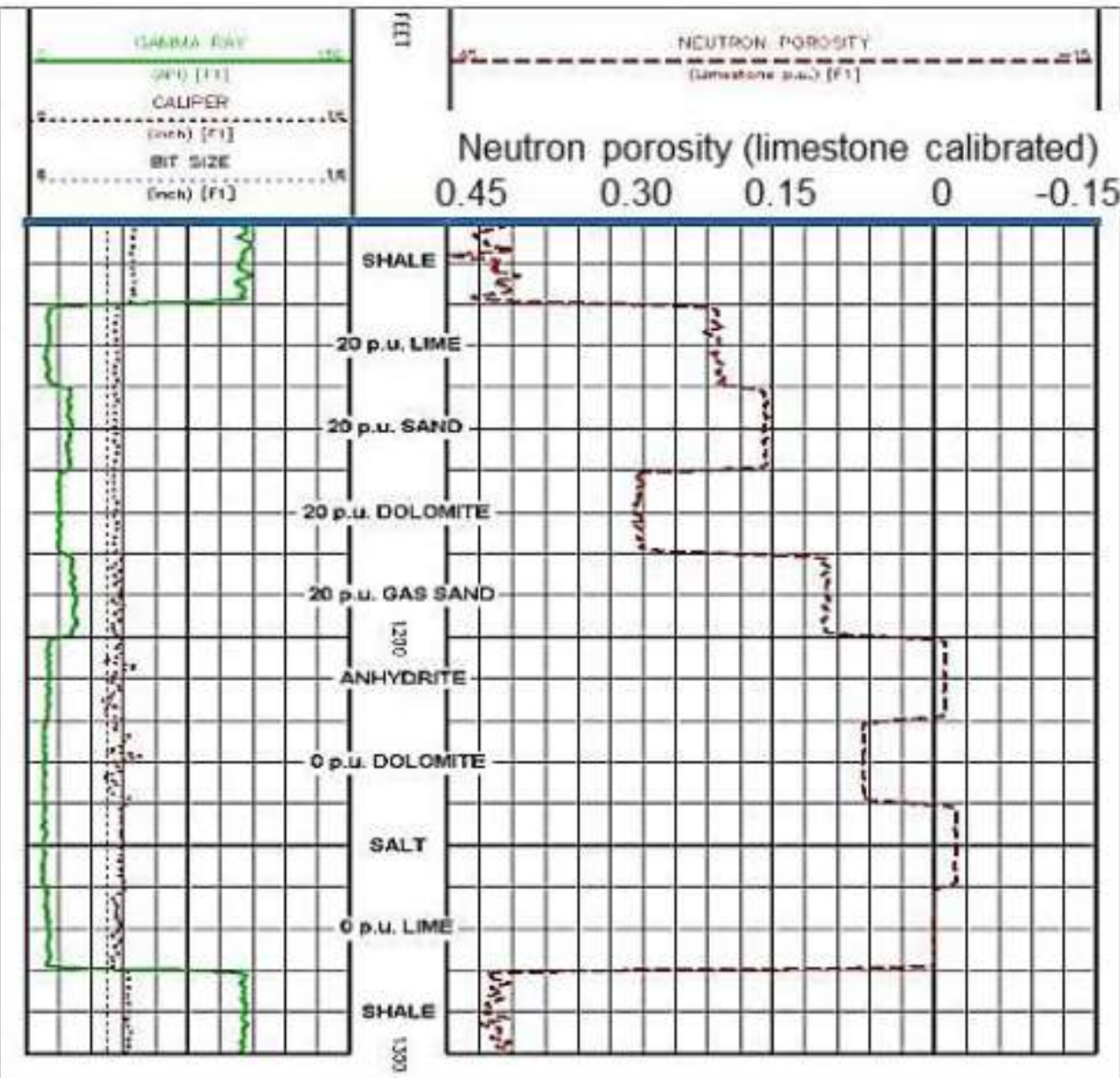
Neutron Tool – Hydrogen Index

- Hence, the partial concentration of hydrogen per unit volume is $1/9$, because the density of pure water is 1.000 g/cm^3 . The Hydrogen Index of a material is defined as the partial concentration of hydrogen per unit volume relative to water. So, if the hydrogen index of water is constrained by the definition to be unity, and water has a partial concentration of hydrogen per unit volume of, the hydrogen index of a material is

$$HI = \frac{9n_H A_H}{\sum_i n_i A_i + n_H A_H} \rho_b$$

- Note the following:
 - If the tool is in 100% water, the $HI = 1$. This is equivalent to a rock of 100% porosity saturated with water. So we have a fixed point that $HI = 1$ represents porosity, $\phi = 1$.
 - If the tool is in a pure limestone rock with zero porosity, the $HI = 0$, because there are no hydrogen atoms in the solid matrix of calcite. So we have a second fixed point where $HI = 0$ represents $\phi = 0$.
 - If the tool is in a pure limestone with a given porosity f , the hydrogen index will be directly proportional to the amount of water in the formation. Hence $HI = \phi$, for completely water saturated limestone.

Neutron Response – Influence of Mineralogy and Pore Fluid



Comments

Shale: indicated by high Neutron-porosity and gamma

Limestone, $S_w = 1$; shows exact porosity

Sandstone, $S_w = 1$; shows lower Neutronporosity

Dolomite, $S_w = 1$; shows higher Neutronporosity

Sandstone shows lower porosity (gas effect)

Anhydrite; small negative deviation from porosity = 0; matrix effect

Dolomite; positive deviation from porosity = 0; matrix effect

Anhydrite; negative deviation from porosity = 0; matrix effect

Limestone porosity = 0; exact reading

Shale: indicated by high neutronporosity and gamma

Interpretation Effects

- **The Hydrocarbon Effect**

- As with the formation density tool, the zone of investigation of the neutron tool is constrained to the flushed zone. Hence, the tool will measure the hydrogen index associated mainly with the mud filtrate, together with whatever remains of the hydrocarbons and formation water in the formation. Imagine, for example that we are in an oil zone in a limestone, the mud filtrate almost completely replaces the formation fluids such that there is a saturation of mud filtrate S_{XO} , and a residual saturation of hydrocarbons $(1 - S_{XO})$. The porosity read by the neutron tool is related to the actual porosity in the formation by

$$\phi_N = \phi [HI_{mf} S_{XO} + HI_{hc} (1 - S_{XO})]$$

- If the hydrocarbon is oil, this equation reduces to $\phi_N \approx \phi$, because it is observed from Table 1 that the hydrogen index of water and oil is similar; $HI_{mf} \approx HI_{hc}$ whether the mud is oil based or water based.

Interpretation Effects

- **Chlorine Effect**

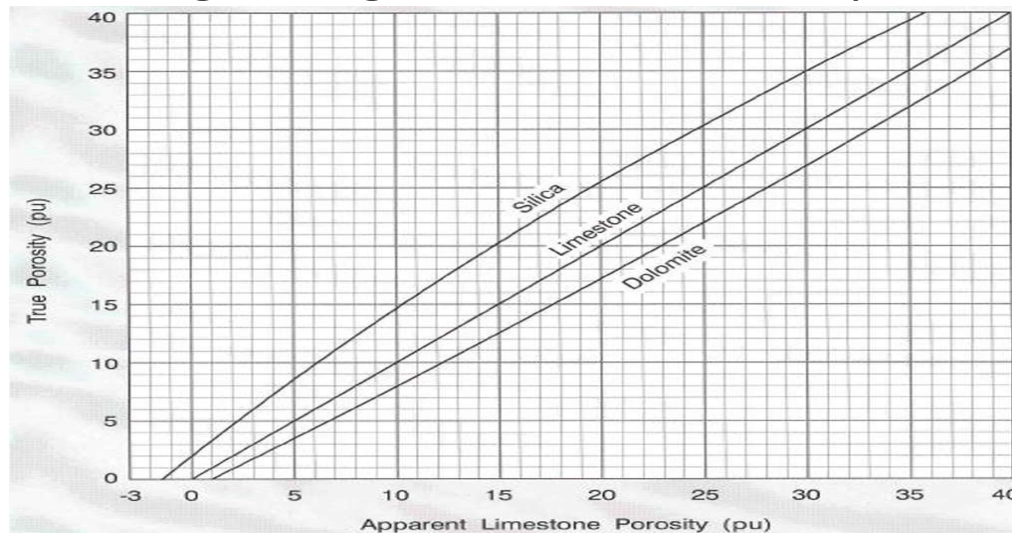
- Some types of neutron tool measure the thermal neutrons and gamma rays produced during the capture of neutrons. There are only two elements that are found in reservoirs that contribute significantly to neutron absorption; hydrogen and chlorine. The presence of hydrogen in the fluids is what we want to measure, so this is not a problem.
- However, if the drilling mud, mud filtrate or formation fluids contain a significant amount of dissolved chloride ions, as is often the case, the tool will measure a lower flux of neutrons and hence overestimate the porosity. This is called the chlorine effect, and is present in wells drilled or logged in the presence of drilling mud containing dissolved chlorine or in formations where the formation waters are particularly salty.

- **Shale Effect**

- Shale contains clays that have a significant amount of surface absorbed (bound) water. Hence shale can contain a significant proportion of hydrogen despite being low porosity. The apparent porosity read from the neutron tool in shale formations is therefore always significantly higher than it really is. This is called the shale effect or the bound-water effect.

Calibration and Correction

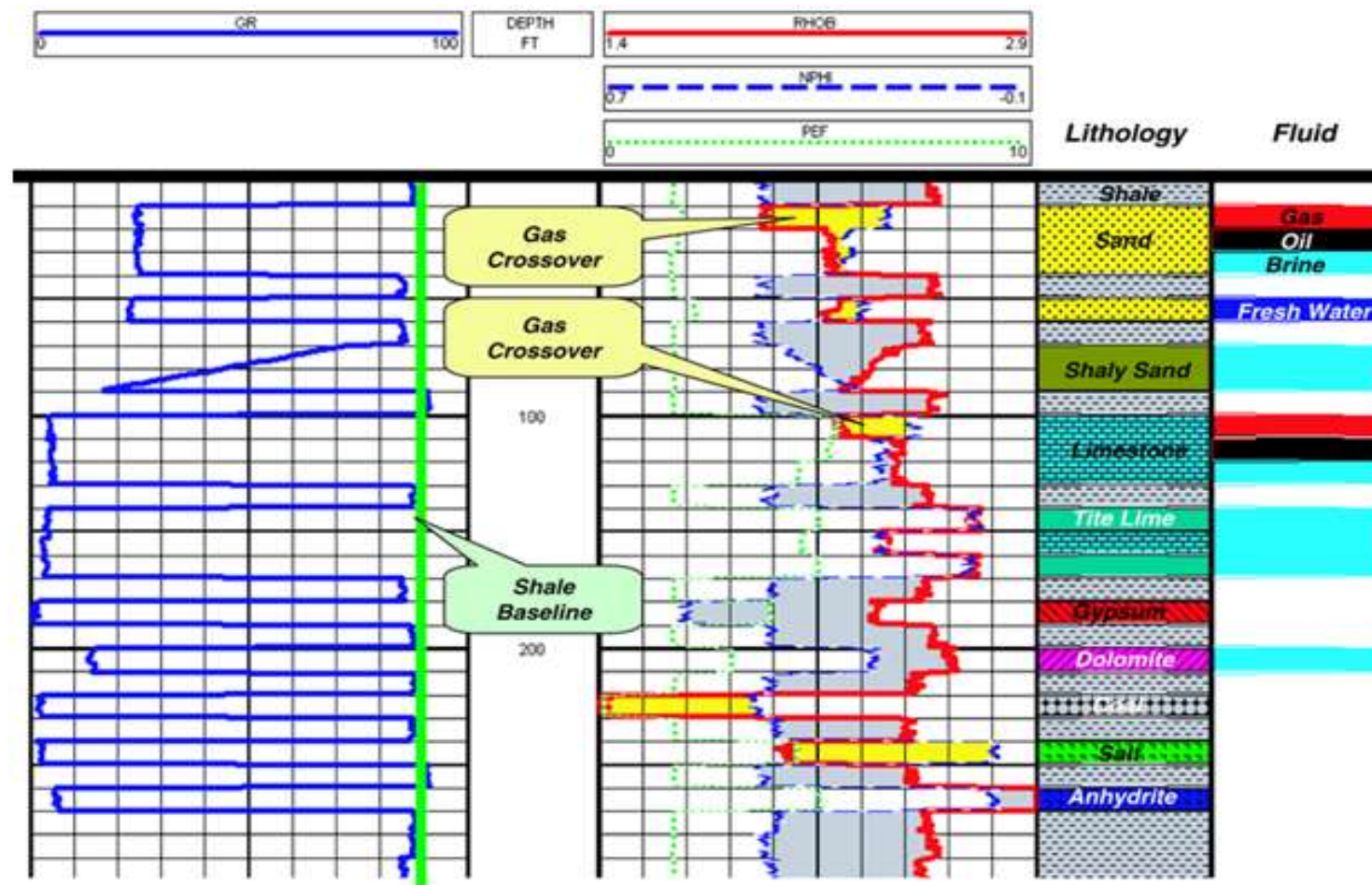
- This tool is calibrated in blocks of limestone, sandstone and dolomite of high purity and accurately known porosity in test pits. The tools are calibrated, not to give readings in API neutron porosity units, but to give the porosity directly in percent. Limestone is chosen as the calibrating lithology, so the readings are given in limestone porosity units.



- The common use of limestone porosity units as the calibrating unit for these logs means that the reading of porosity in a limestone formation will be exact, but the reading of porosity in these units in any other formation lithology will need correction.
- The calibration of the CNL tool is checked at the well site before and after each logging run by the use of a neutron source of accurately known activity placed a standard distance from each detector.

Log Interpretation

- Used in triple- and quad-combo toolstrings, the combination of the CNL and other porosity logs results in more accurate porosity values and lithology identification, including the evaluation of shale content.



CNL Response for Common Formations

Porosity from CNL Log

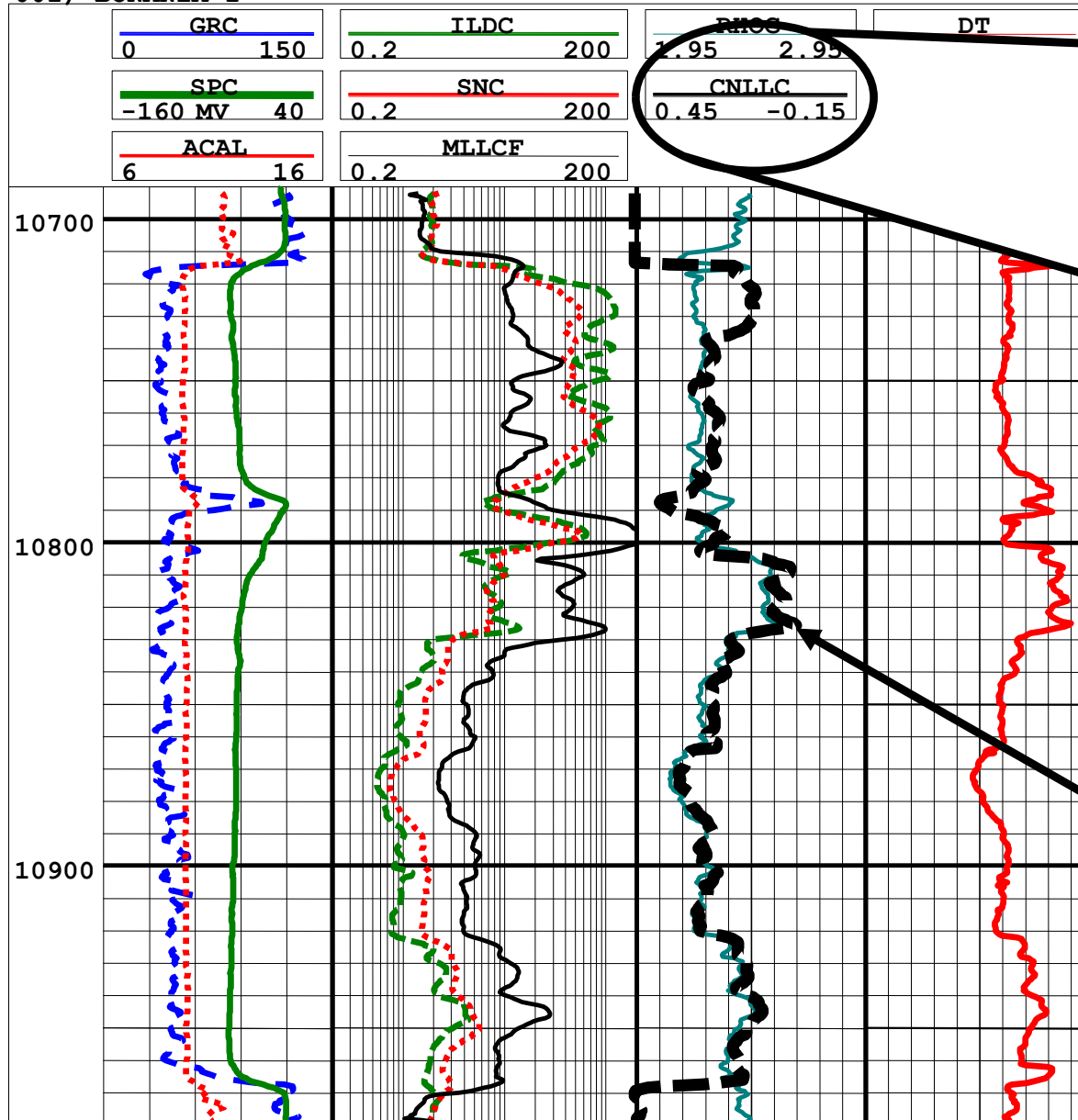
- At fundamental level, the neutron log records the amount of hydrogen in the formation regardless of if it occurs in a gas, solid, or liquid. However, the commercial application of this measurement is to assess the porosity in potential reservoir zones.
- The neutron log is anchored to two calibration points: the matrix and the fluid. The matrix choices are calcite, dolomite, or quartz. These have slightly different neutron moderation characteristics, so that the zero porosity point differs slightly, depending on whether the scale selected is limestone-equivalent, sandstone equivalent, or dolomite equivalent. The units can either be in percent (often written as "porosity units" or "p.u.") or in decimal units (zero to one).
- Because the neutron log is actually a response to the hydrogen in the formation, regardless of its presence in a gas, liquid or solid, hydrogen-rich lithologies will show high neutron porosities, even though there is no effective pore space. Examples include;

Porosity from CNL Log

- Shales
- Water of Crystallization: some evaporitic minerals, such as gypsum have water of crystallization. Gypsum is a hydrated form of calcium sulfate with a formula of $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$. The gypsum response exceeds 60% equivalent limestone porosity on a neutron log.
- Hydroxyls: minerals that have a hydroxyl component (OH). Goethite is an example, with its formula of $\text{FeO}(\text{OH})$, results in a neutron porosity reading in excess of 60%.
- Coal: the high neutron porosities of coals are caused by their high moisture content. Their neutron porosity varies by rank so that anthracite has an expected value of about 38%, while bituminous coals have much higher values.
- In contrast with these high hydrogen sources, the occurrence of gas is marked by anomalously lower neutron porosities. The neutron porosity scale is calibrated to a pore content of water. Both water and oil have a high concentration of hydrogen per unit volume of pore space. However, hydrogen is relatively diffuse in a gas such as methane, so that the recorded neutron porosity is less than the actual pore volume. Together with the density porosity, which becomes anomalously higher with gas saturation, the "gas effect" has commercial implications in the recognition of gas zones.

POROSITY FROM NEUTRON LOG

001) BONANZA 1



SONIC (ACOUSTIC) LOG

Sonic Log

- Acoustic logs or Sonic logs measure the velocity of elastic waves (or slowness, the inverse property) of the formation. Therefore the general elastic properties of the formation are investigated with the measurements
- **Uses**
 - Porosity determination: from interval transit time, DT
 - Lithology Identification: with the Density and/or Neutron.
 - Synthetic Seismograms: with the Density.
 - Formation mechanical properties: with the Density.
 - Detection of abnormal formation pressures
 - Permeability identification: (with waveforms)
 - Cement bond quality
 - Borehole size: (from an attached caliper)

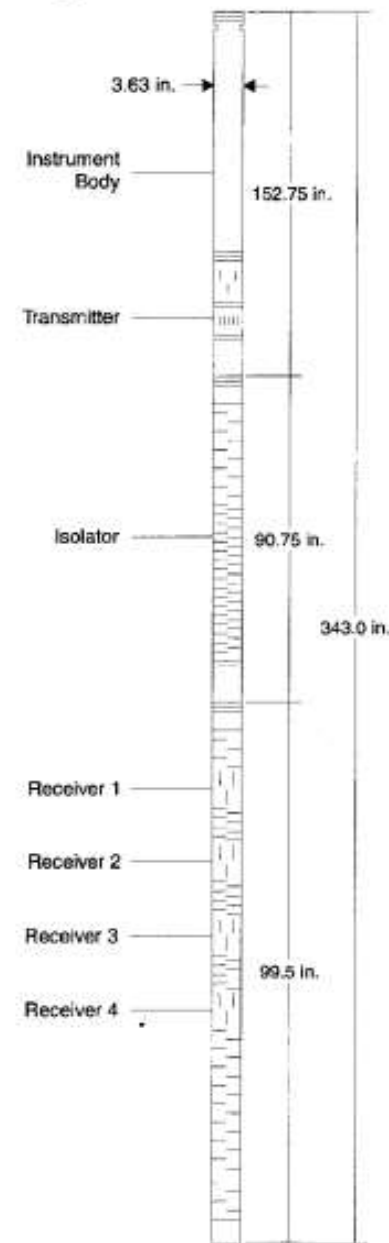
Sonic Logging Tool

- **Physics of Measurement**

- A high frequency acoustic (ultrasonic) pulse from a transmitter is detected at two or more receivers. From the borehole wall, incident wave propagates as reflected and refracted waves. The time of the first detection of the transmitted pulse at each receiver is processed to produce an interval transit time (DT or Δt). The delta t is the transit time of the wave front over one foot of formation. If the entire acoustic waveform is captured, arrival times and attenuations (energy decrease) of several portions of the waveform can be measured including: compressional, shear and Stoneley.
- “Compensated” tools use multiple transmitter-receiver pairs to minimize the effects of borehole size changes. “Array” tools usually have 4 or more receivers, and the data from all receivers is processed to determine the arrival times.

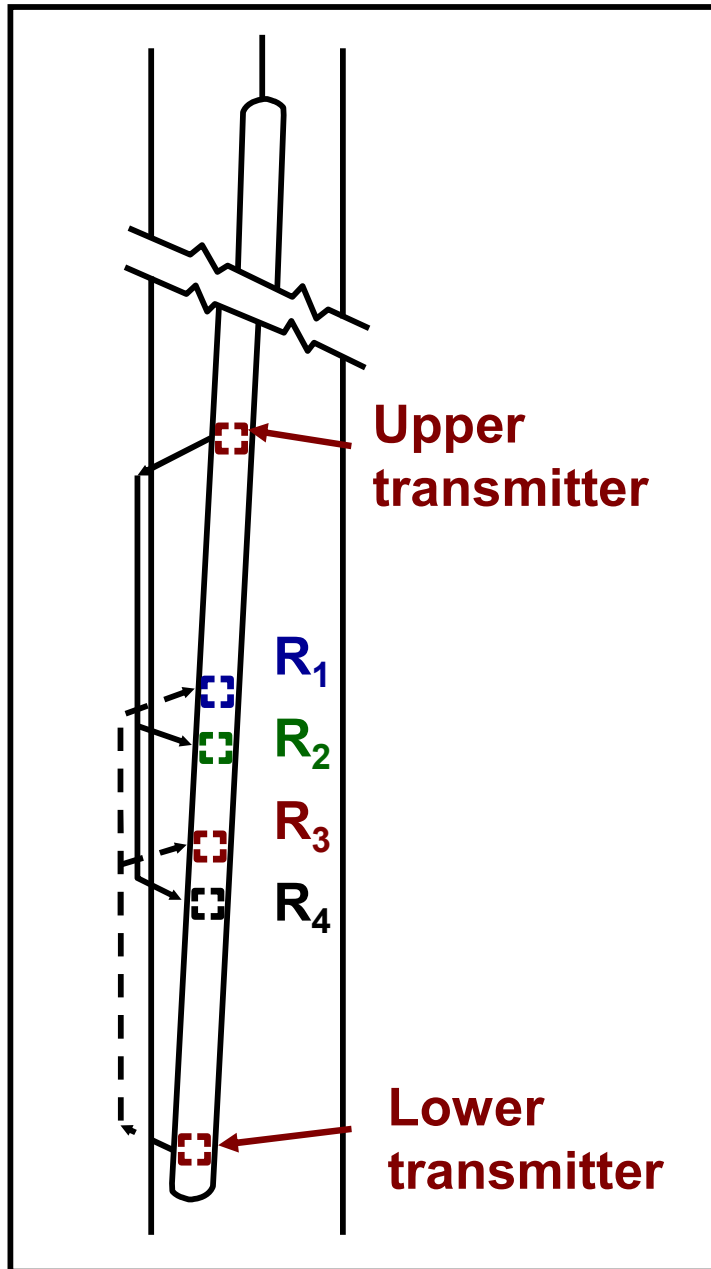
- **Operational Constraints**

- In Open hole, to minimize signal attenuation, the tool should be run centered in boreholes < 16in with a standoff to reduce road noise.
- Can be run in water or oil-based mud



Full Wave Sonic Tool

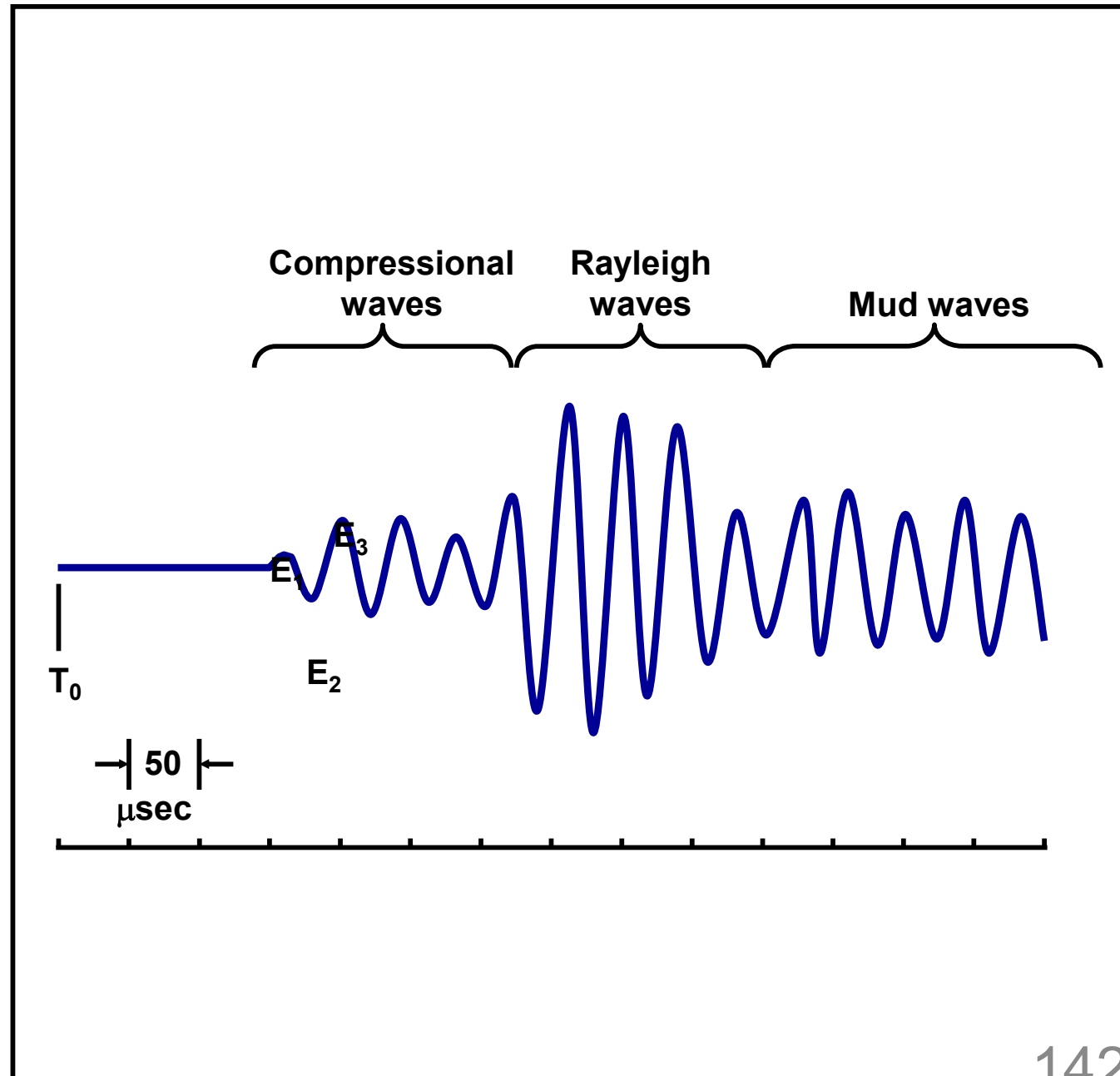
Sonic Logging Tool



- Sonic tools are usually borehole compensated (BHC), which substantially reduces spurious effects at hole size changes as well as errors due to sonde tilt. The BHC transmitters are pulsed alternately, and the Δt readings are averaged. In this way, the tool is compensated for tilt.
- If the travel time for the matrix is known, then porosity can be calculated.
- The speed of sound in the sonic sonde and mud is less than that in the formations. Accordingly, the first arrivals of sound energy the receivers corresponds to the sound-travel paths in the formation near the borehole wall.
- Tool usually consists of one sound transmitter (above) and two receivers (below)
- Sound is generated, travels through formation
- Elapsed time between sound wave at receiver 1 vs receiver 2 is dependent upon density of medium through which the sound traveled

Sonic Log Response

This figure illustrates sonic log response to an acoustic wave transmitted through a compacted formation. The time, T_0 , at which the acoustic wave was initiated at the transmitter, is shown for reference. The first arrival at the receiver is the compressional wave. The Rayleigh wave, traveling at a slower rate, arrives later and is superimposed on the compressional wave. Following the Rayleigh waves are the slower mud waves, transmitted through the mud column and the tool.



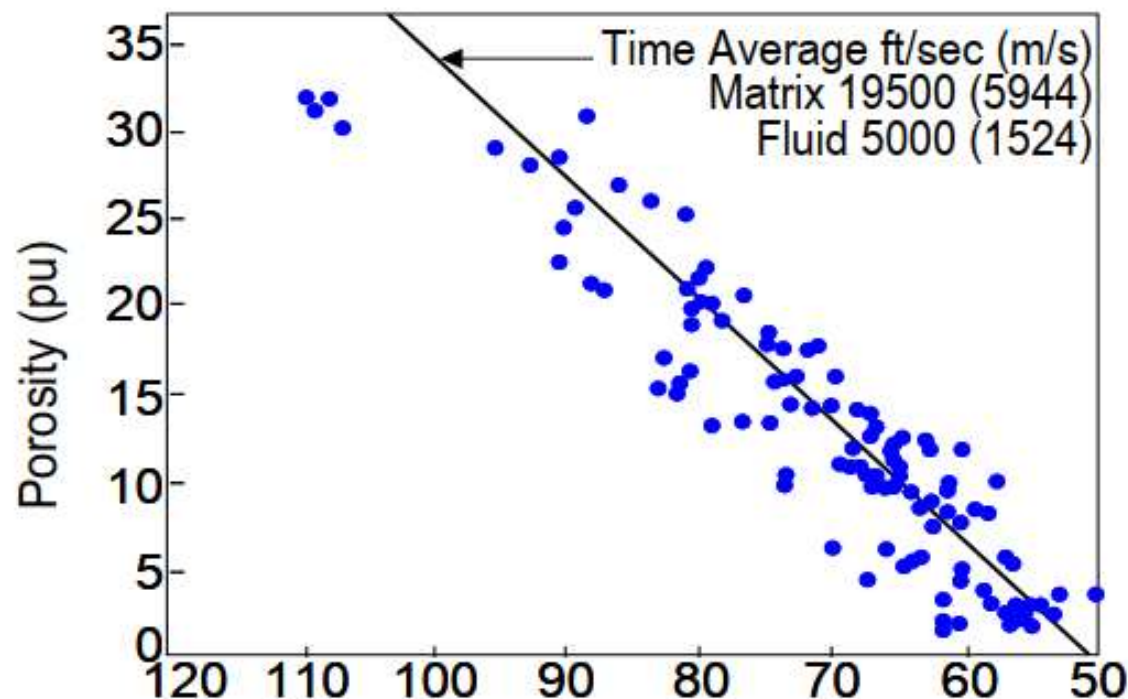
Time Average Relationship

From the strong correlation between compressional wave slowness and porosity, Wyllie et. al (1956) derived their famous time-average equation. Written in terms of slowness:

$$\Delta t_p = \phi \Delta t_{fl} + (1 - \phi) \Delta t_{p,ma}$$

$$\phi = \frac{\Delta t_p - \Delta t_{p,ma}}{\Delta t_{fl} - \Delta t_{p,ma}}$$

This equation works best for water-saturated and well-compacted porous rock, particularly sandstones. The presence of gas can give erroneous results



- Δt_p compressional wave slowness of the porous rock
- $\Delta t_{p,ma}$ compressional slowness of the matrix material
- $\Delta t_{p,ma}$ compressional slowness of the pore fluid (mostly assumed water)
- ϕ is the porosity

Mean Input Parameters for Wyllie's Equation & Corrections

Rock type	$\Delta t_{p,ma}$ $\mu\text{s}/\text{ft}$	$\Delta t_{p,ma}$ $\mu\text{s}/\text{m}$	Fluid	Δt_f $\mu\text{s}/\text{ft}$	Δt_f $\mu\text{s}/\text{m}$
Sandstone $\phi > 0.1$	55.5	182	fresh water mud filtrate	189	620
Sandstone $\phi < 0.1$	51.2	168			
Limestone	47.6	156	Saltwater mud filtrate	185	607
Dolomite	43.5	143			

Anhydrite: $50.0 \mu\text{s m}^{-1}$, Halite: $66.7 \mu\text{s m}^{-1}$.

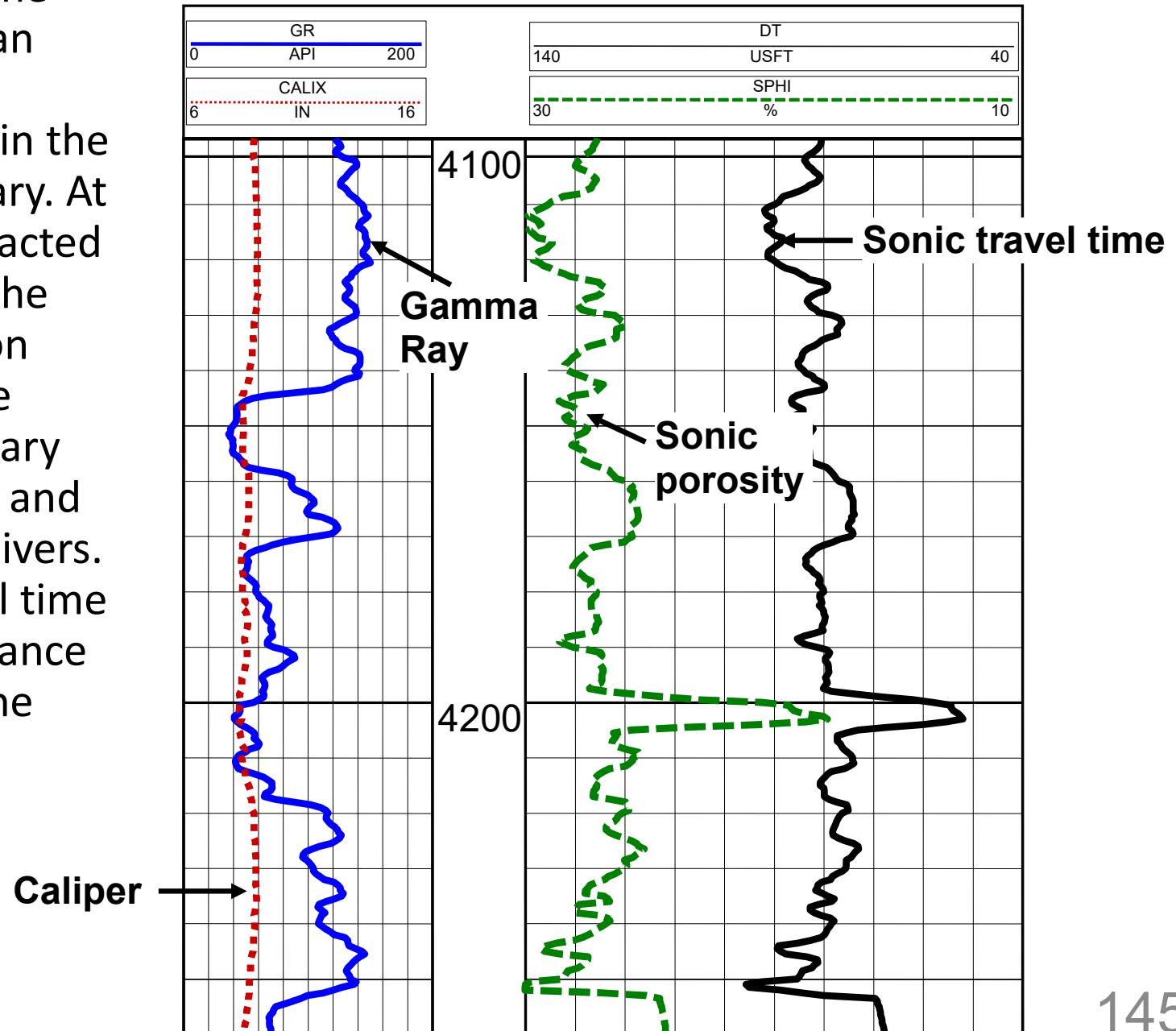
Compaction Correction

- Poor consolidation or poor effective stress results in high slowness values and therefore an overestimate of porosity. In this case a “compaction correction” is recommended. The slowness in an adjacent shale bed is used as “compaction reference”. If measured slowness is $\Delta t_{shale} > 100 \mu\text{s m}^{-1}$, the following equation for a compaction-corrected Wyllie porosity is:

$$\phi_{corrected} = \phi_{Wyllie} \cdot \frac{1}{C_p} = \frac{\Delta t - \Delta t_{ma}}{\Delta t_{fl} - \Delta t_{ma}} \cdot \frac{100}{\Delta t_{shale}}$$

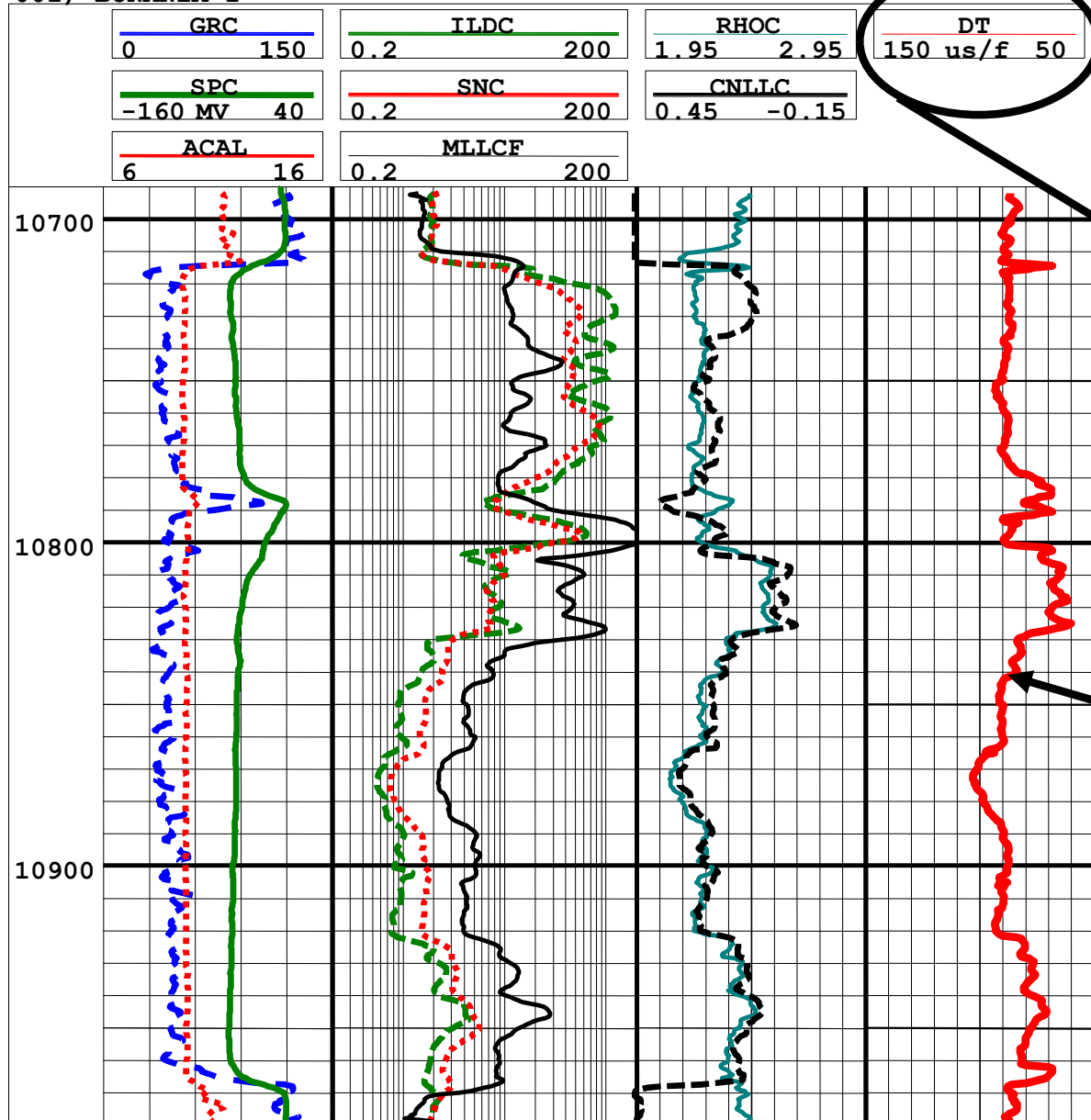
ACOUSTIC (SONIC) LOG

If the wave velocity of the formation is greater than that of the mud, the refraction wave moves in the direction of the boundary. At the critical angle, a refracted wave propagates with the velocity of the formation along the borehole. The wave generates secondary waves passing the mud and arriving at the two receivers. Difference of the arrival time divided by receiver distance gives the slowness of the formation (if the tool is centered).



SONIC LOG

001) BONANZA 1



DT
150 $\mu\text{s}/\text{f}$ 50

Sonic
Log

FACTORS AFFECTING SONIC LOG RESPONSE

- Unconsolidated formations
- Naturally fractured formations
- Hydrocarbons (especially gas)
- Rugose salt sections

RESPONSES OF POROSITY LOGS

The three porosity logs:

- Respond differently to different matrix compositions
- Respond differently to presence of gas or light oils

Combinations of logs can:

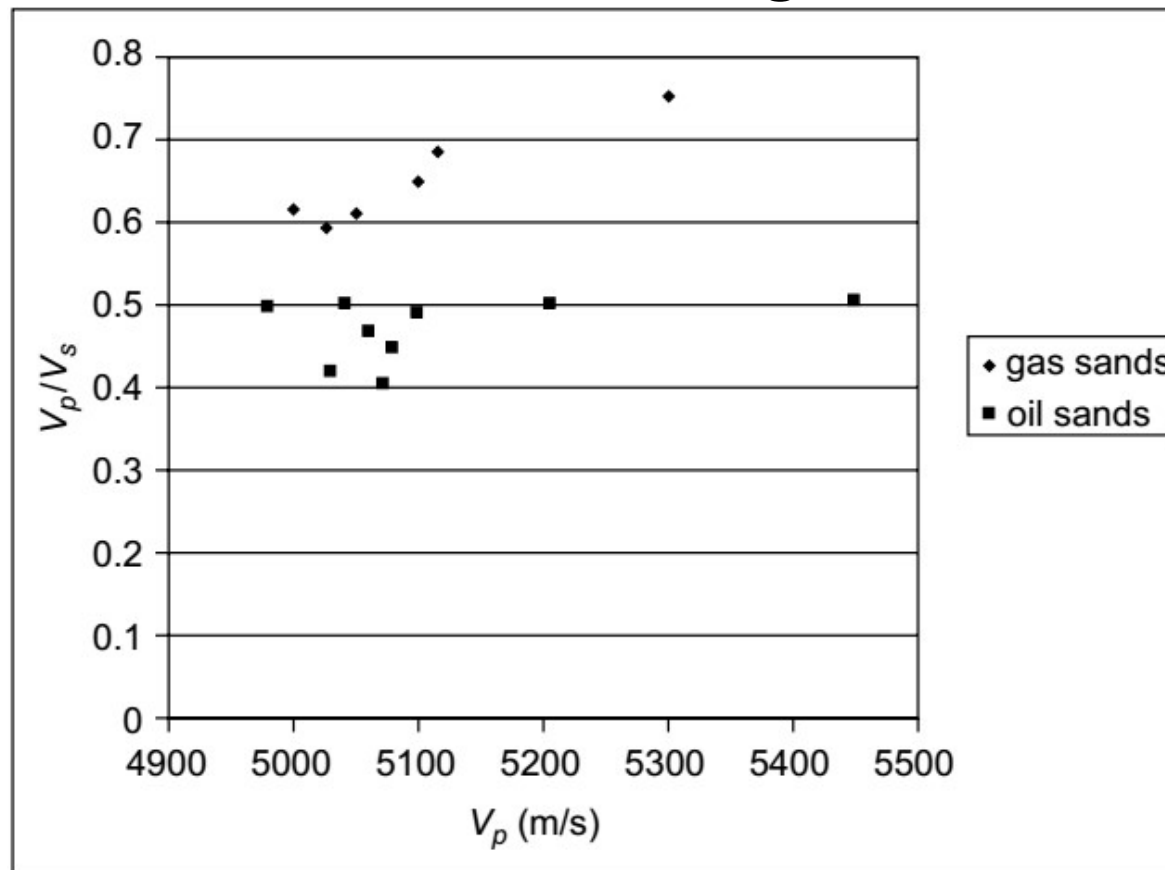
- Imply composition of matrix
- Indicate the type of hydrocarbon in pores

GAS EFFECT

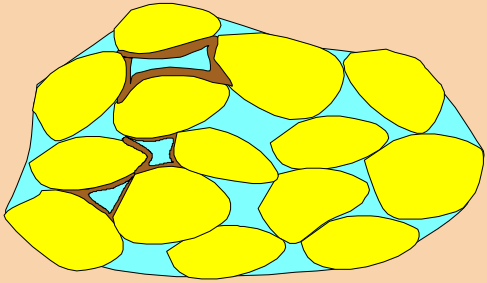
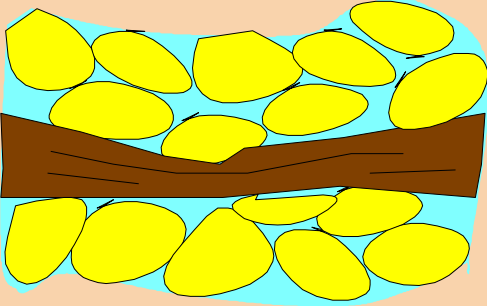
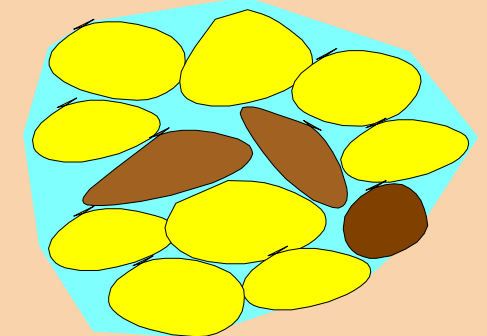
- Density - ϕ is too high
 - Neutron - ϕ is too low
 - Sonic - ϕ is not significantly affected by gas
- Remember that the density log, the neutron log, and the sonic logs do not measure porosity. Rather, porosity is calculated from measurements such as **electron density**, **hydrogen index** and **sonic travel time**.
 - The calculated density porosity is too high only because in the calculation we typically don't account for the fluid density change. In other words, we assume the fluid density is 1 (or completely liquid filled) even though with gas that value is lower, which causes the calculated porosity to be too high.
 - The neutron porosity is too low because the hydrogen index or the hydrogen density of gas is lower; therefore, the liquid-filled porosity is what the neutron log sees. So when gas is present, that value is lower than the actual porosity.
 - And finally, the sonic log is not significantly affected by gas because it reads very near the wellbore and small gas saturations do not impact the overall travel time significantly.

Identifying the Fluid Type and Contacts

- One reliable way to identify gas zones is to use the shear sonic log (if available) combined with the compressional sonic. If ratio of compressional velocity (V_p) to sheer velocity (V_s) is plotted against V_p , then because V_p is much more affected by gas than is V_s , a deviation will be observed in gas zones.



Influence Of Clay-Mineral Distribution On Effective Porosity

	Dispersed Clay • Pore-filling • Pore-lining • Pore-bridging	ϕ_e Clay Minerals
	Clay Lamination	ϕ_e
	Structural Clay (Rock Fragments, Rip-Up Clasts, Clay-Replaced Grains)	ϕ_e

NUCLEAR MAGNETIC RESONANCE (NMR) LOG

Nuclear Magnetic Resonance (NMR)

- Multifrequency NMR measurements are taken to derive a permeability estimate and realise fluid characterisation
- is essentially an electrical measurement, is sensitive to the quantity and distribution of free protons in the formation. Free protons occur uniquely in the fluids, so that their quantity provides another value for porosity (lithology-independent).
- Their distribution, in small pores or large pores, leads to the determination of an average pore size and hence, through various empirical transforms, to the prediction of permeability.
- The viscosity of the fluid also affects the movement of the protons during a resonance measurement, so that the data can be interpreted to give viscosity.
- **Free-fluid index:** The percent of the bulk volume occupied by fluids that are free to flow often measured by the NMR log.

Nuclear Magnetic Resonance

- NMR measurements use the interaction of hydrogen nuclei with an external magnetic field.
- Spin magnetic moment is the magnetic moment caused by the spin of elementary particles
- Magnetic moment is the measure of an object's tendency to align with a magnetic field. Magnetic moment is defined as magnetic strength and orientation of a magnetic or other object that produces a magnetic field

NMR models for Permeability Estimation

- SDR model: $k = A * \varphi^m * T_{2gm}^n = A * \varphi^4 * T_{2gm}^2$
- Where k is permeability (md), A is the formation-dependent correction coefficient (md ms⁻²)

- Coates model

$$k_{Coates} = \left(\frac{\phi}{C} \right)^m \cdot \left(\frac{BVM}{BVI} \right)^n$$

- where

ϕ is porosity in percent,

BVM is bulk volume fluid movable,

BVI is bulk volume fluid non-movable (irreducible),

m, n are empirical parameters and approximately equal to 4 and 2,

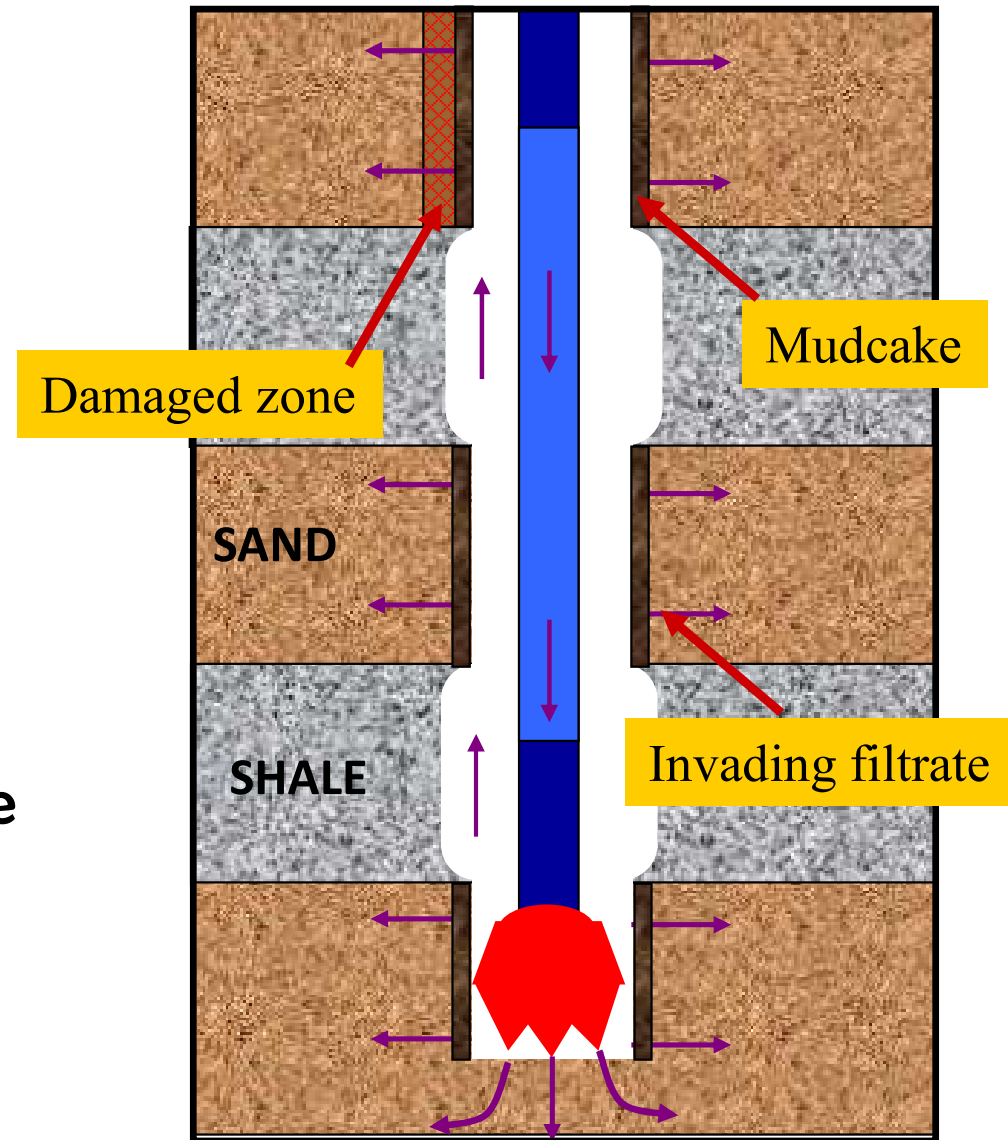
C is also an empirical parameter generally between 6 and 15.

Active Well Logging Tools: Resistivity Measurements – Lecture 5

THE WELLBORE ENVIRONMENT

Drilling Disturbs Formation

- Drilling and rock crushing
 - ✓ Damaged zone
- Mud systems and invasion
 - ✓ Oil based mud
 - Small conductivity mud
 - Shallow invasion
 - Thin cake
 - ✓ Water based mud
 - Moderate to very conductive mud
 - Shallow to deep invasion
 - Thin to thick cake

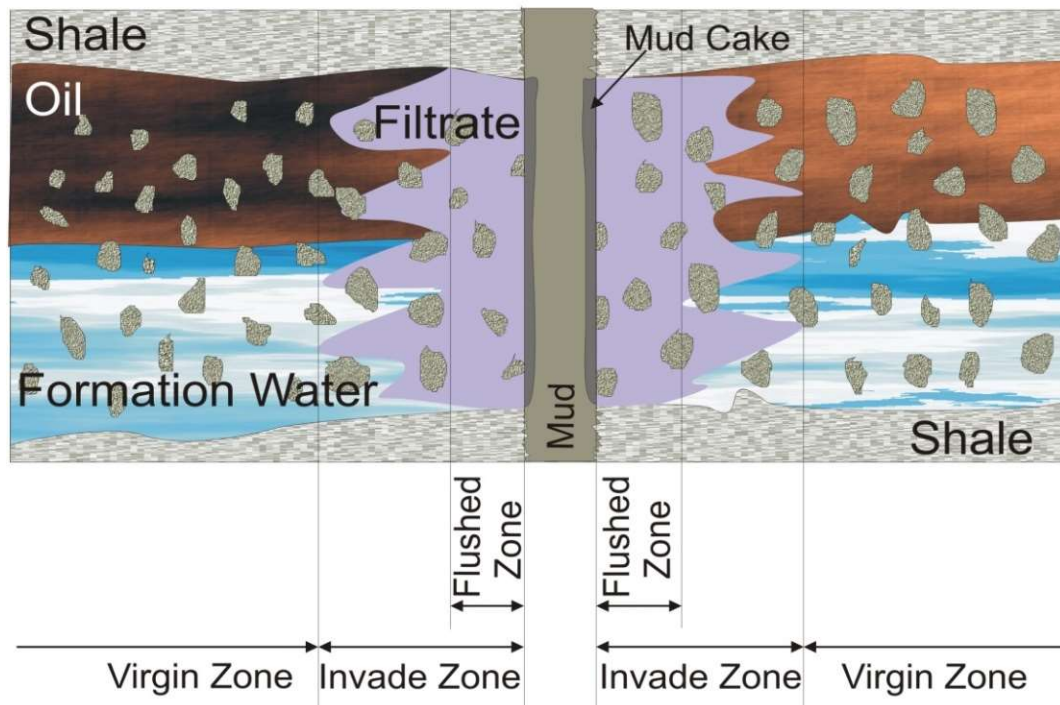
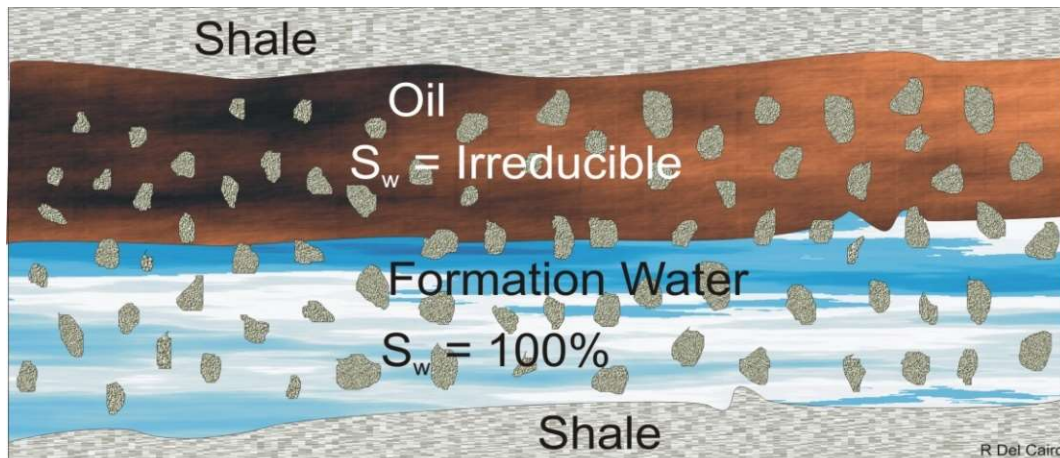


Degradation of the formation during and after drilling. Overpressured mud is indicated to be invading porous and permeable sand formations with the formation of a mudcake. The mud circulation also causes borehole washout in the shale zone

The Invasion Process

- The generally overpressured wellbore causes invasion of a porous and permeable formation by the drilling fluid.
- In the permeable zones, due to the imbalance in hydrostatic pressure, the mud begins to enter the formation but is normally rapidly stopped by the buildup of a mudcake of the clay particles in the drilling fluid. The mudcake usually has a very low permeability (of the order of 10^{-2} to 10^{-4} md) This initial invasion is known as the spurt loss.
- As the well is drilled deeper, further invasion occurs slowly through the mudcake, either dynamically, while mud is being circulated, or statically when the mud is stationary. In addition, the movement of the drill string can remove some mudcake, causing the process to be restarted. Thus, while a typical depth of invasion at the time of wireline logging is 20 in., the depth can reach 10 ft or more in certain conditions.
- Generally, the lower the formation porosity, the deeper the invasion

Invasion Profile



- **Mudcake:** Mud filtrate infiltrates the formation due to pressure differential between the mud column in the borehole and the formation. This causes mud to be deposited on the borehole wall where it forms *mudcake*. Mudcake usually has well low permeability and once developed considerably reduces the rate of further mud filtrate invasion.
- **Flushed zone:** the zone just beyond the mudcake. Original formation water and hydrocarbons have been flushed by Mud Filtrate to residual saturations.
- **Transition zone:** Further out into the wellbore where there is less displacement of mud filtrate. The flushed and transition zones make up the **Invaded zone**.
- **True zone:** the undisturbed zone beyond the transition zone where the fluid type we are interested in measuring is.

Invasion Process

- To account for the distortion which is frequently present with electrical measurements, a simplified model of the borehole/formation in vertical wells with horizontal beds has evolved. It considers the invaded formation of interest, of resistivity R_t , to be surrounded by “shoulder” beds of resistivity R_s .
- The invasion is represented by the profile shown schematically in Fig. 2.1, along with the regions and parameters of interest, starting with the mudcake of thickness h_{mc} and resistivity R_{mc} .
- The next annular region of diameter d_i is the flushed zone whose resistivity is denoted by R_{xo} , determined principally by the resistivity of the mud filtrate. Beyond the invaded zone lies the uninvaded or virgin zone with resistivity R_t .
- A transition zone separates the flushed zone from the virgin zone. The transition may be smooth, but when hydrocarbons are present its resistivity can be significantly lower than either R_{xo} or R_t . This condition is known as an annulus and occurs mainly when the oil or gas is more mobile than the formation water, so that the formation water displaced from the flushed zone accumulates in the transition zone while the oil or gas is displaced beyond it. The annulus disappears with time, but can still exist at the time of logging.

Non Anc

- Resistivity of zone
- Resistivity of the water in the zone
- △ Water saturation in the zone

Borehole

R_m : Borehole mud resistivity;

R_{mc} : Mudcake resistivity

Invaded zone

R_{mf} : Mud filtrate resistivity

R_{xo} : Invaded zone resistivity;

S_{xo} : Invaded zone water saturation

Un-invaded zone

R_w : Interstitial water resistivity;

R_t : Un-invaded zone resistivity;

S_w : Un-invaded zone water saturation

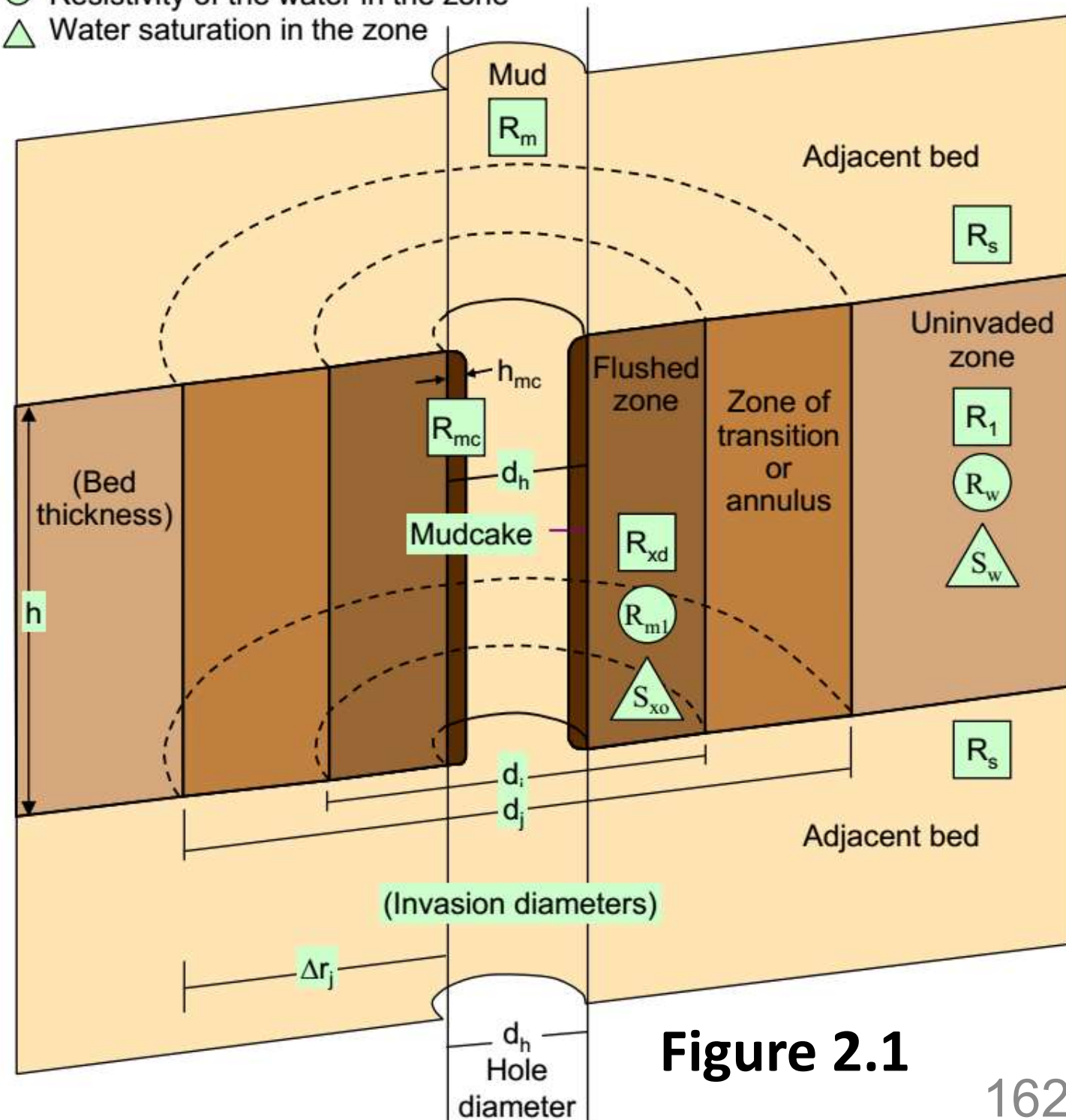


Figure 2.1

Relationship between Mud Filtrate Invasion and Formation Porosity

- The amount of invasion which takes place is primarily dependent on the permeability of the zone and early formation of mud cake. However, before the formation of impervious mud cake, diameter of invasion will be greatest in low porosity rocks.
- Mud cake is the sealing agent which slows down invasion. As a result, high permeability zones which allow quick build-up of mud cake, invade the least and low permeability zones invade the most or deepest.
- Non-permeable zones are not invaded. This sounds counter-intuitive, but that is what happens with a good quality drilling mud. Poor quality or native mud may not form a mud cake, and invasion will proceed in proportion to the permeability.
- The diameter of invasion is often expressed as a ratio: d_i/d_h where d_h is borehole diameter.

Relationship between Mud Filtrate Invasion and Formation Porosity

The radial profile:

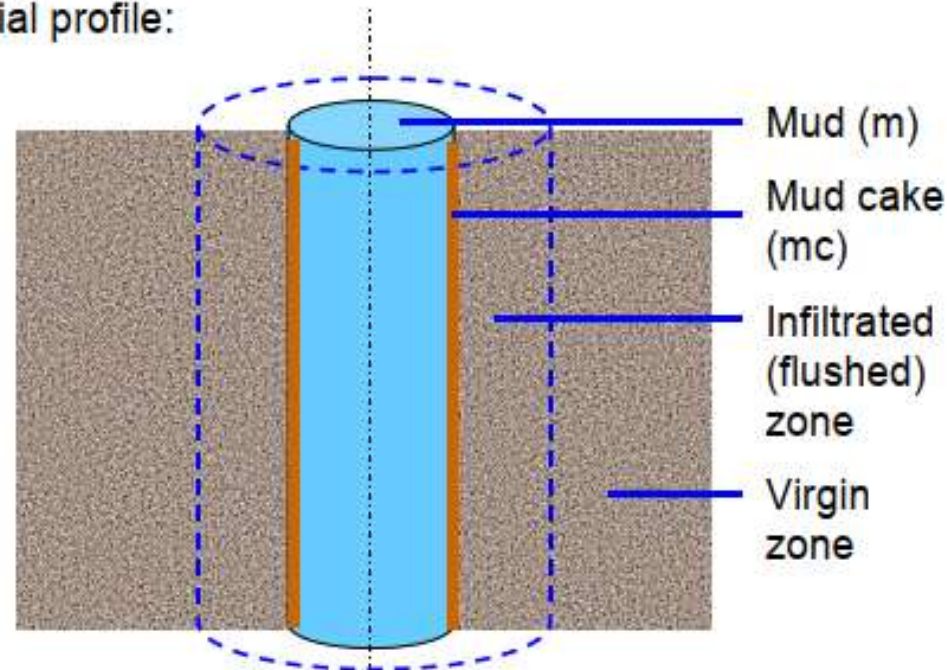


Figure 3-6: Invaded and virgin zone in a permeable section of a borehole.

For the ratio d_i/d_h , i.e. the ratio of the diameter of the invasion d_i to the diameter of borehole d_h (see also Figure 4-6), Asquith and Krygowsky (2004) give the following values as orientation:

$d_i/d_h = 2$	for high porous rocks
$d_i/d_h = 5$	for intermediate porous rocks
$d_i/d_h = 10$	for low porous rocks

A BACKGROUND TO EMPIRICISM (ARCHIE'S WORK)

Formation Factor

- In a clean porous rock (for example sandstone without clay and/or other conductive solid components) the formation water is the only conductive component because matrix (quartz) and HC or air in the pore spaces are insulators. In this case the specific resistivity of the rock is proportional to the specific resistivity of the pore water.
- The specific resistivity of the formation water is a function of salinity (concentration of the salts) and temperature.
- The proportionality between rock resistivity and formation water resistivity is the fundamental idea of the Archie's (1942) equations which describe the relationship of rock resistivity to porosity and water saturation.
- For a water saturated porous rock ($S_w = 1$) the specific rock resistivity R_o is controlled by the resistivity of the pore water thus R_w thus:
$$R_o = F * R_w$$

Formation Factor

- For a given porosity, the ratio R_o/R_w remains nearly constant for all values of R_w below about 1 ohm-m. For fresher, more resistive waters, the value of F may decrease as R_w increases. This phenomenon is attributed to a greater proportionate influence of surface conductance of the rock matrix
- Archie proposed, based on observations, a formula relating porosity, ϕ , and formation factor; which is: **$F = a/\phi^m$**
- Where the exponent **m** is the “cementation exponent” or “porosity exponent” whose value varies with grain size, grain size distribution and the complexity of the paths between the pores i.e. Tortuosity (**m** is in the order of 2 for sandstones). The constant a is also determined empirically.
- Archie went on to suggest that “ **m** has been found to range between 1.8 and 2.0 for consolidated sandstones . . . loosely or partly consolidated sands might have a value of m anywhere between 1.3 and 2.” m values of 2.2 to 3 have been obtained for oolitic rocks

Water Saturation

- For partial water saturation S_w , the specific rock resistivity R_t increases with decreasing water saturation. Archie (1942) formulated this by introducing a “Resistivity Index I ” and referenced the rock resistivity R_t at any water saturation to the rock resistivity at total water resistivity R_o thus:

$$R_t = I * R_o = R_o * (S_w)^{-n}$$

Where n is the empirical saturation exponent

- In the plot I versus water saturation S_w with logarithmically scaled axis, this is also represented by a straight line with the slope n .
- The resistivity index I expresses the magnification of the rock resistivity (compared with water resistivity) as a result of non-conductive fluid components (oil, gas).
- The relationships deliver Archie’s fundamental equation for water saturation calculation

$$S_w = \left(\frac{R_o}{R_t} \right)^{\frac{1}{n}} = \left(\frac{R_w}{R_t} \cdot \frac{1}{\phi^m} \right)^{\frac{1}{n}}$$

Water Saturation

- In many cases (and if no other information is available) $m \approx n \approx 2$ is recommended for a first orientation. In this case the equation results in:

$$S_w = \sqrt{\frac{R_o}{R_t}} = \frac{1}{\phi} \cdot \sqrt{\frac{R_w}{R_t}}$$

- The formation resistivity R_o at $S_w = 1$ can be:
 - Estimated for a quick look interpretation from a water zone in the same reservoir
 - Calculated from Archie's equation with water resistivity R_w , an independent porosity determination, and the exponent m
- The most appropriate application of the above equation is a thick reservoir rock of constant porosity, which has a water column at its base and an oil column at its top. This is because its use assumes that the two beds have similar porosities and formation factors and contain formation waters of similar salinity

Applications of Archie's Saturation Formula

- In a permeable porous formation Archie's equation can be applied
 - On the non-invaded zone in order to determine S_w from formation resistivity R_t . $(1 - S_w) = S_h$ gives HC saturation.

$$S_w = \left(\frac{R_o}{R_t} \right)^{\frac{1}{n}} = \left(\frac{R_w}{R_t} \cdot \frac{1}{\phi^m} \right)^{\frac{1}{n}} \qquad S_{xo} = \left(\frac{R_{mf}}{R_{xo}} \cdot \frac{1}{\phi^m} \right)^{\frac{1}{n}}$$

- On the invaded zone in order to determine the mud filtrate saturation S_{xo} from the resistivity of the invaded zone R_{xo} using mud filtrate resistivity R_{mf} as resistivity of the conductive fluid with the saturation equation.
- The difference $(1 - S_{xo}) = S_{hr}$ gives the non-movable (residual) hydrocarbon saturation (part of the original fluid which is not displaced by infiltration) or residual HC saturation. The fluid displacement is an indication of fluid mobility.
- The bulk volume of moved oil is $\phi^*(1 - S_w)$.
- N.B: Movable Saturation: $S_h - S_{hr} = (1 - S_w) - (1 - S_{xo}) = (S_{xo} - S_w)$ 170

Shaly Rocks

- The conductivity contribution of shale/clay depends on:
 - Shale Content (Volume fraction, V_{sh}) , derived from e.g. Gamma Log
 - Shale Type / properties (Clay Mineralogy) controlling the “shale conductivity”
 - Distribution of shale in the formation (laminar shale or dispersed shale)
- Standard methods for a shale content estimate are: SP-log, Gamma-log, Neutron-Density-Combination.
- Calculation of the shale volume is the **first** step of quantitative interpretation. The calculated shale content is used for:
 - General Reservoir Description
 - Volumetric Rock Model
 - Various “shale corrections”
- Various “shaly sand equations” have been derived in order to model the different types of shale in a rock.

Some Shaly Sand Equations

- Waxman-Smit Eq : $S_w^{-n^*} = \left[\left(\frac{R_t}{R_w} \right) (\varphi^{m^*}) \left(1 + \frac{R_w B Q_v}{S_w} \right) \right]$
- Dual-water model: $\frac{1}{R_t} = \left[\left(\frac{S_w^2 \varphi^m}{R_w} \right) + (B Q_v S_w \varphi^m) \right]$
- Simandoux:: $\frac{1}{R_t} = \left[\left(\frac{S_w^2 \varphi^m}{R_w} \right) + \left(\frac{V_{sh} S_w}{R_{sh}} \right) \right]$
- Indonesian Eq.: $S_w^n = \left(\frac{1}{R_t} \right) \left[\sqrt{\frac{V_{sh}^{(2-V_{sh})}}{R_{sh}}} - \sqrt{\frac{\varphi^{(m)}}{R_w}} \right]^{(2)}$

A BACKGROUND TO RESISTIVITY MEASUREMENTS

Resistivity

- The Electrical resistivity of a substance is its ability to impede the flow of electrical current through the substance as opposed to resistance, which is associated with the geometric form of the material.

- The reciprocal of resistivity is conductivity

$$\text{Resistivity} = \frac{1}{\text{Conductivity}}$$

- Units are ohm-m² / m; usually ohm-m (Ω.m)

$$R = \frac{r \times A}{L}$$

- Factors Affecting Resistivity:

- Resistivity of water
- Porosity of the formation,
- Pore geometry - tortuosity
- Lithology of the formation
- Degree of cementation, and
- Type and amount of clay in the rock.



Applications of Resistivity Measurements

- Of all the rock parameters measured by today's logging tools, resistivity is of particular importance
- Two major applications of resistivity measurements:
 - Determination of water saturation
 - Characterization of the invasion process.
- Formation resistivity covers a broad range of magnitudes depending on porosity, water saturation, salinity and shaliness.
- The specific resistivity of the formation water is a function of salinity (concentration of dissolved salts) and temperature.
- The temperature dependence of NaCl-based water can be described by Arps' equation.

$$R_w(T_2) = R_w(T_1) \cdot \frac{T_1 + 21.5}{T_2 + 21.5}$$

Where temperatures T_1 & T_2 are in °Celsius

$$R_w(T_2) = R_w(T_1) \cdot \frac{T_1 + 6.77}{T_2 + 6.77}$$

Where temperatures T_1 & T_2 are in °Fahrenheit

Resistivity logs

- Resistivity tools can be grouped into two:
 - **Electrode tools:** electrical current sent by electrodes into formation. *Requires water-base muds*
 - **Induction tools:** generate a magnetic field that induces a current in the formation (Oil-base, air, or fresh-water muds)

Electrode Tools

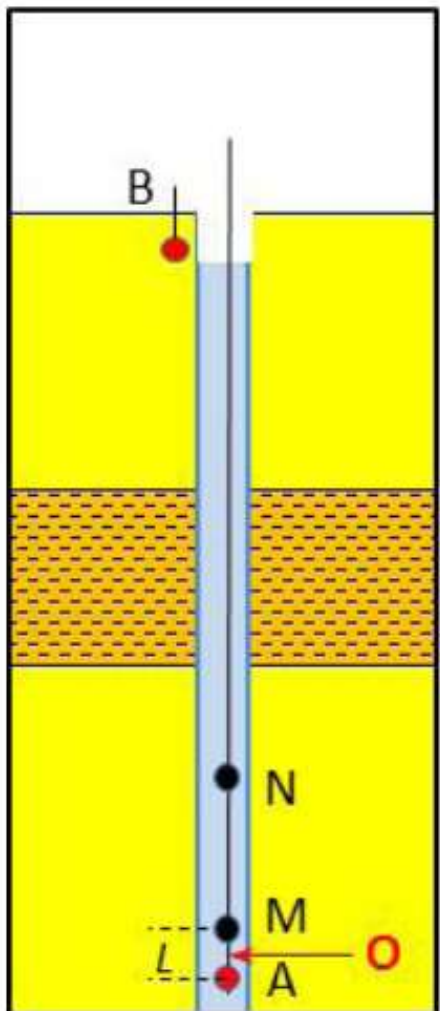
- **Wireline (conveyed on line)**
 - ✓ Dual Laterolog (LLD and LLS)
 - ✓ Azimuthal tool (deep and shallow)
 - ✓ Spherically-focused
 - ✓ Micro-resistivity
- **LWD (Logging-While-Drilling)**
 - ✓ Resistivity at bit (or at least close)
 - ✓ Side-scanning electrodes

Induction Tools

- **Wireline (conveyed on line)**
 - ✓ Dual Induction (Deep; Medium)
 - ✓ High Resolution Induction
 - ✓ Dielectric induction
 - ✓ Array Induction
 - ✓ EWR/CDR
- **LWD (Logging-While-Drilling)**
 - ✓ x

Early Resistivity Logging Tools

- This first resistivity tools used 4 electrodes (current electrodes A & B, voltage electrodes M & N) - The Normal tool and Gradient tool.
- Conventional tools have moderate depth of investigation (depth increases with spacing) and no sharp vertical resolution (vertical resolution decreases with spacing)

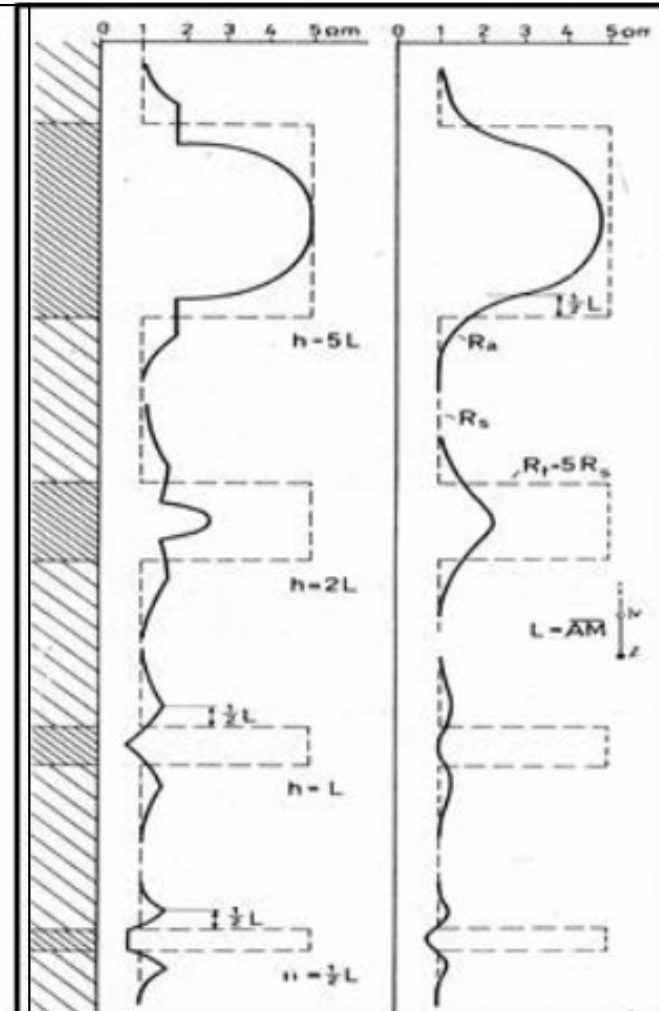


Electrode configuration of a Normal sonde.

There are 2 standard configuration of 4-electrode normal tools

(a) Short normal with the spacing $L=16$ inch ≈ 0.40 m

(a) Long normal with the spacing $L=64$ inch ≈ 1.60 m



Shape of resistivity curves (Normal) and influence of spacing L & bed thickness h .

Left trace without mud influence, Right trace with mud influence

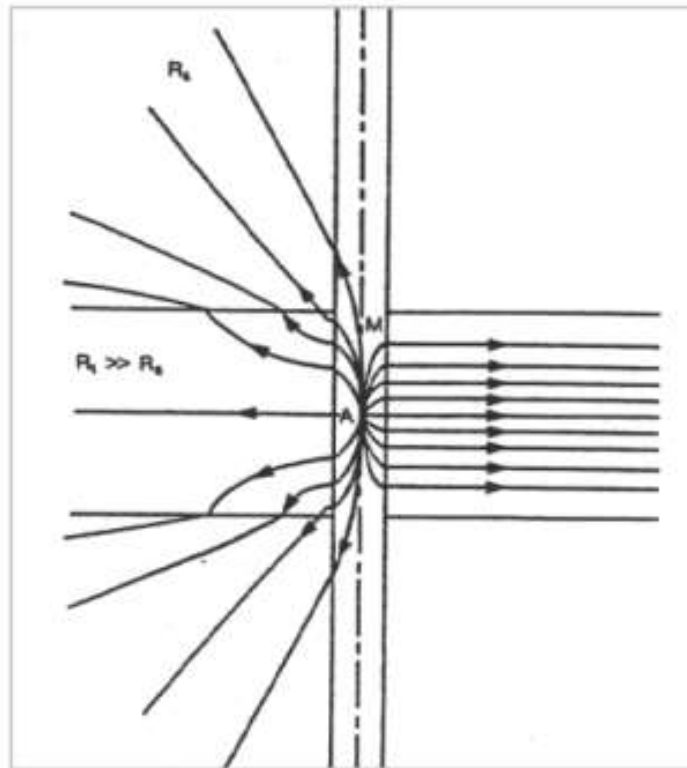
Issues with Early Resistivity Measurements

- The ratio of bed thickness to spacing controls the logged resistivity curve. There is no sharp step at the boundary but a smoothed shape called “shoulder bed effect”.
- In highly resistive formations the current from the electrode A has the tendency to miss the formation and prefer adjacent better conductive zones and the mud column.
- Focusing electrode systems or laterologs were developed. Additional bucking or guard electrodes above and below the measuring current electrode A emit current to focus the measuring current laterally into the formation. The guard current is regulated such that there is no potential difference between additional monitoring electrodes.
- With this technique three goals are attained: (a) Focusing measuring current flowing perpendicular to the tool (b) Deep penetration of the focused current (into the invaded zone) and simultaneously (c) Sharp vertical resolution

Issues with Early Resistivity Measurements

Non-focusing tool (normal device):

For non-focusing tools the “free current flow” avoids the highly resistive layer and takes the way of “lowest resistivity”.



Focusing tool (Laterolog):

The additional bucking current from the guard electrodes presses the measuring current into the highly resistive formation.

- Current lines in a highly resistive bed for conventional (a) non-focusing (b) focusing tools

INDUCTION LOG

Induction Log - Introduction

- Induction log was originally developed to measure formation resistivity in boreholes containing oil-based and fresh water-based drilling muds.
- Electrode devices (conventional electric logs) do not function in non-conductive mud systems.
- Induction logging devices are focused to minimize influence of borehole and surrounding formations.
- Designed for deep investigation to determine R_t .
- New induction log devices are being developed using improved electronics, telemetry, and computer processing.

Induction - Interpretation Goals

- **True (undisturbed) formation resistivity, R_t :**
 - True formation resistivity can be determined from the deep induction curve by applying corrections for borehole size, mud resistivity and bed thickness. The medium and shallow curves are needed for invasion correction along with the proper algorithms and/or charts. In most cases, the deep induction curve is sufficiently close to R_t thus not requiring correction.
- **Geopressure (overpressure) Detection:**
 - Deep resistivity values in the shales in a well are plotted on a reduced scale, and a trend with deep is defined. The trend is usually one of increasing resistivity with depth. An abrupt decrease in shale resistivity with depth usually indicates an increase in formation pressure.
- **Diameter of Invasion:**
 - is determined as a by-product of the correction of the Deep Induction curve using a graphical “tornado chart” (or mathematical equivalent).
- **Correlation:**
 - Curves are scanned for similarities in shape and magnitude. The logs are often displayed at scales of 1 inch per 100 feet or 2 inches per 100 feet. Correlation of resistivity is really a correlation of fluid volume and fluid salinity.

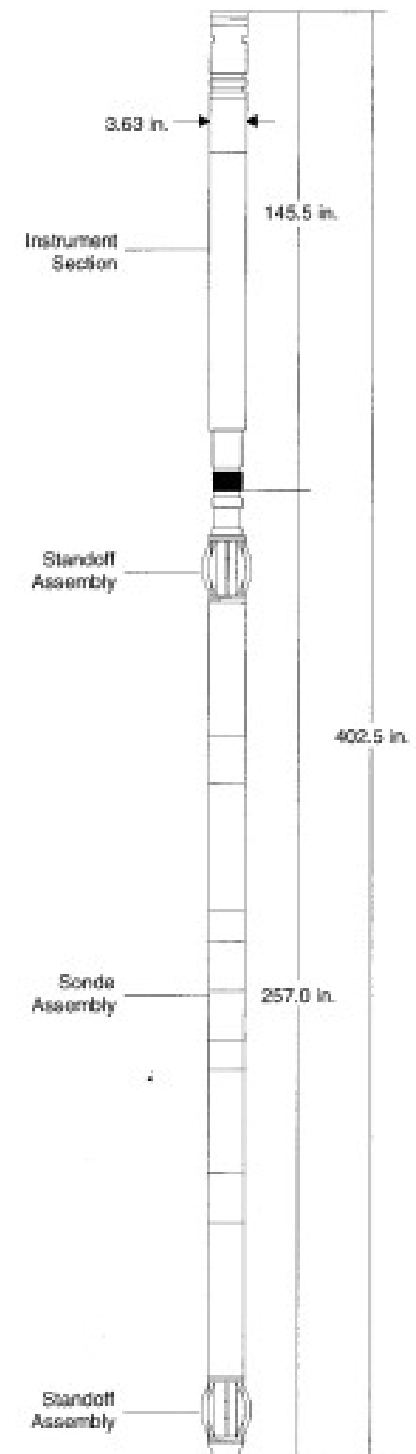
Induction

- **Physics of Measurement**

- Transmitter coils induce an alternating current in the formation. Receiver coils sense the response of the formation, both in magnitude and phase. This response is proportional to the formation conductivity (the inverse of resistivity). Multiple transmitter and coils are used in the effort to minimize borehole and invasion effects on the tool.
- Measurements of the in-phase and out-of-phase parts of the signal are digitally recorded and the tool operates at different frequencies in order to improve its accuracy. Accuracy is further enhanced by environmental corrections done in real time.
- Array tools have many receivers, usually at small spacings, and rely on signal processing to create a common vertical resolution for all measurements.

- **Operational Constraints**

- In Open hole, the tool is run centered
- Can be run in air, water or oil-based mud
- Logging speed: 30 to 100 feet/minute depending on tool type



Array Induction
Logging Tool

Induction

Fundamentals:

● Transmitter coils:

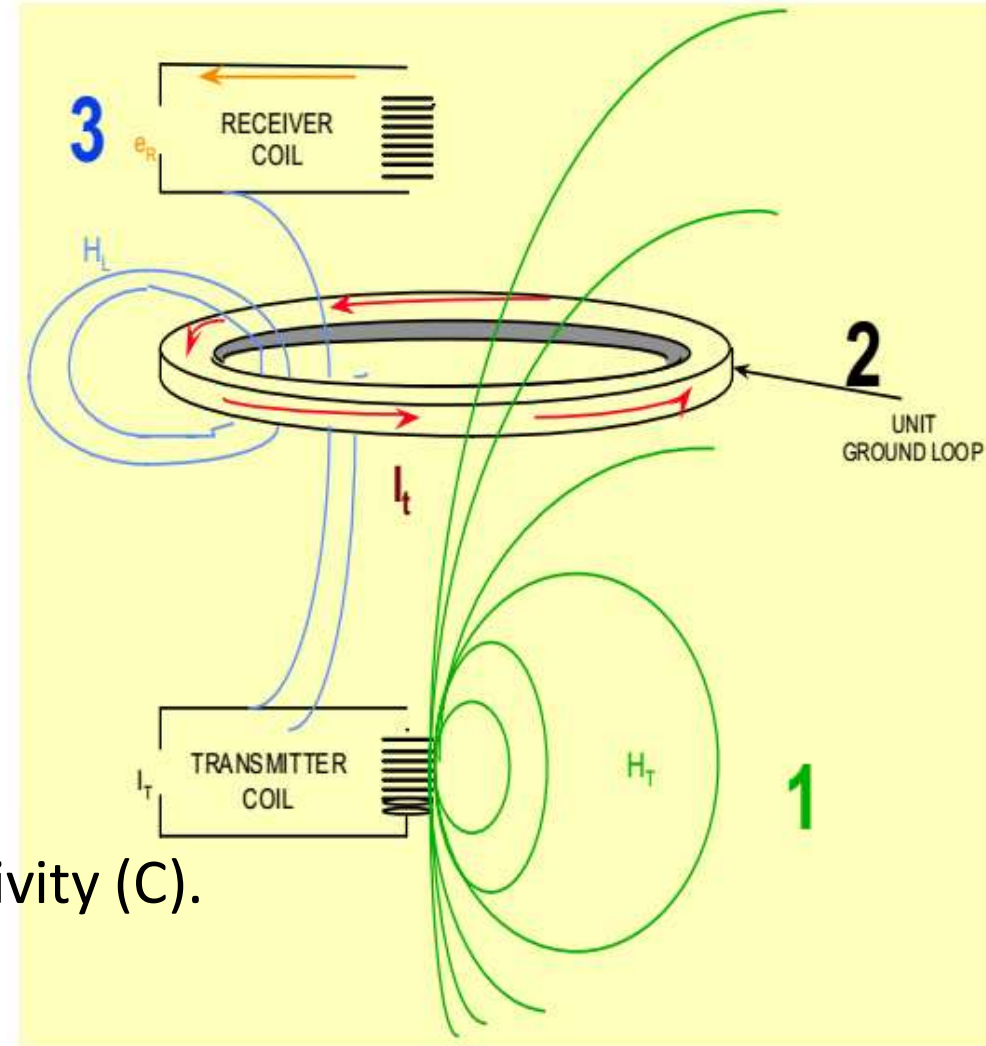
- Current (I_T).
- Creates magnetic field (H_T).

● Formation currents, I_t :

- Caused by magnetic field (H_T).
- Create magnetic field (H_L).

● Receiver coils:

- Senses magnetic field (H_L).
- Records output voltage (e_R).



Estimation of Resistivity Response

- Voltage (e_R) proportional to conductivity (C).
- Resistivity computed as:
- induction tools are focused.
- Depth of measurement depends on:
 - Frequency: Multiple frequencies
 - Number and position of coils

$$R(\text{ohm} - \text{m}) = \frac{1000}{C (\text{mmho/m})}$$

Induction Log – Quality Control

- All three resistivity curves should stack (have the same value) in impermeable beds once they are corrected for borehole and bed thickness effects.
- Curves should stack at about 2000 ohm-m in impermeable, non-conductive beds (e.g. anhydrite) greater than 20 feet thick.
- The proper invasion profile in water-bearing permeable beds is:
 - ✓ If $R_{mf} < R_w$ then Shallow < Medium < Deep;
 - ✓ If $R_{mf} > R_w$ then Shallow > Medium > Deep;
- Shale values should be similar to those of nearby wells.
- Induction curves should be free of spikes; especially the high resistivity range.
- Check repeatability: curves should have the same values and character as those previous runs or repeat sections.
- Cross-check the curve character with other curves from the same logging run.

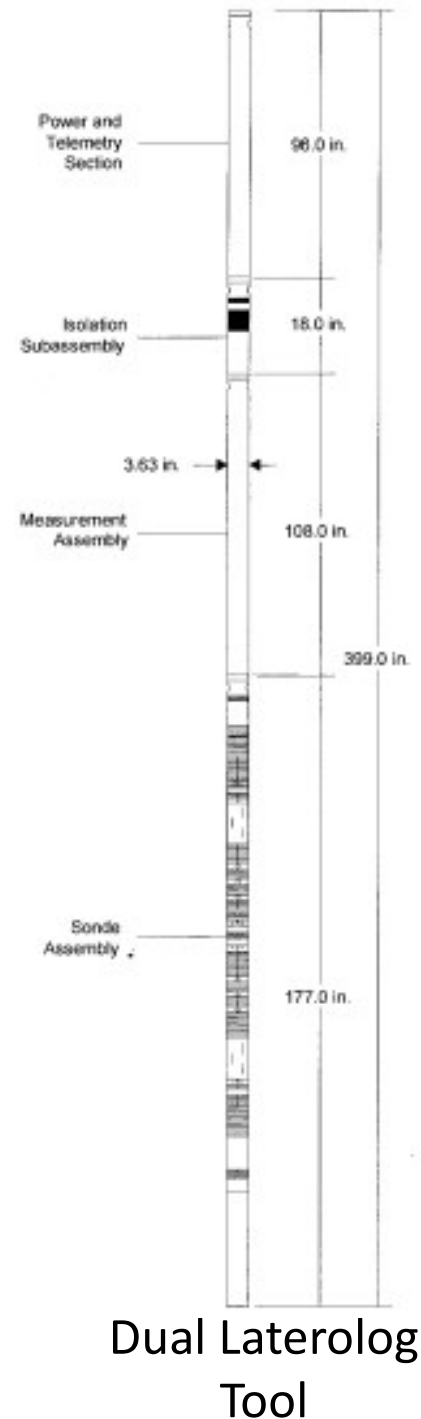
LATEROLOG

Laterolog - Interpretation Goals

- **True (undisturbed) formation resistivity, R_t :**
 - True formation resistivity can be determined from the Deep Laterolog curve by applying corrections for borehole size and bed thickness. The Shallow Laterolog and R_{xo} curves are needed for invasion correction along with the proper algorithms and/or charts. In most cases, the Deep Laterolog curve is sufficiently close to R_t thus not requiring correction.
- **Geopressure (overpressure) Detection:** Same as for Induction Log.
- **Correlation:** Same as for Induction Log.
- **Fluid Saturation, S_w , via Archie's:** Same as for Induction Log.
- **Diameter of Invasion:**
 - is determined as a by-product of the correction of the Deep Induction curve using a graphical “tornado chart” (or mathematical equivalent). Using the chart, ratios of the Deep/Shallow and Deep/ R_{xo} resistivity measurements are entered on the x and y axis, and values of invasion diameter (d_i), R_t/R_{xo} , and R_t/R_{deep} are read from the three families of curves which comprise the chart. R_t is determined from the R_t/R_{deep} ratio and R_{xo} from R_t/R_{xo} ratio
 - Invasion can be determined qualitatively by the separation of the three resistivity curves especially if R_w and R_{mf} are significantly different.
Deep Laterolog ~ Shallow Laterolog < R_{xo} resistivity indicates shallow invasion;

Laterolog - Physics of Measurement

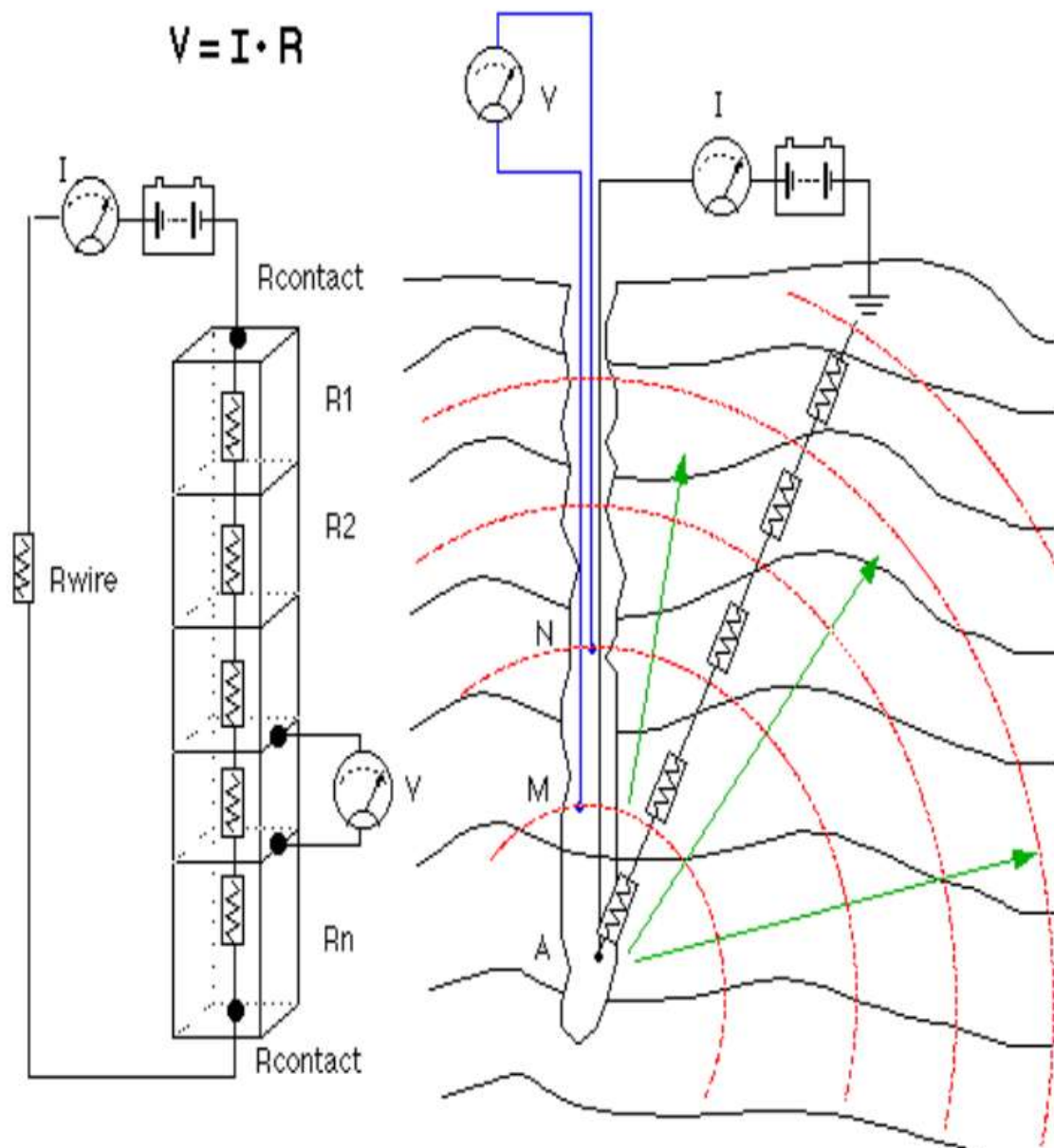
- A very low frequency current flows from the tool through the borehole, into the formation. Electrode arrays on either side of the source electrode force the measurement current into a horizontal disk-shaped pattern around the borehole. Formation resistivity is determined by monitoring the amount of current flowing from the tool.
- The tool must make contact with the formation.
- **Operational Constraints**
 - In Open hole, the tool is run centered
 - Run only in water-based mud
 - Logging speed: 100 feet/minute; slower if an Rxo tool is run in combination



Laterolog – Principles of Electrode Tools

Elements/Features:

- Electrode emits current, I :
 - Electrode A .
 - Green lines give current flow.
- Electrodes sense voltage, V :
 - Electrodes M and N .
 - Red lines show equipotentials.
- Formation resistivity, R :
 - $R = kV/I$.
 - k is tool constant.
- Simple electrode model ignores:
 - Current does flow up borehole.
 - Radial changes in R (invasion).



Dual Laterolog Tool

Deep and Shallow Current Patterns

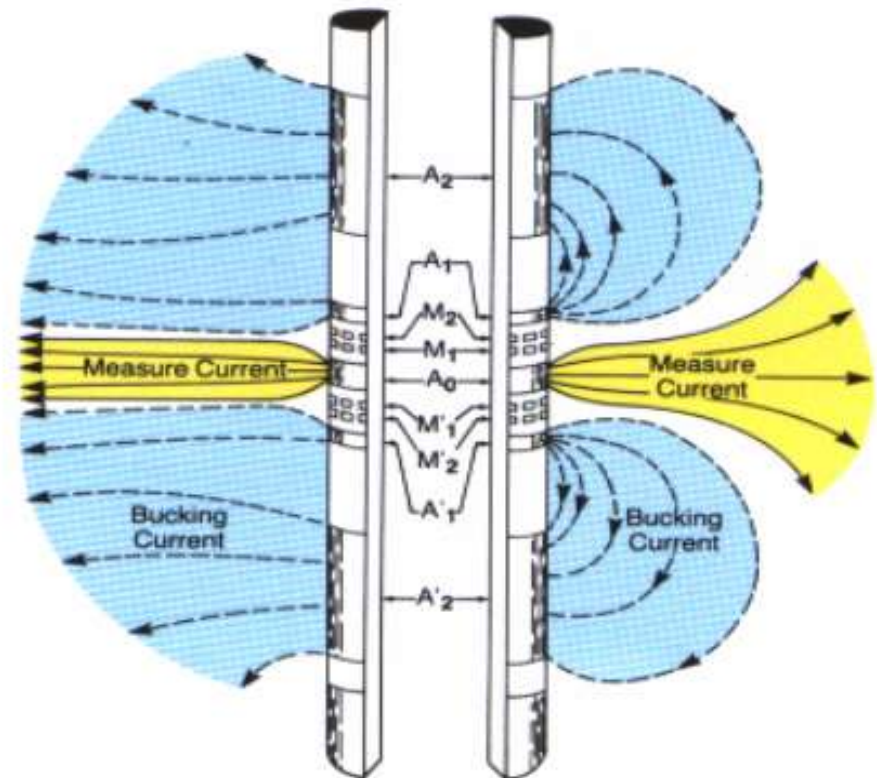
Characteristics:

- Multiple currents:

- Measure.
- Bucking (or guard).

- Objectives:

- R_t : Laterolog Deep (LLD).
- R_{xo} and R_t : Laterolog Shallow (LLS).
- Eliminate borehole effect.
- Eliminate shoulder effect.



LLD

A₀ Measure Current Electrode
M₁ } Monitoring Electrode
M₂ }
A₁ Bucking Current Electrode
A₂ Bucking Current Electrode

LLS

A₀ Measure Current Electrode
M₁ } Monitoring Electrode
M₂ }
A₁ Bucking Current Emitting Electrode
A₂ Bucking Current Return Electrode

LateroLog – Quality Control

- All three resistivity curves should stack (have the same value) in impermeable beds once they are corrected for borehole and bed thickness effects.
- The proper invasion profile in water-bearing permeable beds is:
 - ✓ If $R_{mf} < R_w$ then Shallow < Medium < Deep;
 - ✓ If $R_{mf} > R_w$ then Shallow > Medium > Deep;
- Shale values should be similar to those of nearby wells.
- Check repeatability: curves should have the same values and character as those previous runs or repeat sections.
- Cross-check the curve character with other curves from the same logging run.

MICRORESISTIVITY, R_{x0}

Resistivity logs in the invaded zone

- For investigation of the invaded zone (and determination of R_{xo}) another tool type is applied – the family of micrologs.
- The electrode system is fixed on a pad and pressed against the borehole wall. The electrodes have a very short distance (microlog 2.5cm) and are therefore characterized by:
 - Extreme vertical resolution, able to detect thin layers, but also fractures,
 - Extremely small depth of investigation (R_{mc} , R_{xo})
- There are unfocused systems (classic Microlog ML) as well as focused systems (e.g. Microlaterolog MLL, MicroSphericalFocusing log MSFL).

Microresistivity - Interpretation Goals

- **Flushed zone formation resistivity, R_{xo} :**
 - Can be determined with minimal environmental corrections, from any of the microresistivity measurements, except for the MicroLog. This value is used in Archie's equation to determine flushed zone water saturation S_{xo} , or to indicate moved fluids by comparison to the undisturbed formation resistivity, R_t
- **Flushed zone water Saturation, S_{xo} , via Archie's Equation:**
 - In the flushed zone form of the Archie's equation R_t is replaced by R_{xo} , and R_o by R_{mf} , with the assumption that the original water has been replaced by drilling filtrate. Comparison of S_{xo} and S_w (using the same form of the Archie's equation) gives some qualitative indication of permeability, and the amount of hydrocarbons which will be moved during production.
- **Indication of Permeability:**
 - For MicroLogs: The micronormal resistivity is greater than the microinverse resistivity ("positive separation"). There should be mudcake, as shown in a decrease in the caliper reading.
 - For Other R_{xo} tools: Compare the reading with the resistivity from deeper reading tools. The relationship between the readings will depend on the contrast between the formation water resistivity and mud filtrate resistivity.

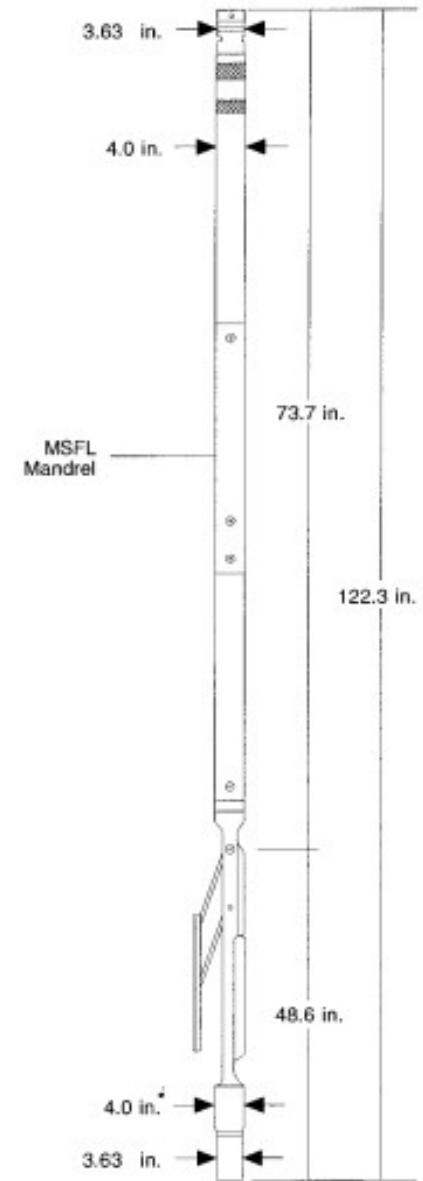
$$S_{xo} = \left(\frac{a \cdot R_{mf}}{\phi^m \cdot R_{xo}} \right)^{\frac{1}{n}}$$

Microresistivity - Interpretation Goals

- **Thin Bed definition:** Same as for Induction Log.
 - These tools will identify very thin beds. The bed definition can be used qualitatively to estimate the effect on deeper reading tools. Also useful for thin bed corrections to other resistivity and porosity tools
- **Fracture Identification:**
 - Rapid curve movement or “hashiness”, may be an indication of fractures as the tools see conductive mud-filled fractures alternating with less conducting beds. Rough hole may also cause the same response. Other information is required to establish presence of fractures.
- **Invasion corrections to other resistivity measurements:**
 - Rxo tools are usually run in combination with two deeper reading tools (e.g. deep and shallow laterolog, deep and medium induction log). Using the combination of three measurements, corrections can be made for invasion using “tornado chart” or equivalent algorithms.

Microresistivity - Physics of Measurement

- Electrical current is focused into the formation by closely placed electrodes mounted on pads pressed against the borehole wall.
- Some designs like Micro Spherically Focused log, use focusing electrodes similar to the laterolog, while older designs like MicroLog, do not focus the current.
- **Operational Constraints**
 - In Open hole, the tool is run centered
 - Run only in water-based mud
 - Logging speed: 60 feet/minute;
- **Operational Constraints**
 - Mudcake corrections may need to be made if the measurements are to be used quantitatively.

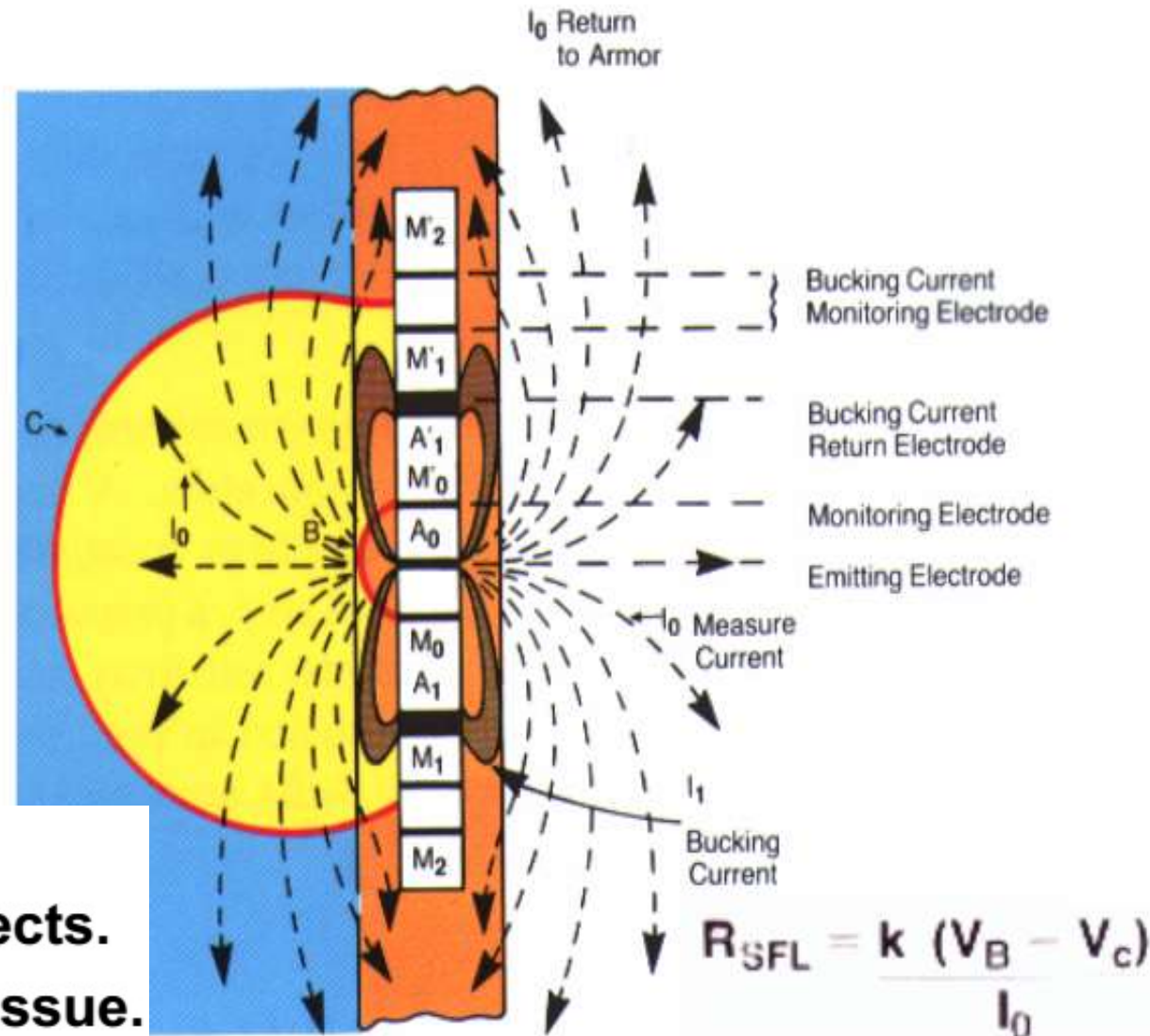


Micro Spherically
Focused log

Spherically Focused Resistivity Tool

Tool Characteristics:

- SFL tool gives *shallow resistivity*.
- Usually run in combination with induction logs.
- Good tool for thin-bed detection/analysis.
- Has a different order of electrodes than the laterolog tool.
- Pad-type tools:
 - Pads reduce borehole effects.
 - Mud cake may still be an issue.
- Very shallow resistivity:
 - 2 to 5 cm profile is typical.



11.

Principles of Log Interpretation – Lecture 6

Log Interpretation – Required Data

Once the section TD (Total Depth) of the hole has been reached, the petrophysicist will be expected to make an interpretation of the open hole logs that have been acquired. Prior to the log interpretation, the petrophysicist should have:

- 1) All relevant daily drilling reports, plus latest well deviation from the well, last casing depth, & mud data (R_m , R_{mf} @ Formation Temp.)
- 2) All the latest mud-log information, including cuttings description, shows, gas reading, ROP (Rate of Penetration) and Core Data
- 3) Logs and interpretations on hand from nearby wells and regional wells penetrating the same formations, in particular where regional or field-wide values of m , n , R_w , ρ_g and fluid contacts are available.
- 4) A copy of the contractor's chart book

This will allow for compilation of all information about lithology, particularly expected rock type and mineral composition, presence and type of clay/shale, pore types etc

Log Interpretation – An Introduction

- The general goal of interpretation is the determination of:
 - Lithological profile and the detection and characterization of the potential reservoir zones with exact depth,
 - Shale content and shale type (e.g. laminated, or dispersed shale)
 - Porosity (and mineral composition),
 - Fluid saturation, water saturation, hydrocarbon saturation, moveable hydrocarbons, type of hydrocarbon (oil, gas)
 - Permeability
- Interpretation delivers a solution in two steps:
 - 1) Quick look interpretation for qualitative description of lithology and zones of interest
 - 2) Detailed quantitative analysis in terms of the goals above.

Log Interpretation – Prior Steps

- Log interpretation requires the synthesis of logging tool response physics, geological knowledge, and auxiliary measurements or information to extract the maximum petrophysical information concerning subsurface formations.
- Knowledge of characteristics log responses, empirical and model derived equations for quantitative interpretation, integration of all available information and the skill of the interpreter is crucial for effective log evaluation.
- The first step in log evaluation is the quality control of the logs followed by correlation of these logs with the mud log and logs of surrounding wells to determine significant events, such as markers, hydrocarbon contacts etc

Log Interpretation – Basic Quality Control

- 1) Check that the logger's TD and last casing shoe depths roughly match those from the last daily drilling report.
- 2) Check that the derrick floor elevation and ground level (or seabed) positions are correct.
- 3) Check that the log curves are on depth with each other. The tension curve can be used to identify possible zones where the tool string has become temporarily stuck, which will put the curves off depth and result in "flatlining."
- 4) Check that the caliper is reading correctly inside the casing (find out the casing ID) and that it is reading the borehole size in non-permeable zones that are not washed out.
- 5) Check the density borehole correction curve. It should not generally exceed 0.02g/cc, except in clearly washed out sections (>18in.), for which the density curve is likely to be unusable.
- 6) Check the sonic log by observing the transit time in the casing, which should read 47 μ s/ft.

Log Interpretation – Basic Quality Control

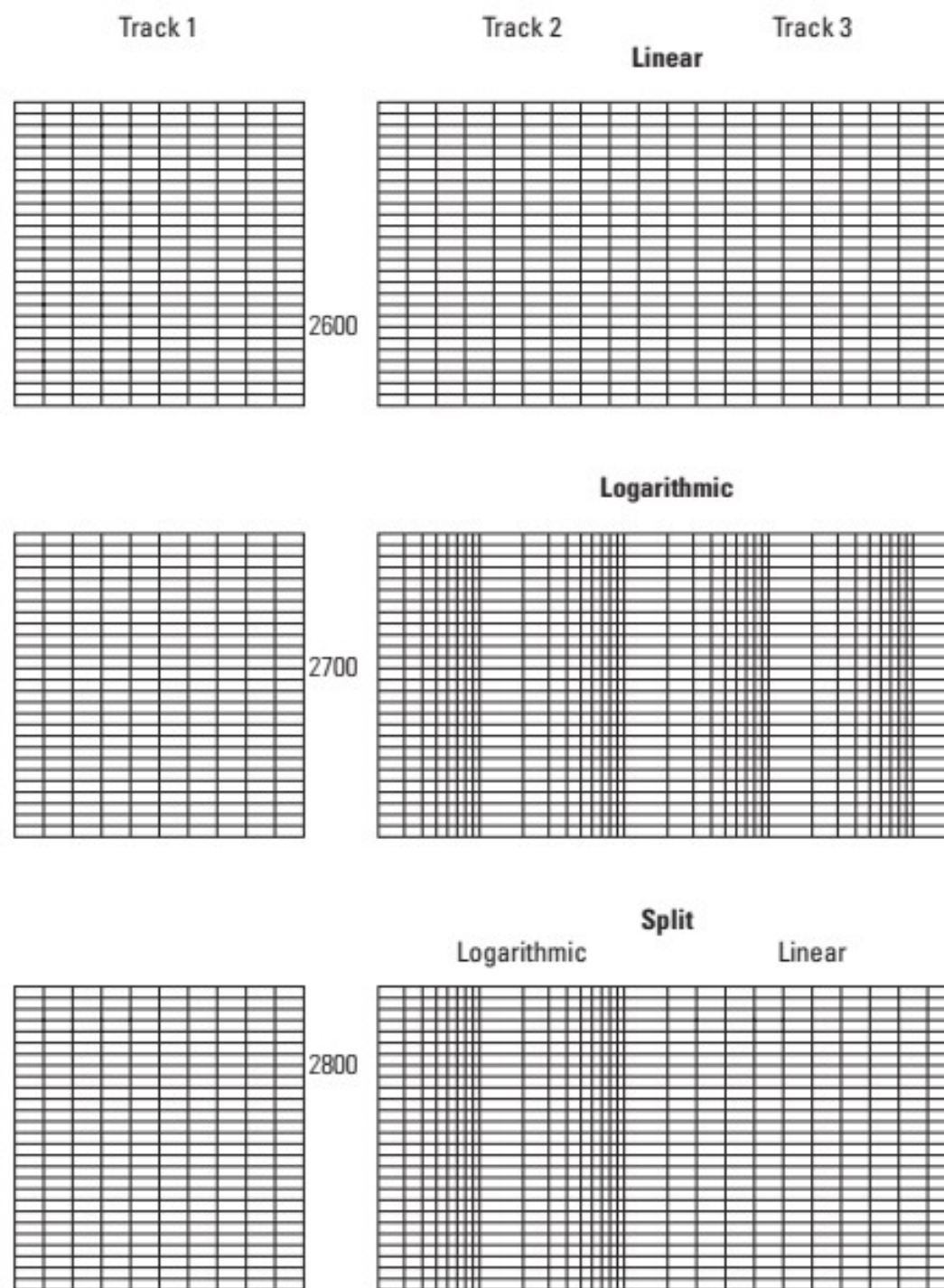
- 7) Inspect the resistivity curves. If oil-based mud (OBM) is being used, the shallow curves will usually read higher than the deep curves (except in highly gas or oil saturated zones). Likewise, with water-based mud (WBM) the shallow curves will read less than the deep curves, providing $R_{mf} < R_w$, or in hydrocarbon-bearing zones. In theory, the curves should overlie each other in non-permeable zones such as shales. However, in practice this is often not the case, due to either anisotropy or shoulder-bed effects.
- 8) Look out for any cycling-type behaviour on any of the curves, such as a wave pattern. This may be due to corkscrewing while drilling, causing an irregular borehole shape. However, it is necessary to eliminate any possible tool malfunction.
- 9) Check that the presentation scales on the log print are consistent with other wells or generally accepted industry norms. These are generally:
 - ☐ GR: 0–200 API
 - ☐ Caliper: 8–18"
 - ☐ Resistivity: 0.2–2000ohmm on log scale
 - ☐ Density: 1.95–2.95 g/cc (solid line)
 - ☐ Neutron: -0.15 ± 0.45 (porosity fraction) (dashed line)
 - ☐ Sonic: 140–40 $\mu\text{s}/\text{ft}$

Information from the Log Header

- Measured log data are give mostly as a data file in a special format - Log ASCII Standard (LAS-file).
 - Hole location and name
 - Depth – Driller, Depth – Logger, Casing Depth, Bit size
 - Logged interval
 - Fluid type in the hole, Density/Viscosity, pH/Fluid Loss
 - Fluid Resistivities: R_m (mud resistivity) at measured temperature, R_{mf} (mud cake resistivity) at measured temperature, R_{mc} (mud cake resistivity) at measured temperature, Source of R_{mf} , R_{mc}
 - Temperature: Bottom hole temperature, maximum recorded temperature
 - Time since circulation

Reading a Log

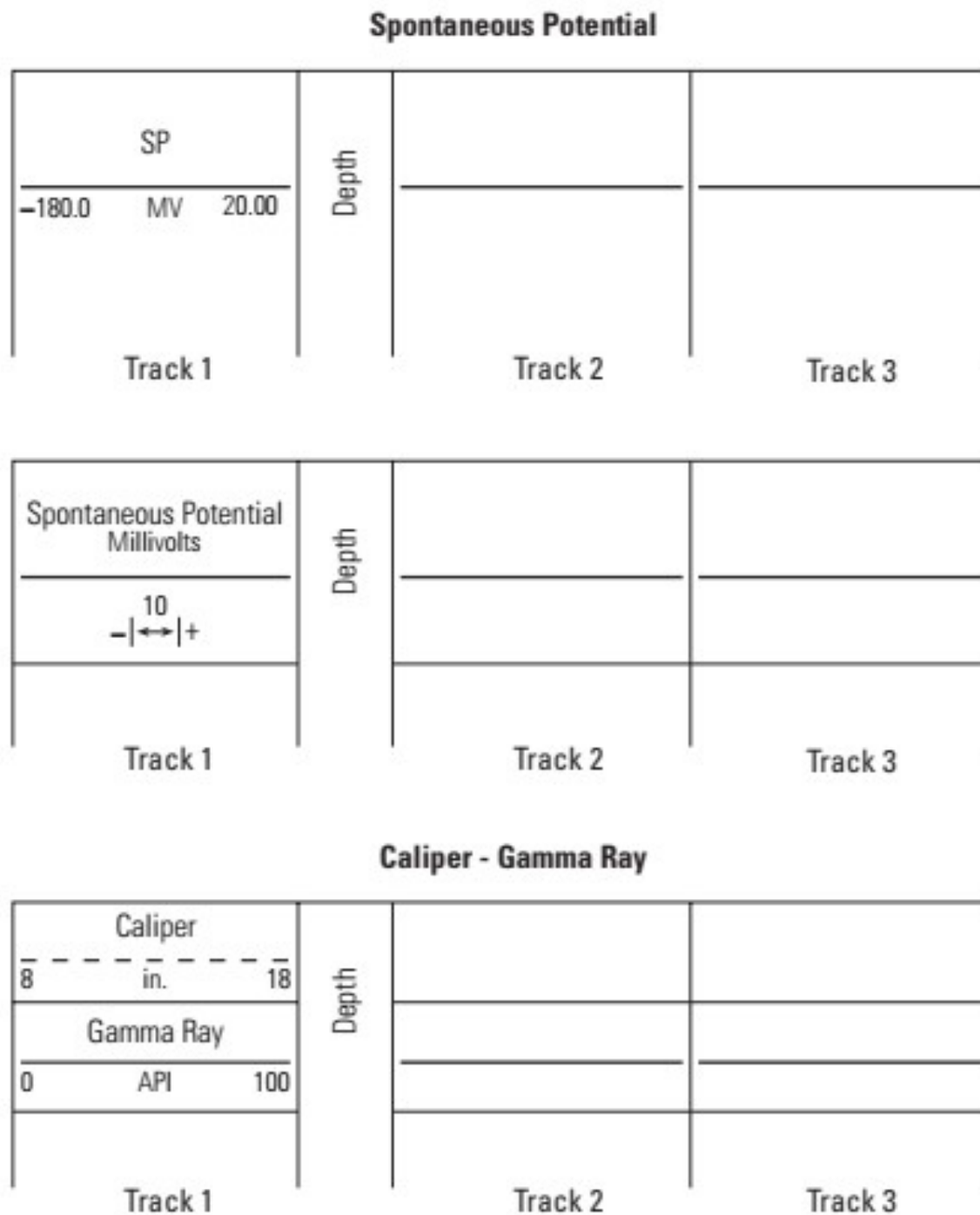
- Reading a log with ease requires familiarity with some of the standard log formats. The formats for traditional logs and most field logs are shown in Fig. LI.1 and can be seen to contain three tracks.
- A narrow column containing the depth is found between track 1 and tracks 2 and 3. The latter are contiguous.
- The top illustration shows the normal linear presentation, with the grid lines in all three tracks having linear scales each with ten divisions. The middle figure shows the logarithmic presentation for tracks 2 and 3. Four decades are drawn to accommodate the electrical measurements, which can have large dynamic ranges.
- The bottom illustration is a hybrid scale with a logarithmic grid on track 2 and a linear one in track 3. Electrical measurements that may spill over from track 2 into track 3 will still be logarithmic even though the indicated scale is linear.
- The depth is shown by the numbers in the centre track and the horizontal lines. In Fig. LI.1 the depth scale is 1/240, or 1 ft of log for 240 ft of formation. The logs have a thin horizontal line every 2 ft, a medium thick line every 10 ft, and a thick line every 50 ft.



- **Fig. LI.1** Standard log presentation formats.

Reading a Log

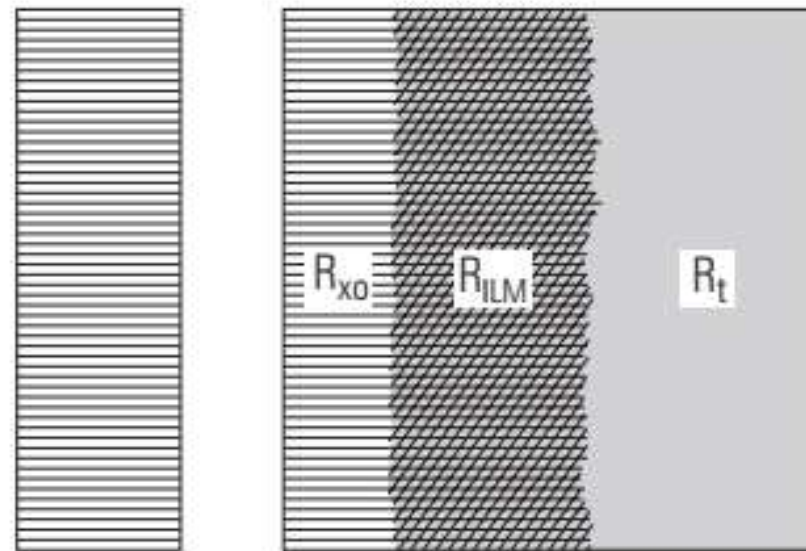
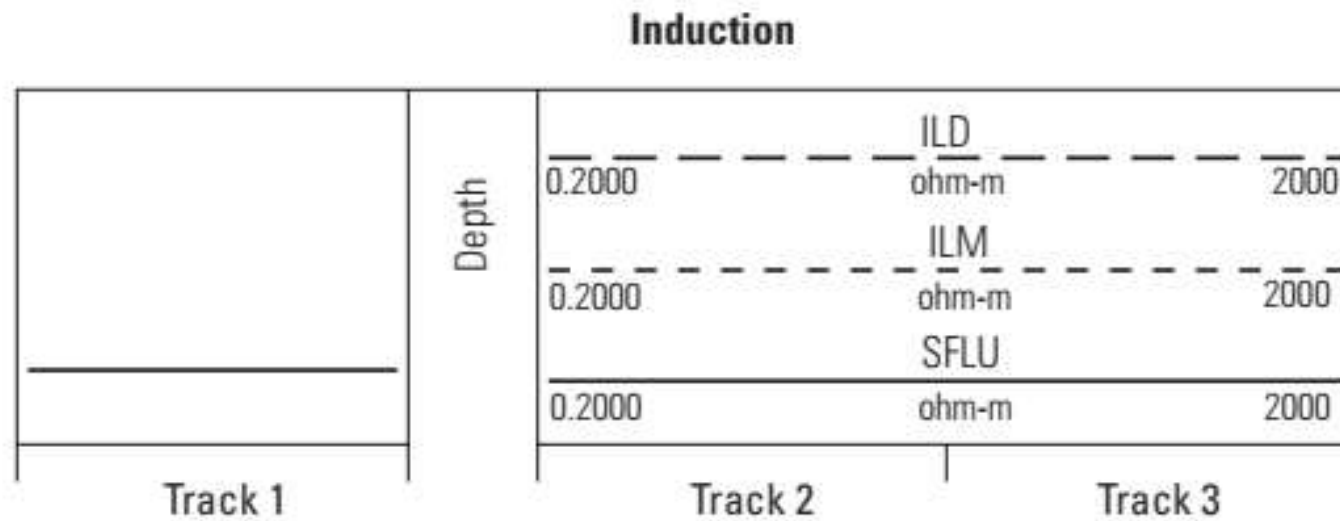
- Figure LI.2 shows the typical log-heading presentation for several of the basic logs that will be used. The upper two presentations show two variations for the SP, which is always presented in track 1.
- The bottom presentation shows the caliper, a one-axis measurement of the borehole diameter, and the gamma ray, which are also generally presented in track 1. Note that the SP decreases to the left. The rule given for finding clean sections was that the SP becomes less negative for increasing shale, so that deflections of the SP trace to the right will correspond to increasing shale content.
- The GR curve, as it is scaled in increasing activity (in American Petroleum Institute (API) units) to the right, will also produce curve deflections to the right for increasing shale content. Thus the two shale indicators can be expected to follow one another as the shale content varies.



- **Fig. LI.2** Standard log presentation formats.

Reading a Log

- Although modern resistivity tools have a larger selection of curves with different depths of investigation, the displays are similar.
- A traditional resistivity log heading is shown in Fig. LI.3, along with a schematic indication of the zones of investigation. The particular tool associated with this format is referred to as the dual induction-SFL and will normally show three resistivity traces (the units of which are ohm-m). The trace coded ILD (Induction Log Deep) corresponds to the deepest resistivity measurement and will correspond to the value of R_t when invasion is not severe. The curve marked ILM (Induction Log Medium) is an auxiliary measurement of intermediate depth of penetration and is highly influenced by the depth of invasion.
- The third curve, in this case marked SFLU (spherically focused log), is a measurement of shallow depth of investigation and reads closest to the resistivity of the invaded zone R_{xo} . By combining the three resistivity measurements, it is possible, in many cases, to compensate for the effect of invasion on the ILD reading.



- **Fig. LI.3** The induction log heading and schematic of the formation, with three zones corresponding approximately to the simultaneous electrical measurements of different depths of investigation.

Reading a Log

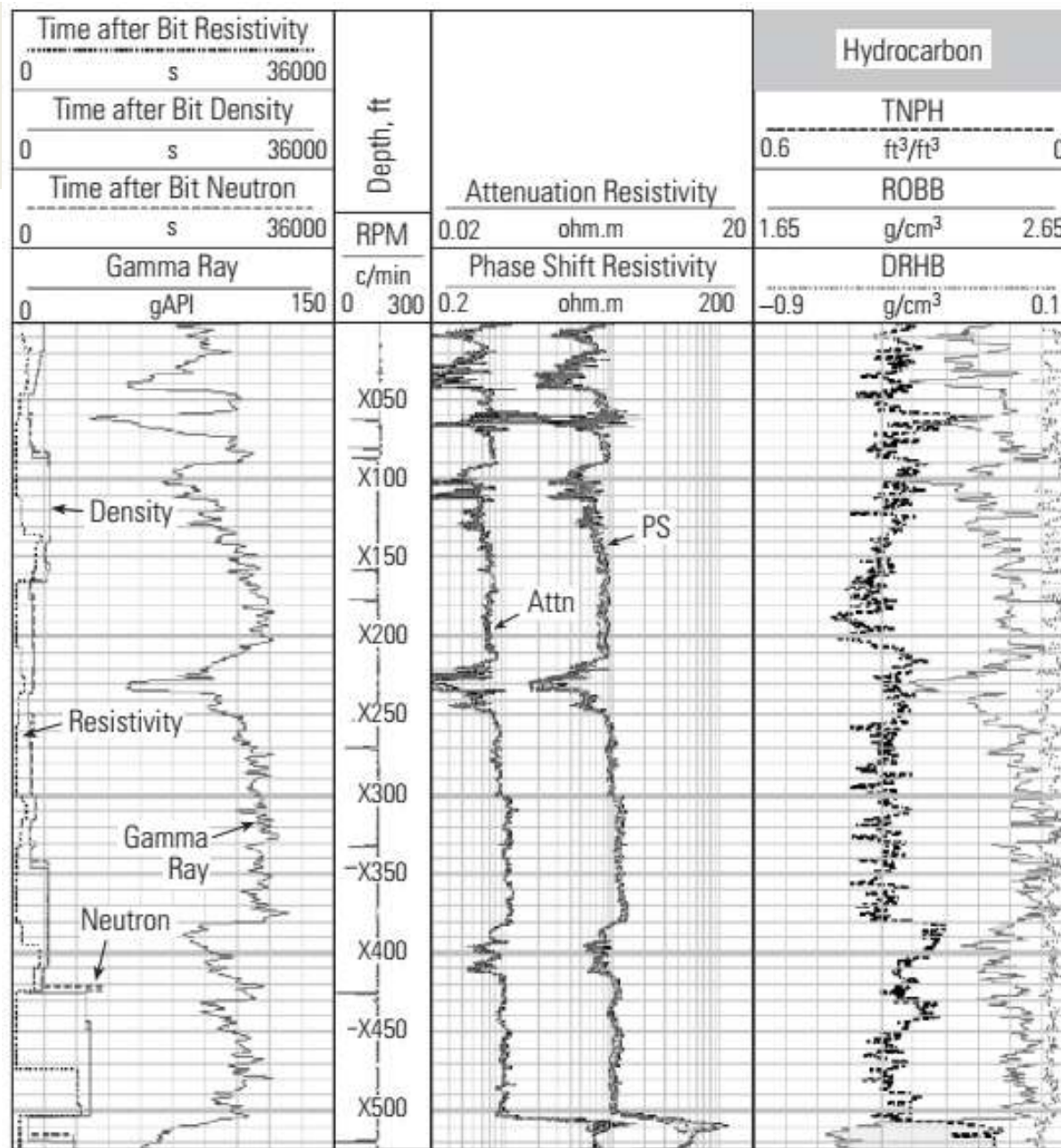
- In Fig. LI.4, three typical headings for the three types of porosity devices are indicated. The porosity is expressed as a decimal (v/v) or in porosity units (p.u.), each of which corresponds to 1% porosity. The top heading shows the format for porosities derived from neutron and density measurements simultaneously.
- In this example, although the scale may vary depending upon local usage, porosity is shown from -0.15 to 0.45 v/v. The middle example shows an additional correction curve for the density log, which can be used to get some idea of the mudcake and rugosity of the borehole encountered during the density measurement.
- It is also common practice to present the density measurement in g/cm^3 with a dynamic range of 1 g/cm^3 on the density trace across the full track (or sometimes two tracks), as is done in Fig. LI.5 below. It is easy to show that a change in bulk density of 1 g/cm^3 in a water-filled formation corresponds to a porosity change of about 60 p.u. Consequently the neutron trace is usually shown with a dynamic range of 60 p.u. across the density track.

Neutron - Density			
Track 1	Depth	NPHI (SS)	
		0.4500	-0.150
		DPHI (SS)	
		0.4500	-0.150
		Track 2	Track 3

Track 1	Depth	Porosity Index %				SS	Matrix	
		Compensated Formation Density Porosity						
		60	45	30	15		0	
		Compensated Neutron Porosity						
		60	45	30	15		0	
		Correction				g/cm ³		
						-0.25	0	+0.25
		Track 2		Track 3				

Acoustic			
Track 1	Depth	Interval Transit Time	
		Microseconds per foot	
		150	ΔT 50
		Track 2	Track 3

- **Fig. LI.4** Log headings for three porosity devices. The top two correspond to two possible formats for simultaneous density and neutron logs. The bottom is the sonic log format.



• **Fig. LI.5** Example LWD log in a horizontal well. In track 1 is the familiar GR along with three curves indicating the time delay between drilling and the three types of measurements made; the tool rotation rate appears in the depth track. Track 2 - two types of resistivity measurements, each with multiple depths of investigation that overlay in this example. The third track contains the LWD versions of the neutron measurement (TNPH), the density measurement (ROBB), and the density correction (DRHB).

Reading a Log

- The density trace is then shifted so that the zero point on the neutron trace corresponds to the density of the matrix. Two such compatible scales are in common use: one for sandstone, in which case the zero porosity point is 2.65 g/cm^3 as in Fig. LI.5; the other for limestone, in which case the zero porosity point is 2.7 g/cm^3 . The bottom heading of Fig. LI.4 is for the traditional sonic log with the apparent transit time Δt increasing to the left. In all three presentations, the format is such that increasing porosity produces curve deflections to the left.
- For the neutron and density logs, another point to be aware of is the matrix setting. This setting corresponds to a rock type assumed in a convenient pre-interpretation that establishes the porosity from the neutron and density device measurements. In both examples shown in Fig. LI.4, the matrix setting is listed as SS, which means that the rock type is taken to be sandstone. If the formations being logged are indeed sandstone, then the porosity values recorded on the logs will correspond closely to the actual porosity of the formation. However, if the actual formation matrix is different, say limestone, then the porosity values will need to be shifted or corrected in order to obtain the true porosity in this particular matrix.

GUIDES TO QUICK LOOK WELLSITE LOG INTERPRETATION

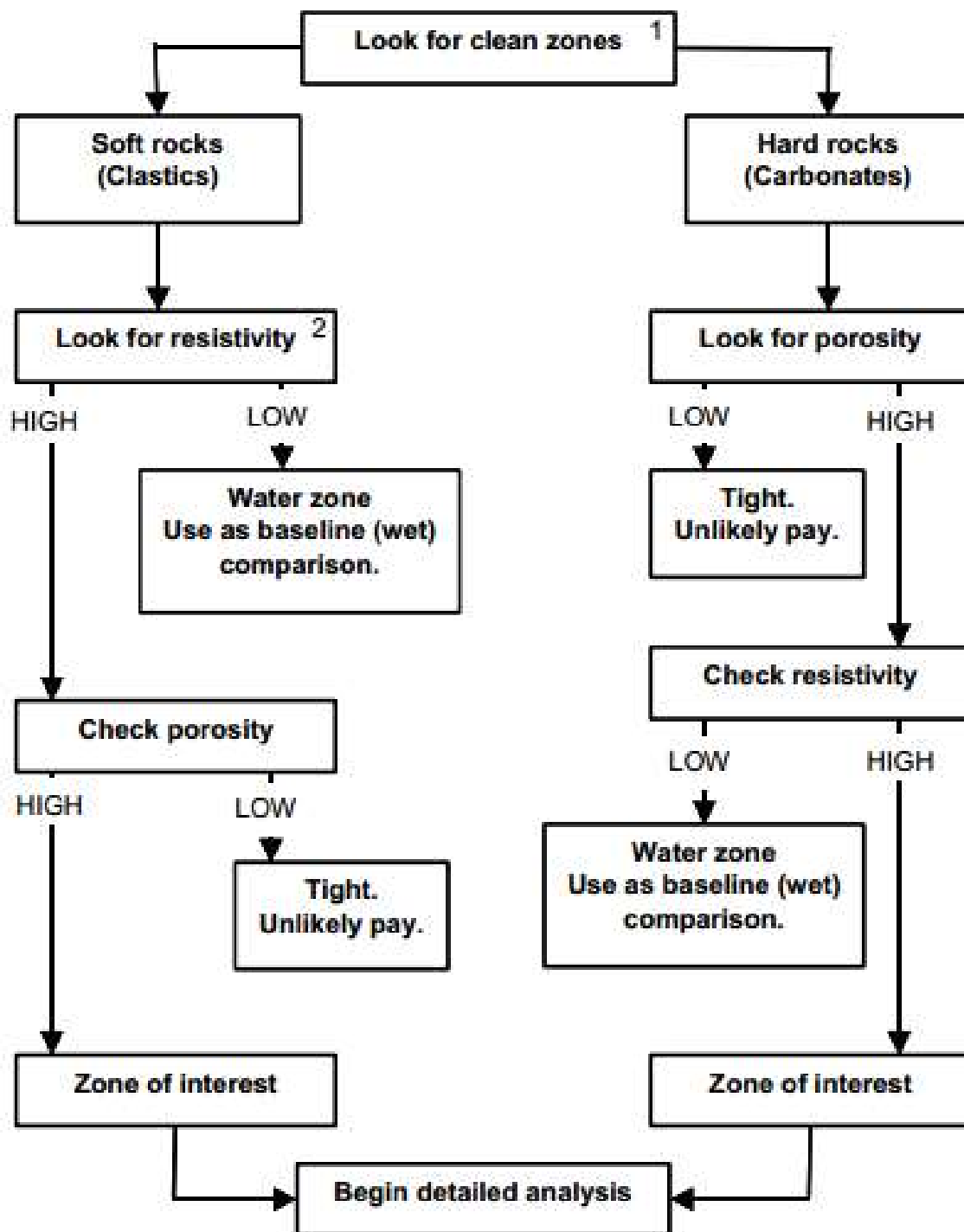
Log Interpretation – Wellsite Evaluation

- Prior to detailed log interpretation, wellsite interpretation is undertaken. This refers to the rapid and somewhat cursory approach to scanning an available set of logging measurements, and the ability to identify and draw some conclusion about zones of possible interest.
- These zones, probably hydrocarbon-bearing, will warrant a closer and more quantitative analysis, which is possible only by the inclusion of additional knowledge and measurements.
- The three most important questions to be answered by wellsite interpretation are:
 1. Can the porous, clean formation be identified?
 2. Does the formation contain hydrocarbons?
 3. What is the formation porosity? And Saturations?
- The three questions above will enable us determine the quantity of hydrocarbons present. N.B.: Question 3 would be addressed under Quantitative Interpretation.

Quick Look Log Interpretation

Asquith and Krygowsky (2004) recommended the following flowchart for quick evaluation to identify zones of interest:

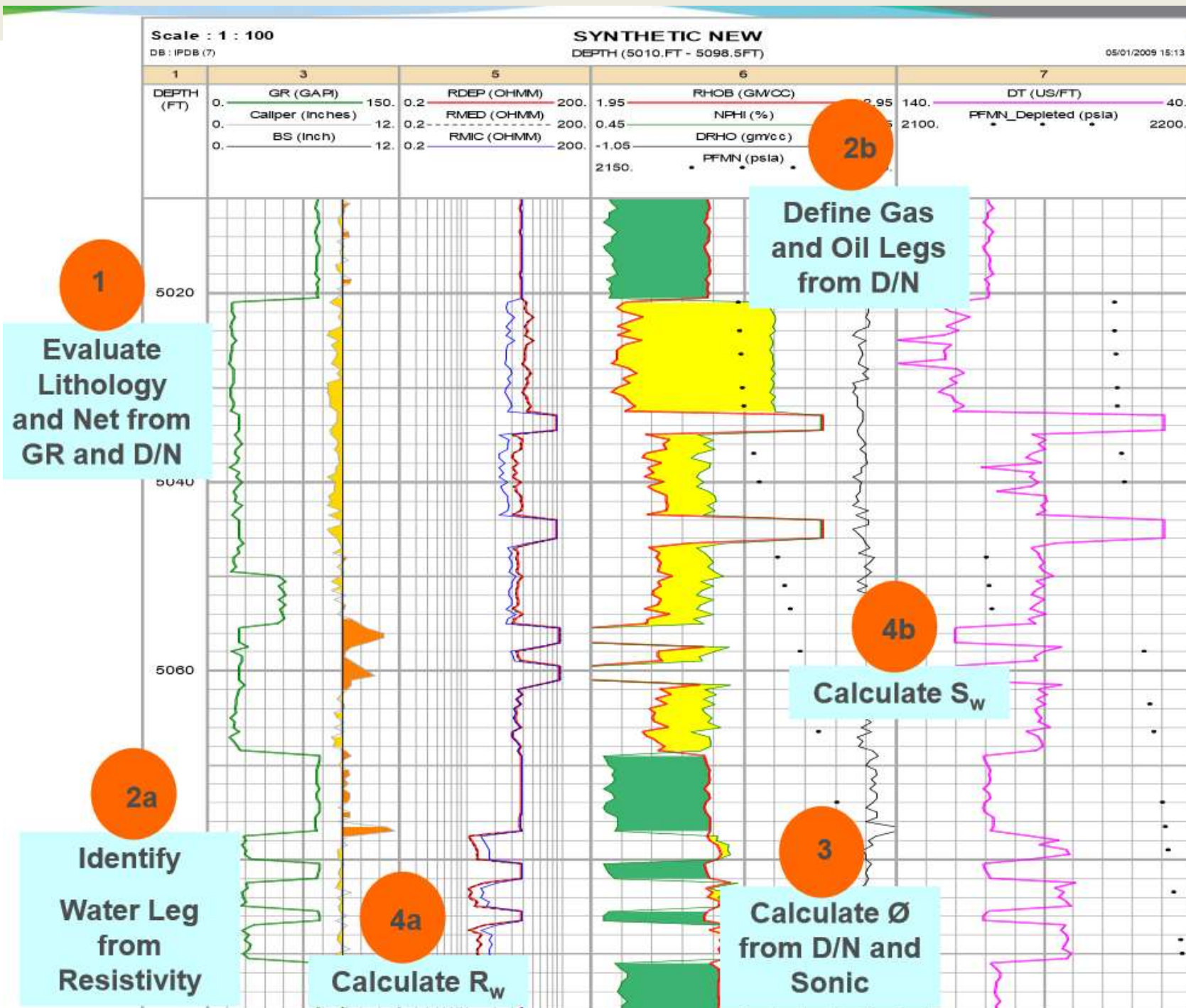
- 1) Separate the shaly sections, select “clean sections” then
- 2) If clastics are logged, look at the (deep) resistivity first – if it is high, hydrocarbons may be present; then look for porosity (only in the case of extremely low porosity the rock with the high resistivity is tight no reservoir)
- 3) If carbonates are logged, look at the porosity first – if it is sufficient, look at the resistivity – if it is high hydrocarbons may be present.



Interpretation

- Clastics: Because the Shales surrounding the clastic reservoirs tend to have slowly varying resistivity with depth, the interpretation process targets formation resistivity to identify zones of interest.
- Carbonates: Because the resistivity of a sequence of carbonate formations may vary over a wide range, the interpretation process targets formation porosity to identify zones of interest.
- N.B.: Zones which appear to be shales may be radioactive productive zones¹. Shaly zones may produce even when having low resistivity².

Recommended Quick look Interpretation Sequence



- 1) Define Reservoir / non-reservoir
- 2) Identify fluid type & contacts
- 3) Calculate Porosity
- 4) Calculate Water Saturation
- 5) Recognize minerals

Wellsite Evaluation – Identifying Clean Zones

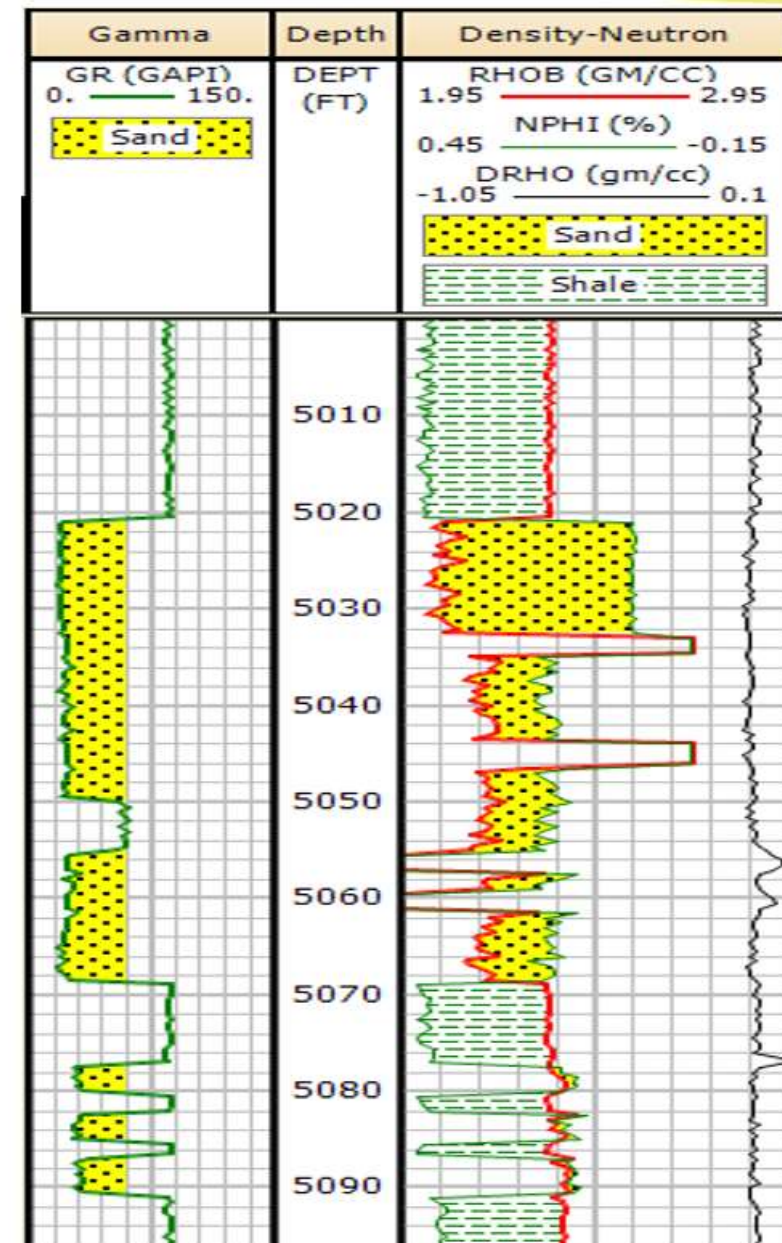
- Formations with low shale content are much more likely to produce accumulated hydrocarbons. Thus the first task is to identify the zones with a low-volume fraction of shale (V_{shale}), also known as clean zones.
- This task has traditionally been accomplished via two measurements: the gamma ray, and the spontaneous potential (SP). The qualitative behaviour of the SP (a voltage measurement reported in mV) is to become less negative with increases in formation shale content. The gamma ray signal will generally increase in magnitude according to the increase in shale content.
- In clastic reservoirs, in nearly all cases, indication of reservoir rock will correspond to a fall in the gamma ray (GR) log (except for the presence in sands of radioactive minerals).
- Other techniques have been used recently, for example the separation between the neutron and density measurements, the nuclear magnetic resonance (NMR) distribution, and elemental spectroscopy analysis.

Wellsite Evaluation – Identifying Clean Zones

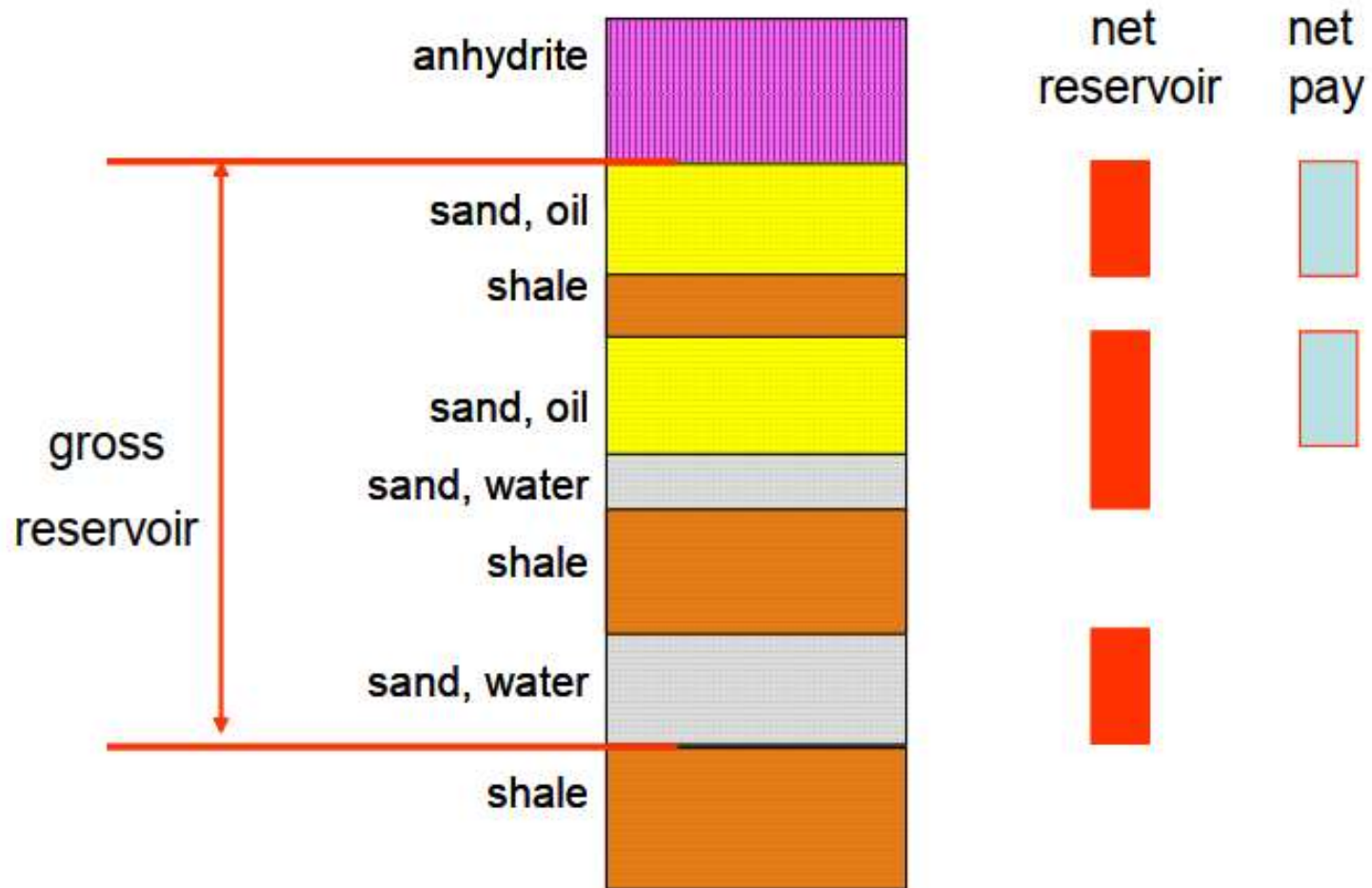
- The most reliable indicator of reservoir rock will be from the behaviour of the density/neutron logs, with the density moving to the left (lower density) and touching or crossing the neutron curve (Darling, 2003).
- Shales can be clearly identified as zones where the density lies to the right of the neutron, typically by 6 or more neutron porosity units.
- The greater the crossover between the density and neutron logs, the better the quality of the reservoir. However, gas zones will exhibit a greater crossover for a given porosity than oil or water zones.
- The following approach is recommended (Darling, 2003).
 - 1) Determine an average GR reading in clean sands (GR_{sa}) and a value for shales (GR_{sh}). For GR_{sh} , do not take the highest reading observed, but rather the mode of the values observed.
 - 2) Define the shale volume, V_{sh} , as $(GR - GR_{sa}) / (GR_{sh} - GR_{sa})$. By comparing V_{sh} with the density/neutron response, determine a value of V_{sh} to use as a cut-off. Typically 50% is used.

1) Define reservoir / non-reservoir

- Using the GR
 - Choose a cut-off to differentiate Sand/Shale.
 - Draw a line in the GR track at this value.
 - Shade between the line and the GR to delineate potential Net Sand.
- Using the Neutron-Density
 - If Density left of Neutron shade “sand”.
 - If Neutron left of Density shade “shale”



Net Pay Definition

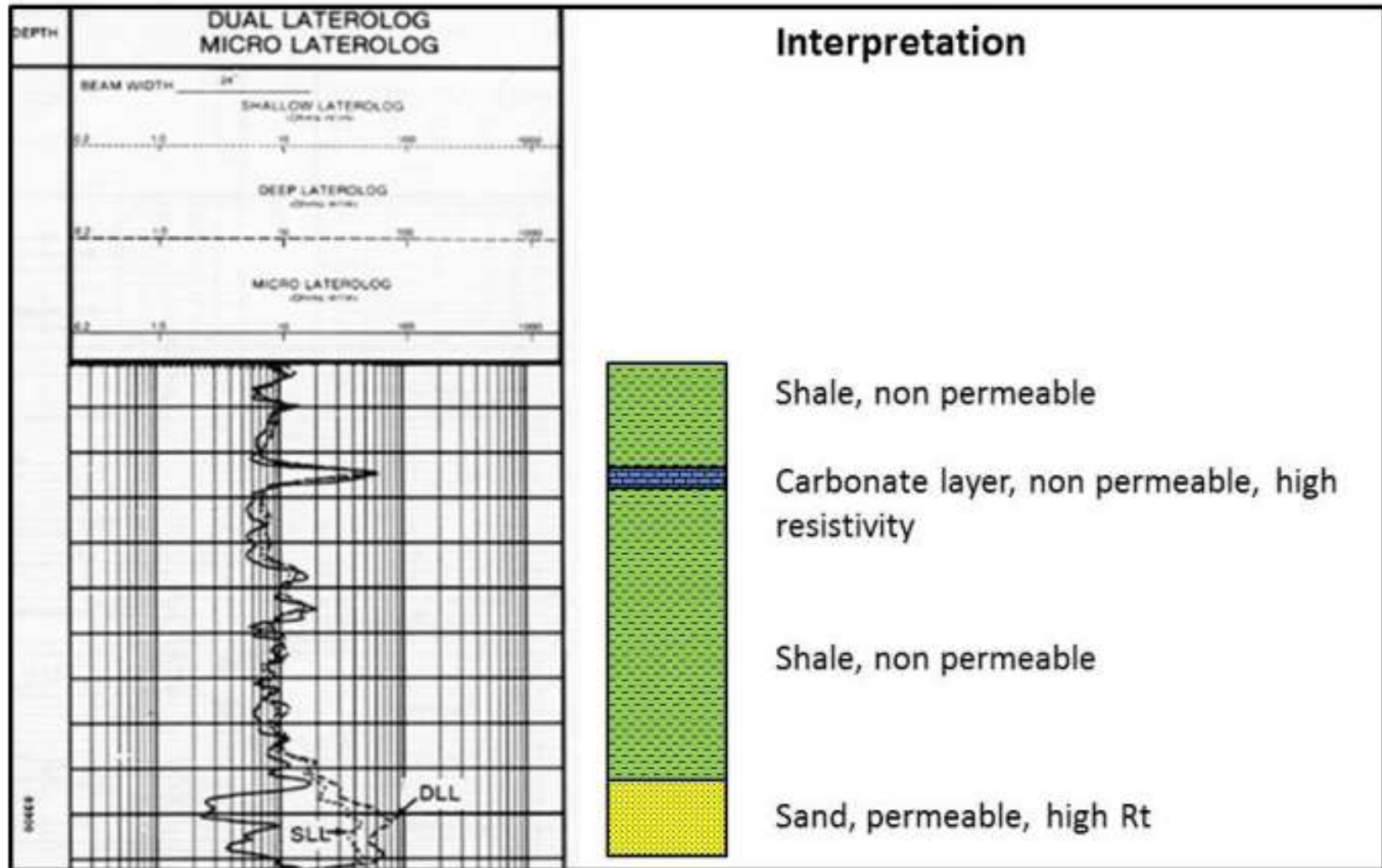


Wellsite Evaluation – Permeable Bed Identification

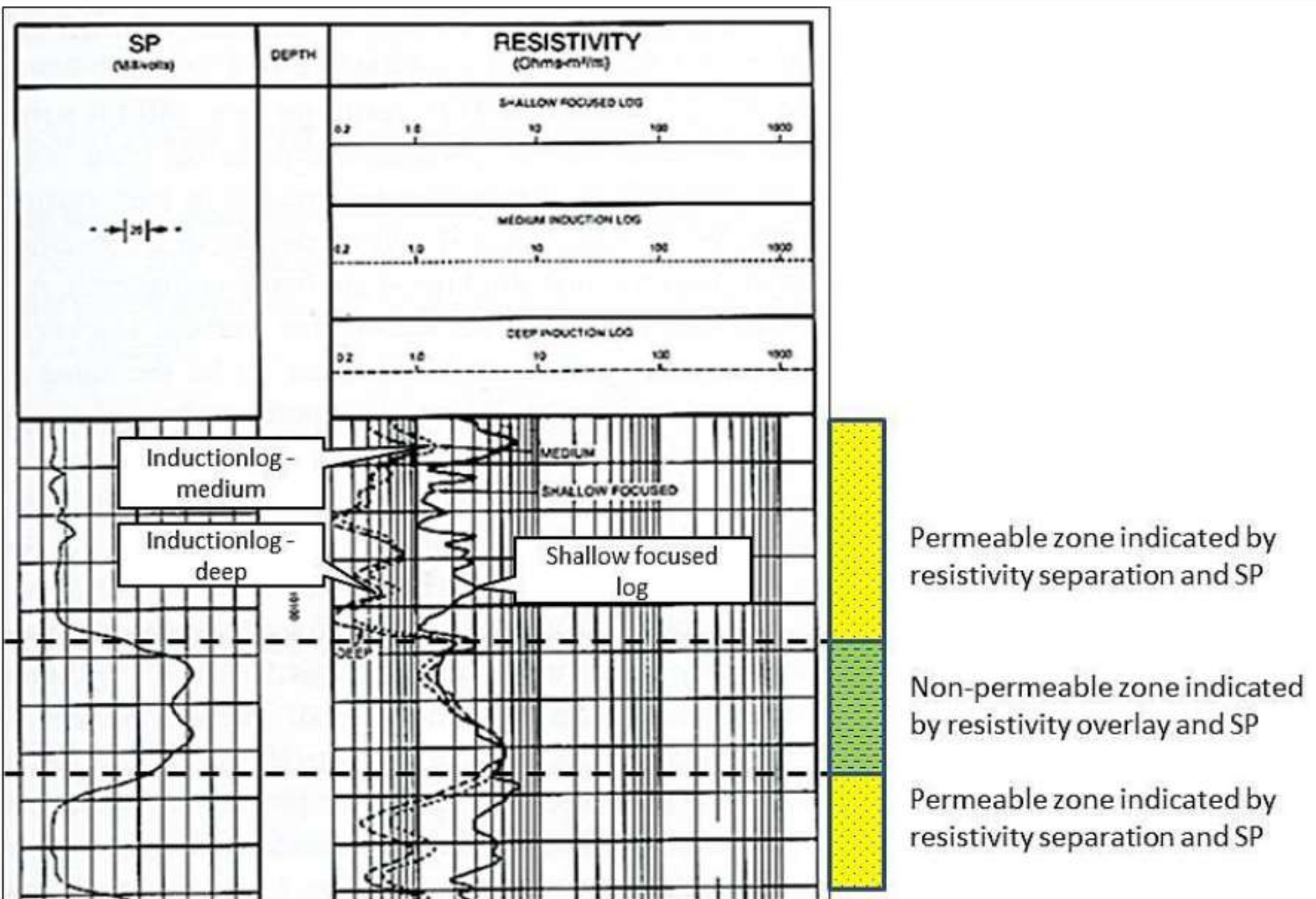
- Permeable beds can be identified quickly using the following indicators:
 - Spontaneous Potential (SP); the Diffusion Potential directly indicates a permeable formation,
 - Mud Filtrate Invasion detected by multiple resistivity measurements,
 - Presence of mudcake indicated by Caliper log,
- Note: The GR Log gives an indication related to mineral composition, but not directly to permeable and non-permeable zones
- For example Figure WE3 shows a section of sand-shale profile with $R_{mf} > R_w$. The three resistivity curves fit well in the non-permeable shale beds (no invasion) and separate in permeable sandstone. In the sandstone the Micro-Latero Log (MLL) – controlled by R_{xo} – shows a low resistivity compared to with the Laterolog Deep (LLD) and Laterolog Shallow (LLS) which are strongly influenced by the non-invaded zone (R_t). Because the

Wellsite Evaluation – Permeable Bed Identification

- Because the $R_{MLL} < R_{DLL}$ and R_{SLL} it can be concluded that the sandstone is hydrocarbon-bearing.



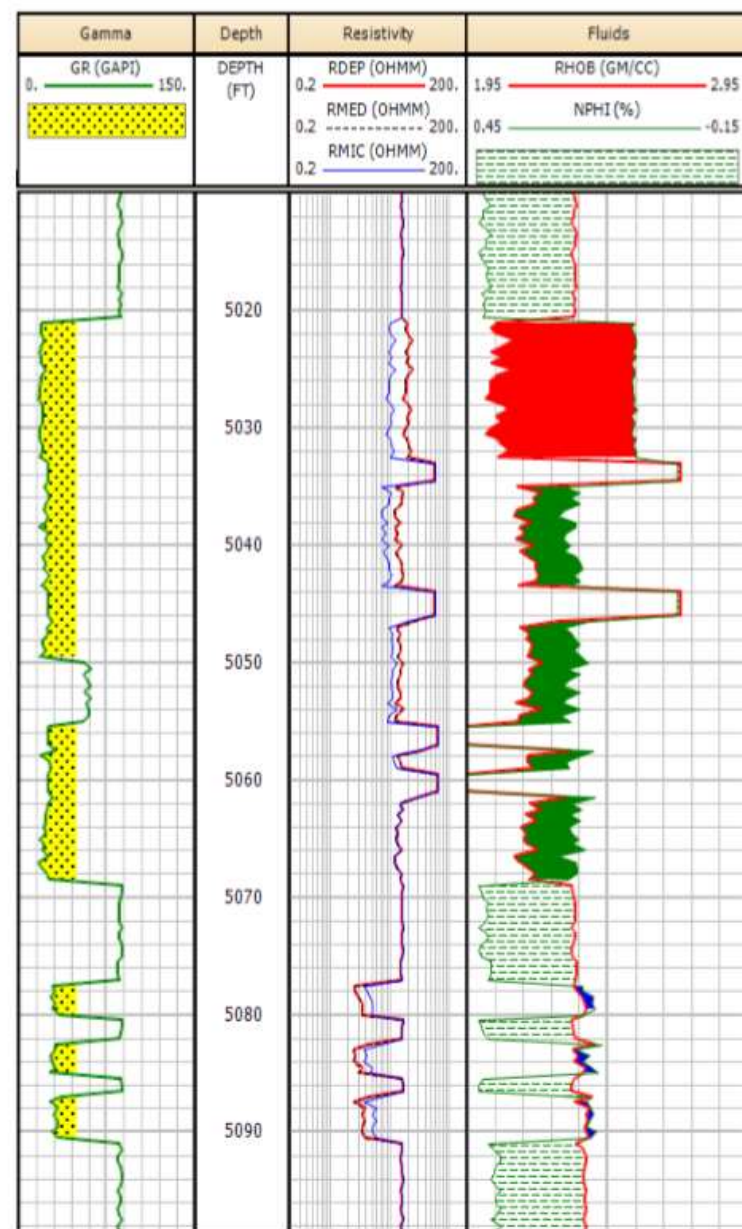
- Fig. WE3** Permeable zone identification



- Separation of the resistivity logs results from an invasion effect and indicates permeable zones (sandstone), whereas a fit of the three indicates a non-permeable zone (shale)

2) Identify fluid types and contacts

- Examine the resistivity logs.
 - Note change from high resistivity to low resistivity in Net Sand.
 - High resistivity: Hydrocarbon
 - Low resistivity: Water
 - Look at resistivity separation:
 - $R_{mic} < R_{DEP}$: Hydrocarbon
 - $R_{mic} \geq R_{DEP}$: Water
- Examine neutron-density separation in Net Sand.
 - Recognise small sand separation in water leg.
 - Large separation at top of hydrocarbon leg; wide separation indicating gas.
 - Middle section indicating oil.
- In this case the contacts fall within Non-Net and cannot be picked.
 - Pick Gas Down to/Oil Up to
 - Pick Oil Down to/Water Up to



Wellsite Evaluation – Identifying the Fluid Type

- The second step especially for clastics is to answer the question: “Does the formation contain hydrocarbons?”
- Once a porous, clean formation is identified, the analyst is faced with deciding whether it contains hydrocarbons or not. This analysis is done in quite an indirect way, using the resistivity R_t of the formation.
- Basically, if the porous formation contains conductive brine, its resistivity will be low. If, instead, it contains a sizable fraction of non-conducting hydrocarbon, then the formation resistivity will be rather large.
- The hydrocarbons may be oil or gas. The distinction is most easily made by comparing the formation density and neutron porosity measurement.

Wellsite Evaluation – Identifying the Fluid Type

- To at least have a working assumption regarding the fluids, start by comparing the density and deepest reading resistivity log for any evidence of hydrocarbons. In the classic response, the resistivity and density (and also GR) will be seen to “tramline” (i.e., follow each other to the left or right) in water sands and to “Mae West” (i.e., be a mirror image of each other) in hydrocarbon sands (see Figure WE.2).
- However, some hydrocarbon/water zones will not exhibit such behaviour, the reasons being:
 - 1) When the formation-water salinity is very high, the resistivity may also drop in clean sands.
 - 2) In shaly sand zones having a high proportion of conductive dispersed shales, the resistivity may also fail to rise in reservoir zones.
 - 3) If the sands are thinly laminated between shales, the deep resistivity may not be able to “resolve” the sands, and the resistivity may remain low.
 - 4) If the well has been drilled with very heavy overbalance, invasion may be such as to completely mask the hydrocarbon response.
 - 5) When the formation water is very fresh (high R_w), the resistivity may Mae West even in water-bearing zones.

Wellsite Evaluation – Identifying the Fluid Type

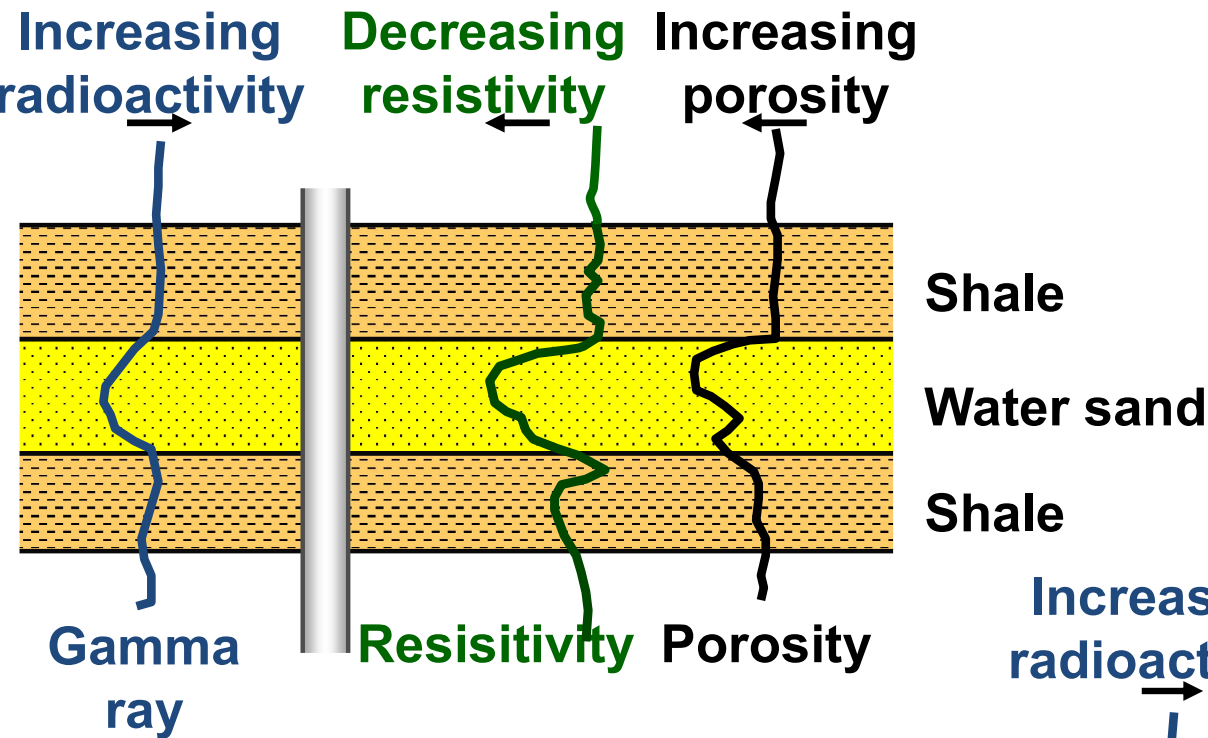
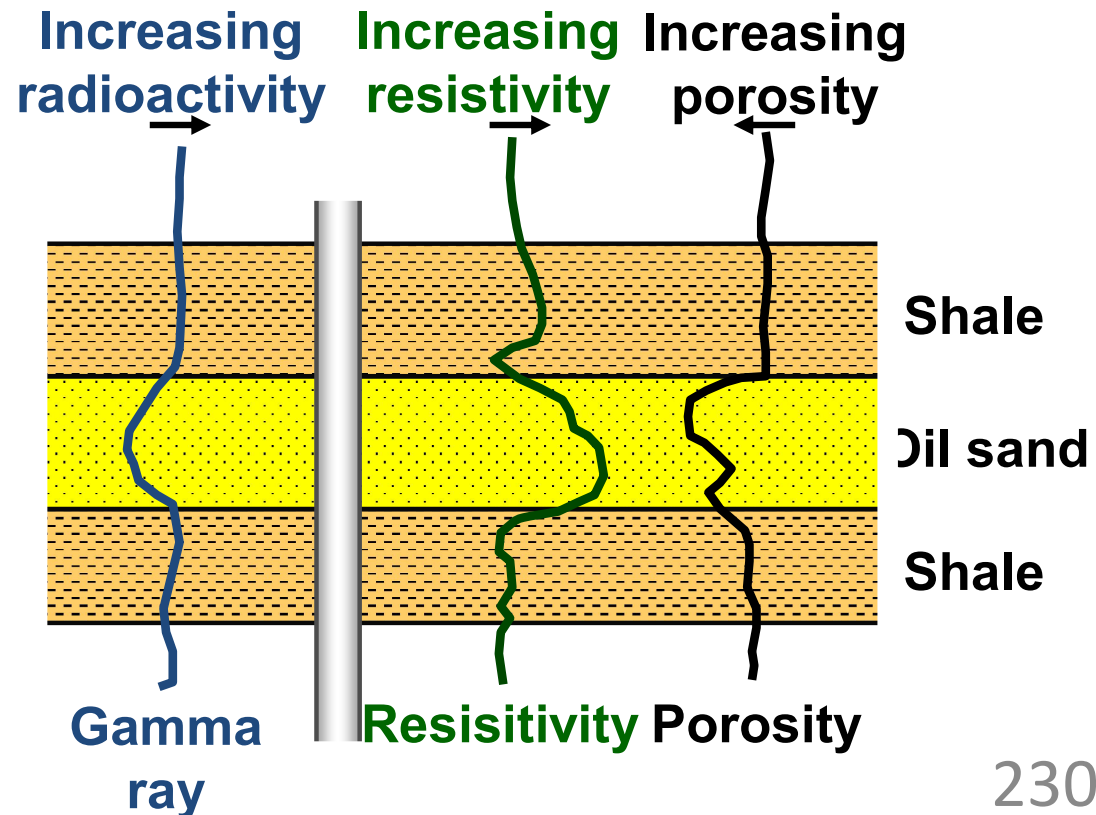


Figure WE.2

Typical Logging Signatures in
Water-bearing sand vs.
Oil-bearing sand



Wellsite Evaluation – Identifying the Fluid Type

- When either of the first two situations arises, it is very important to look at the absolute value of the deep resistivity, rather than at only the behaviour compared with the density.
- As long as a known water sand has been penetrated in the well (or a neighbouring well), a good idea of what the resistivity ought to be for a water-bearing sand should exist. If the resistivity is higher than this value, whatever the shape of the curve, then hydrocarbons should be suspected.
- There is also an effect of porosity on the resistivity. As porosity increases, the value of R_t will decrease if the water saturation remains constant.
- Gas zones will exhibit a greater density/neutron crossover than oil zones. In a very clean porous sand, any GOC can be identified on the log relatively easily.

Wellsite Evaluation – Identifying the Fluid Type

- Another common resistivity measurement, R_{xo} , corresponds to the resistivity of the flushed zone, a region of formation close to the borehole, where drilling fluids may have invaded and displaced the original formation fluids.
- The measurement of R_{xo} is used to get some idea of the recoverability of hydrocarbons in the following way. If the value of R_{xo} is found to be the same as the value of R_t , then it is most likely that the original formation fluids are present in the flushed zone, indicating that no formation fluid displacement has taken place. However, if R_{xo} is different than R_t , then some invasion has taken place, and the fluids are movable.
- This can be taken one step further. If the ratio of R_{xo} to R_t is the same as the ratio of the water resistivities in the two zones (R_{mf} and R_w), then the flushed and non-flushed zones have either the same quantity of hydrocarbons or none. Any hydrocarbons are unlikely to be producible in this case. If the ratio of R_{xo} to R_t is less than that of R_{mf} to R_w , then some hydrocarbons have been moved by the drilling fluid and will probably be producible.

3) Calculate Porosity

- Calculate the porosity in the Gas, Oil and Water legs
- If logging a number of runs you may only have a sonic porosity initially.
 - Assume the matrix transit time for the lithology drilled (sandstone or limestone) and transit time for mud filtrate (water in this case).
 - Refine the porosity calculation as described below when neutron-density data is available.
- If you have all the logs in one pass:
 - Estimate porosity from neutron-density by drawing a line halfway between the two curves and reading off the neutron scale in the oil and water legs.
 - Estimate porosity in the gas leg by plotting sample data points from the log on a chart book neutron-density cross-plot.

Wellsite Evaluation – Reservoir Porosity

Recall that four logging devices yield estimates of porosity.

- In the case of the density tool, the measured parameter is the formation bulk density ρ_b . As porosity increases, the bulk density ρ_b decreases.
- The neutron tool is sensitive to the presence of hydrogen. Its reported measurement is the neutron porosity φ_n , which reflects the value of the formation hydrogen content.
- The acoustic tool measures the compressional wave slowness or, interval transit time Δt (reported in $\mu\text{s}/\text{ft}$). It will increase with porosity.
- The total NMR signal depends on the amount of hydrogen and therefore increases with porosity.

N.B.: *In all three presentations, the format is usually such that increasing porosity produces curve deflections to the left.*

Porosity from a single log

- Porosity as a fundamental reservoir property can be determined indirectly from Gamma-Gamma-Density log, Neutron log, and Acoustic -/Sonic log.
- Porosity calculation requires the matrix and fluid properties as expressed in the equations as (left side with Greek symbols, right side with Greek symbols) additional input.

$$\phi = \frac{\rho_{ma} - \rho_b}{\rho_{ma} - \rho_{fl}}$$

$$PHI = \frac{RHO_{ma} - RHOB}{RHO_{ma} - RHO_{fl}}$$

$$\phi = \frac{\phi_N - \phi_{N,ma}}{\phi_{N,fl} - \phi_{N,ma}}$$

$$PHI = \frac{NPFI - NPFI_{ma}}{NPFI_{fl} - NPFI_{ma}}$$

$$\phi = \frac{\Delta t_{ma} - \Delta t}{\Delta t_{ma} - \Delta t_{fl}}$$

$$PHI = \frac{DT - DT_{ma}}{DT_{fl} - DT_{ma}}$$

$\rho_b(RHOB)$,
 $\phi_N(NPFI)$,
 $\Delta t(DT)$,
 are the measured
 (logged) data

Porosity from a single log

- Matrix properties can be determined from:
 - Known lithology/mineral composition; in this case representative values are recommended. The most critical is matrix slowness.
 - Core laboratory measurements (for example grain density)
 - Cross plot techniques

Component	ρ_{ma}, ρ_{fl} in g/cm ³	$\phi_{N,ma}, \phi_{N,fl}$	$\Delta t_{ma}, \Delta t_{fl}$ in μ s/m	$\Delta t_{ma}, \Delta t_{fl}$ in μ s/ft
Quartz	2.65	- 0.02 ... - 0.04	180	55
Calcite	2.71	0	156	47.4
Dolomite	2.87	0.02.....0.04	143	43.5
Water	1.00	1.00	620	189

Recommended properties for matrix and fluid

Porosity from a single log – invasion rules

- Invasion tends to force gas out of the formation and replace it with borehole fluid. Porosity calculation from NEU & DEN measurement requires fluid properties. The influence of the two porosity logs depends on their radial sensitivities and their depths of investigation (DOI)
- If the invasion fluid front gets deep into the reservoir, the NEU & DEN Porosity measurements approach the true porosity for the assumption of mud filtrate density and the Neutron response as fluid properties.
- For shallow invasion (shallow with respect to the depth of investigation) the tools' responses are spatially weighted averages of the invaded and non-invaded regions of the formation. The problem in deriving the porosity in the presence of shallow invasion comes from the fact that the neutron and density logging devices have different radial response:
 - The 50% DOI of the thermal neutron porosity tool is 6 to 12 inches (15 to 30 cm).
 - The 50% DOI of the density tool is about 2 to 3 inches (5 to 8 cm).

Porosity from a single log – invasion rules

- In the presence of borehole invasion into a gas-bearing reservoir, some specific conditions must be considered.
 - When the invasion front is greater than 12 inches (30 cm), both tools see only water-filled (mud filtrate) formations and the two porosity estimates agree and read true porosity.
 - When the invasion front is less than 12 inches (30 cm) but greater than 6 inches (15 cm), the density tool on sees the invaded formation, while the neutron tool is sensitive to both the invaded and the non-invaded region. Under these conditions, the density porosity estimate is the true value while the Neutron porosity estimate is still low.
 - Below 6 inches (15 cm) of invasion, both tools are sensitive to both the invaded and non-invaded regions. Thus, for a certain range of invasion, accurate determination of formation porosity becomes very difficult. A Root Mean Square (RMS) is recommended for gas reservoirs:

$$\phi_{ND,gas} = \sqrt{\frac{\phi_N^2 + \phi_D^2}{2}} \approx \frac{\phi_N}{3} + 2 \cdot \frac{\phi_D}{3}$$

4) Calculate water saturation S_w

- Read RDEP in the water leg as approximate R_t .
- Assume that the sands are clean and the Archie equation applies:

$$S_w = \left[\frac{1}{\phi^2} \cdot \left(\frac{R_w}{R_t} \right) \right]^{0.5}$$

- If $S_w=1$ this becomes $1 = \frac{1}{\phi^2} \cdot \left(\frac{R_w}{R_t} \right)$
which can be solved for R_w : $R_w = R_t \cdot \phi^2$

Where, S_w = Water saturation (v/v)

ϕ = Calculated Porosity (v/v)

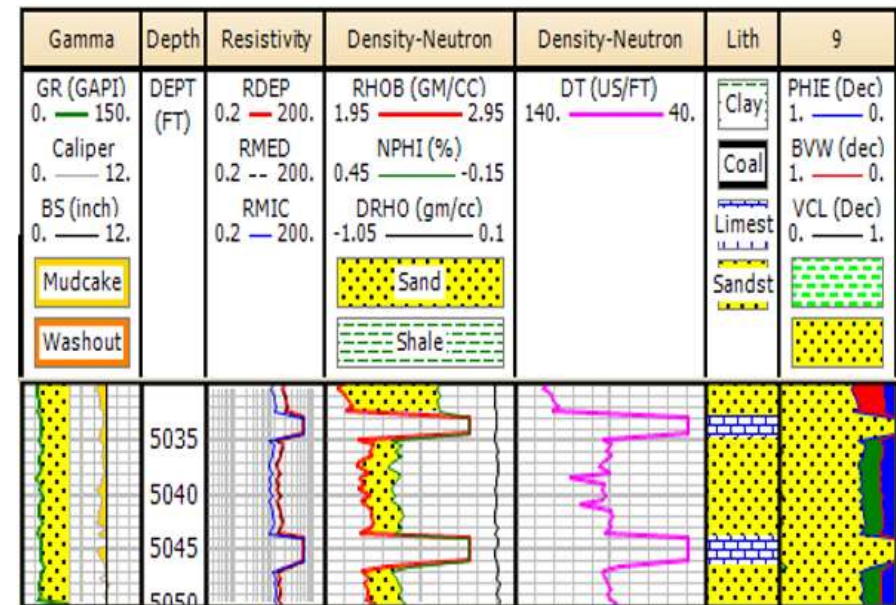
R_t = True formation resistivity (ohmm)

R_w = Formation water resistivity (ohmm)

- Determine R_w using this equation
- Read R_{DEP} in the gas and oil legs and calculate S_w using the Archie equation and the value of R_w you found in the water leg.

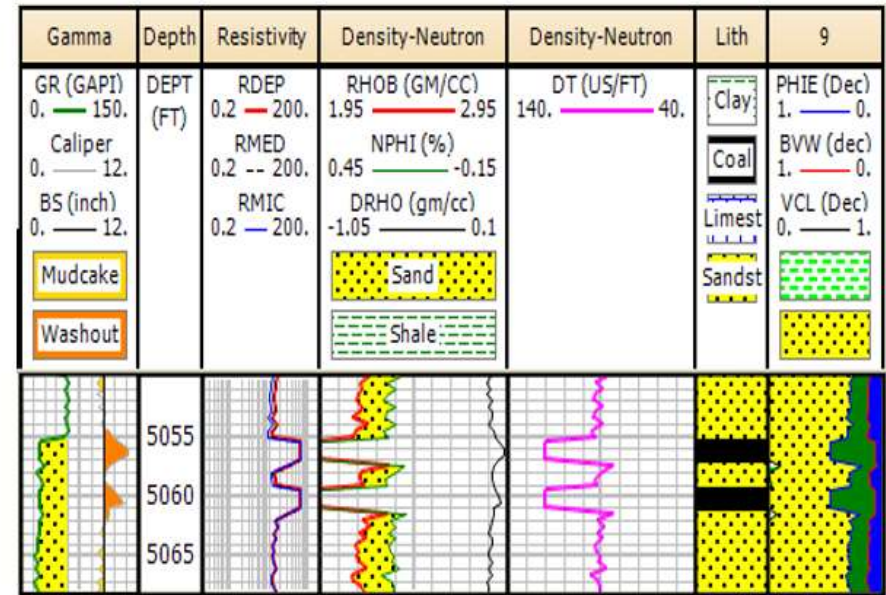
5) Recognize minerals: Calcite Stringers

- Calcite (tight limestone) stringers can be recognised @ 5,034 and 5,045 Ft MD.
 - Neutron and Density logs overlay at low porosity.
 - Resistivity is high, because of low porosity.
 - Gamma Ray is low.
 - Sonic transit time is short.
- Calculated porosity is very low and these intervals would be excluded from Net Reservoir by a porosity cut-off.

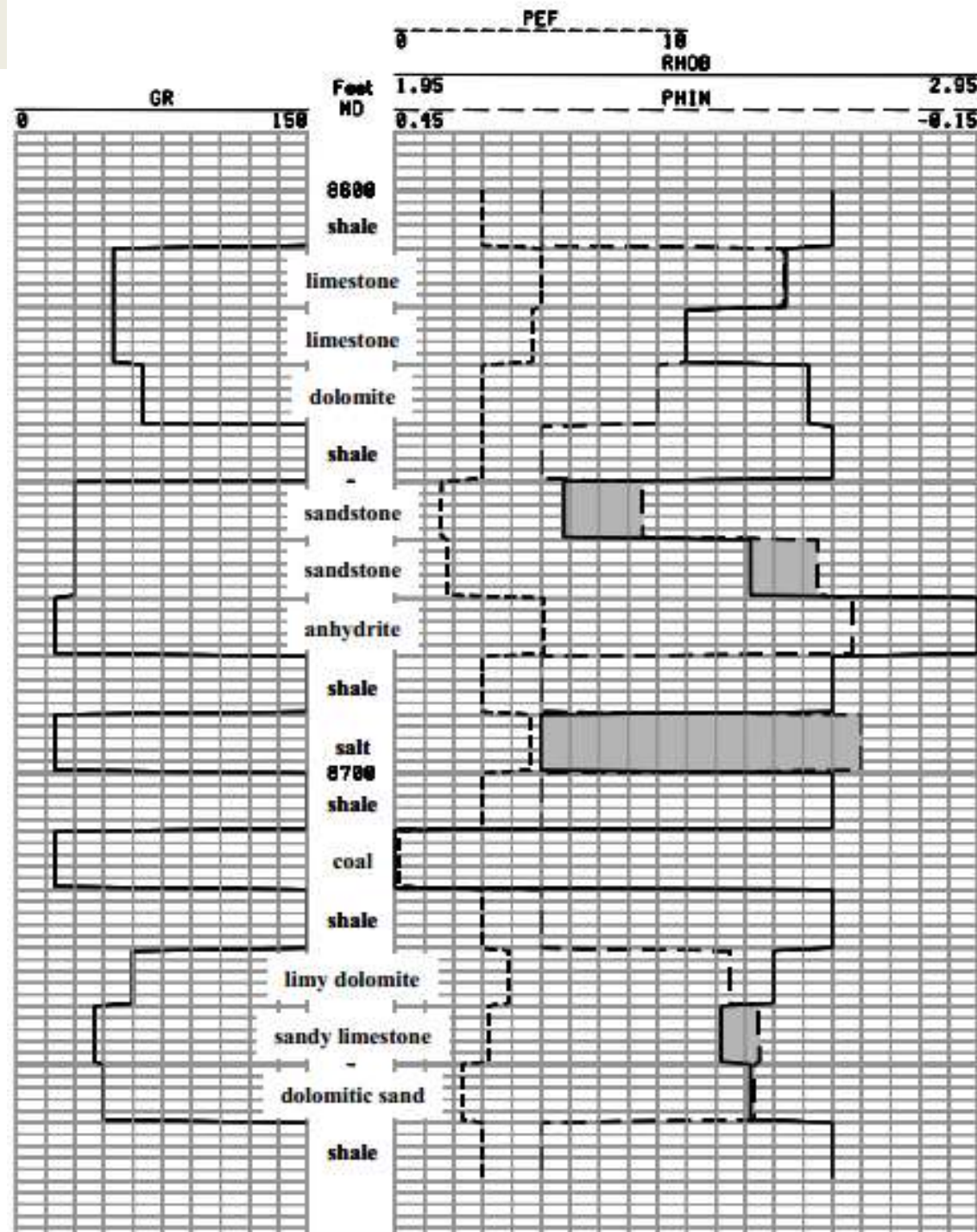


5) Recognize minerals: Coal

- Around 5,056 Ft MD and 5,060 Ft MD.
 - Apparent very high porosity:
 - Neutron density.
 - Sonic.
 - Apparent oil saturation.
 - High resistivity.
- Recognised as coals because:
 - Neutron and density indicate unrealistically high porosities.
 - Sonic shows long transit times.
 - Hole is washed out as it often is in friable coals.
- These intervals should be flagged as non Net Reservoir, porosity set to 0 and S_w to 1.



Lithology Ambiguities



- While the positions on the log will vary with varying changing porosity, the relative position of the neutron and density curves will remain fairly constant with lithology.
- The Photoelectric Effect (PE) curve may be useful to resolve some ambiguities which occur with some lithologic mixtures. (N.B.: The PE curve captures low energy gamma rays returning from the formation).
- It is assumed that (a) the Neutron porosity and Density curves (or bulk density) are scaled by limestone matrix (b) formations are assumed to be clean (i.e. No shales/clays) (c) filled with liquid only (no gas present)

Wellsite Evaluation – Summary Relations

Descriptor	Measurement	Functional behavior		
Clean/shaley	SP	$V_{shale} \uparrow$	$\rightarrow$	SP $\uparrow$
	GR	$V_{shale} \uparrow$	$\rightarrow$	GR $\uparrow$
Porosity (ϕ)	Density	$\phi \uparrow$	$\rightarrow$	$\rho_b \downarrow$
	Neutron	$\phi \uparrow$	$\rightarrow$	$\phi_n \uparrow$
	Acoustic	$\phi \uparrow$	$\rightarrow$	$\Delta t \uparrow$
Hydrocarbon	R_t	$S_w \uparrow$	$\rightarrow$	$R_t \downarrow$
		$(S_o \uparrow$	$\rightarrow$	$R_t \uparrow)$
		$\phi \downarrow$	$\rightarrow$	$R_t \uparrow$
Recoverable/ movable	R_{xo} vs. R_t (shallow vs. deep)	$R_{xo} = R_t$	$\rightarrow$	No invasion
		$R_{xo} = R_t$	$\rightarrow$	If $R_{mf} = R_w$, no movable hydrocarbon
		$\frac{R_{xo}}{R_t} \neq \frac{R_{mf}}{R_w}$	$\rightarrow$	Moved fluid

A summary of phenomenological interpretation

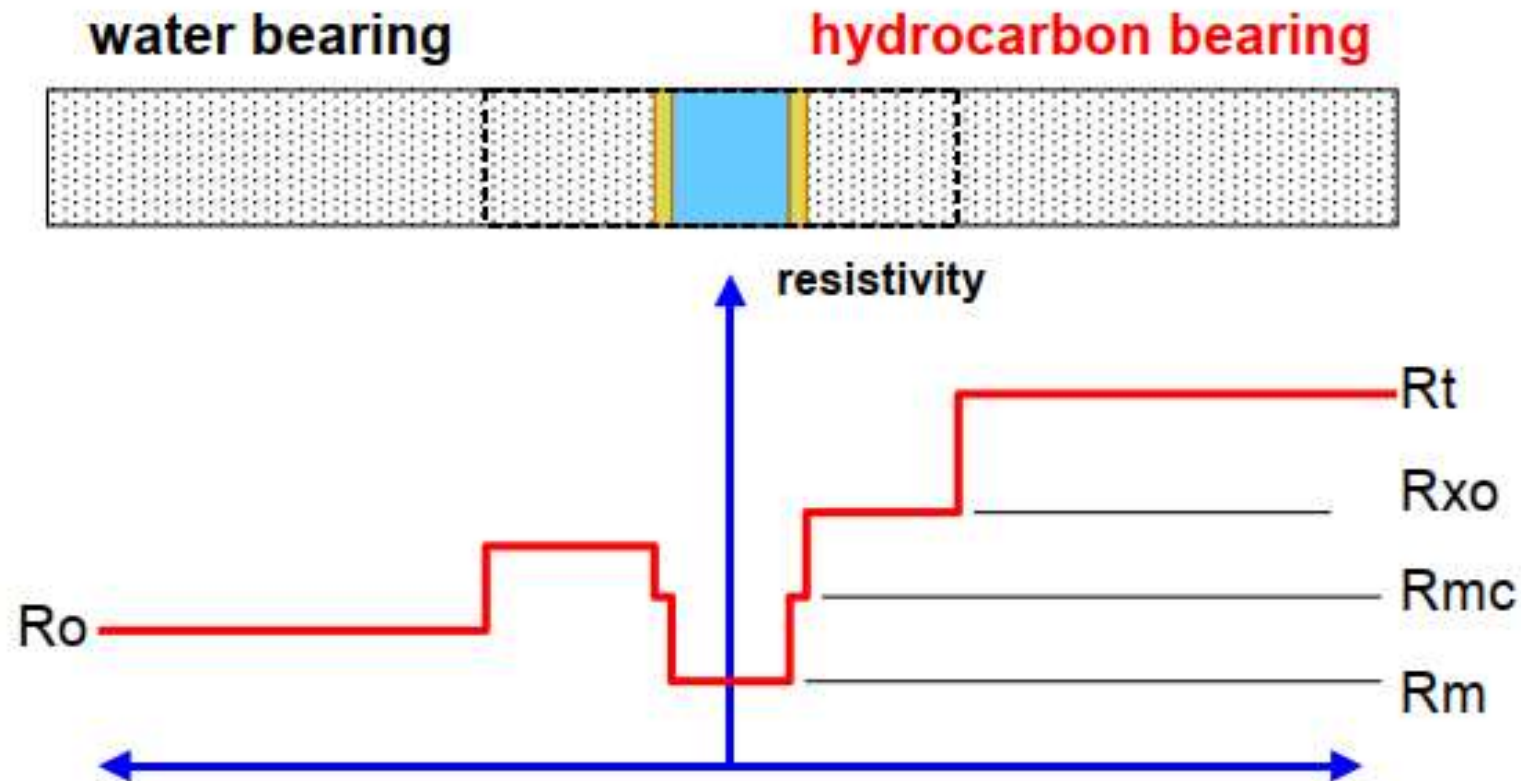
SOME THOUGHTS ON QUANTITATIVE LOG EVALUATION

Determination of True Resistivity

- In a water-bearing formation with low formation water resistivity R_w , the infiltration of the mud filtrate with higher resistivity R_{mf} results in an increase in rock resistivity in the infiltrated (R_{xo}) zone compared to the non-invaded zone (R_o)
- In a hydrocarbon-bearing formation the resistivity R_t is relatively high as a result of lower water saturation S_w . Infiltration displaces a part of the original fluid (water and hydrocarbon) and results in a decrease in of rock resistivity in the infiltrated (R_{xo}) zone compared to the non-invaded zone (R_o)
 - The R_{xo} level in the hydrocarbon-bearing formation is somewhat higher than the corresponding level in the water-saturated formation. This is the effect of a residual hydrocarbon volume in the pore space which is not totally displaced by the mud filtrate.
 - The magnitude of R_t in the hydrocarbon-bearing formation is controlled by porosity, water saturation, and formation saturation. Therefore can be low for highly saline formation water also in case of high HC saturation

Determination of True Resistivity

radial profile of fluids $\rightarrow$ radial profile of properties



Resistivity profile for water-bearing (left) and a HC-bearing zone

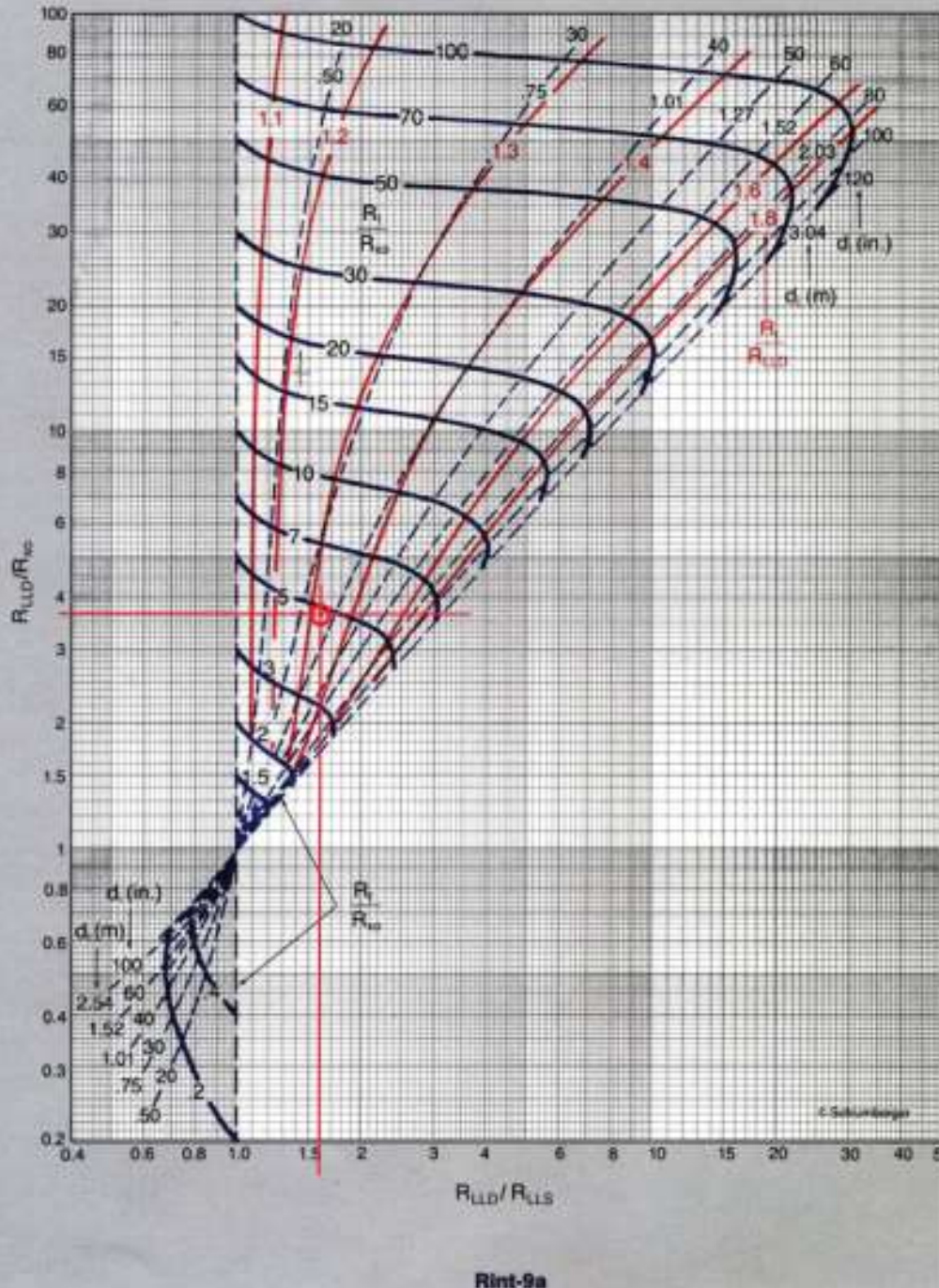
Determination of True Resistivity

- For a “quick-look” estimate of R_{xo} a microresistivity tool is used in many cases and R_t is approximated by a “deep-reading” focused tool.
- A more exact technique is based on the graphic or numerical solution of the problem to derive “true resistivities” R_t , R_{xo} , and d_i from a set of measured “apparent resistivities”.
- This can be demonstrated using “Tornado charts” which are the result of a forward modelling of the response answer of tools with different formation resistivity models (given in terms of R_t , R_{xo} , d_i). The chart for a dual laterolog- R_{xo} device combination for thick beds and 8 inch borehole is shown on the next slide.
- The charts are plotted in logarithmic scale for resistivity ratios in general:
 - The x-axis shows the ratio of the forward calculated ratio R_{LLD}/R_{LLS} and the y-axis the forward calculated ratio R_{LLD}/R_{xo}
 - The parameter of the individual curves correspond to the model input (ratio R_t/R_{xo} , ratio R_t/R_{LLD} and depth of invasion d_i) attributed to the calculated curves.

Dual Laterolog- R_{xo} Device

DLT-B LLD—LLS— R_{xo} Device

Thick Beds, 8-in. (203-mm) Hole,
No Annulus, No Transition Zone,
Use Data Corrected for Borehole Effect



True Resistivity

- The chart is applied in the following steps
- 1) Take the measured resistivities R_{xo} , R_{LLS} , R_{LLD}
- 2) Calculate the ratios R_{LLD}/R_{LLS} and R_{LLD}/R_{xo}
- 3) Enter the abscissa and ordinate and plot the ratios on the chart
- 4) The intersection of the lines is the solution of the problem in terms of R_t/R_{xo} , R_t/R_{LLD} & d_i
- 5) Calculate the true resistivities R_t and R_{xo} .

See Page 59 of the main text for the example

Det. of True Resistivity - Example

- Log Readings:

- $R_{LLD} = 16 \text{ Ohmm}; R_{LLS} = 10 \text{ Ohmm}; R_{MSFL} = 4.5 \text{ Ohmm}$

- Calculate the ratios:

- $R_{LLD} / R_{MSFL} = 3.56; R_{LLD} / R_{LLS} = 1.6;$

- At the intersection:

- $R_t / R_{LLD} = 1.33; R_t / R_{xo} = 4.7; d_i = 36 \text{ inch}$

- **Results:**

- $R_t = 21.3 \text{ Ohmm}; R_{xo} = 4.53 \text{ Ohmm}.$

- Please Note:

- The true resistivity of the non-invaded formation R_t , is distinctly higher than the measured resistivity of the deepest reading tool, R_{LLD} . This results from the influence of the lower resistivity of the invaded zone. Neglecting this effect would result in an overestimate of S_w .
 - The true resistivity of the invaded zone R_{xo} is close to the measured resistivity of the micro tool R_{MSFL} .

Porosity and Mineral Composition

– Multiple Porosity Methods

- Two or more independent methods for porosity are applied to determine an exact porosity value. Combined interpretation makes derivation of porosity and rock composition possible. The combination of DEN and NEU Log is applied most frequently.
- If the lithology (mineralogy) and also the water saturation can be assumed for a formation, then porosity can be calculated with both methods. If both independent calculations deliver the same porosity it may be concluded that:
 - 1) The used interpretation model (limestone/calcite & water) is correct.
 - 2) The calculated porosity is correct.
- If the calculated porosities are different, the conclusion is that the used interpretation model (matrix mineralogy, fluid) is not correct. The conflict of the two answers can be resolved by derivation of representative mineral composition and fluid type using any of these practical techniques.
 - 1) Overlay Technique
 - 2) Cross plot technique
 - 3) Numerical solution of a system of response equations

Porosity and Mineral Composition

– Overlay technique

- Two porosity curves are plotted in the same log trace and referenced to an identical porosity scale and a defined rock model (matrix and fluid). If for example, Neutron porosity (limestone-calibrated tool) and Density-derived porosity (calculated under assumption of calcite as matrix with density 2.71 g/cm^3) are plotted then in a:
 - Water-saturated limestone both tools read same porosity,
 - Sandstone or dolomite both tools read a somewhat different porosity,
 - Shaly section neutron reads higher porosity than density resulting from clay-bound water,
 - Gas zone neutron reads lower porosity than density resulting from the lower hydrogen concentration measured by the neutron log.
- Important for overlay analysis is to understand the scales for both methods (whether identical scales are used or increasing Neutron porosity and decreasing density). If the formation is composed of different lithologies and this overlay technique is applied, it is more comfortable to convert or transform both porosity logs to a sandstone matrix and expect a fit for both logs in such cases.

Porosity and Mineral Composition

– Crossplot technique (Clean Matrix)

- While overlays present the logs of a section, cross plots technique combines individual data sets of defined depth steps. For the combination of Neutron and Density log, the Neutron readings from the LAS file are plotted on the x-axis and the measured Gamma-Gamma-Density in reverse on the y-axis for each depth step.
- Plots start in the lower left corner with the “matrix-points” and go up to the upper right corner to the “water-point” with linear scaled porosity. For practical use only the part for porosity < 0.45 (45%) is plotted:
 - The coordinates $x = \text{NPHI} = 0.0$ and $y = \text{RHOB} = 2.71 \text{ g cm}^{-3}$ describe a dense (no porosity) calcite and therefore is called “matrix-point limestone”.
 - Increasing porosity (pore fluid water) moves the position linearly upward from this matrix point in the direction of a “fluid-point water” at $x = \text{NPHI} = 1.0$ and $y = \text{RHOB}_{\text{fl}} = \text{RHOB} = 1.0 \text{ g cm}^{-3}$. The connecting straight line represents a pure limestone with linear scaled porosity.
 - Different rocks (sandstone/quartz, dolomite) have a matrix point defined by matrix density and matrix neutron response.
- The result are three lines for limestone, dolomite and sandstone. The deviation from a straight line of the dolomite and sandstone curve is contributed for better approximation and depends on tool characteristics.

Porosity and Mineral Composition

– Crossplot technique (Clean Matrix)

- Combining the information from multiple log types can provide important information, both in terms of identifying possible errors and understanding subsurface formations.
- Neutron-Density crossplots provide an alternative method for lithology-porosity interpretation and have several advantages.
- More precise estimates of true porosity can be made and the location of zones on the crossplot gives indications of the quantitative composition of mineral components in mixed lithologies.
- If zone points tend to be clustered, this is an additional indication of the presence of distinctive facies, which may be related to depositional and/or diagenetic controls.

Porosity and Mineral Composition

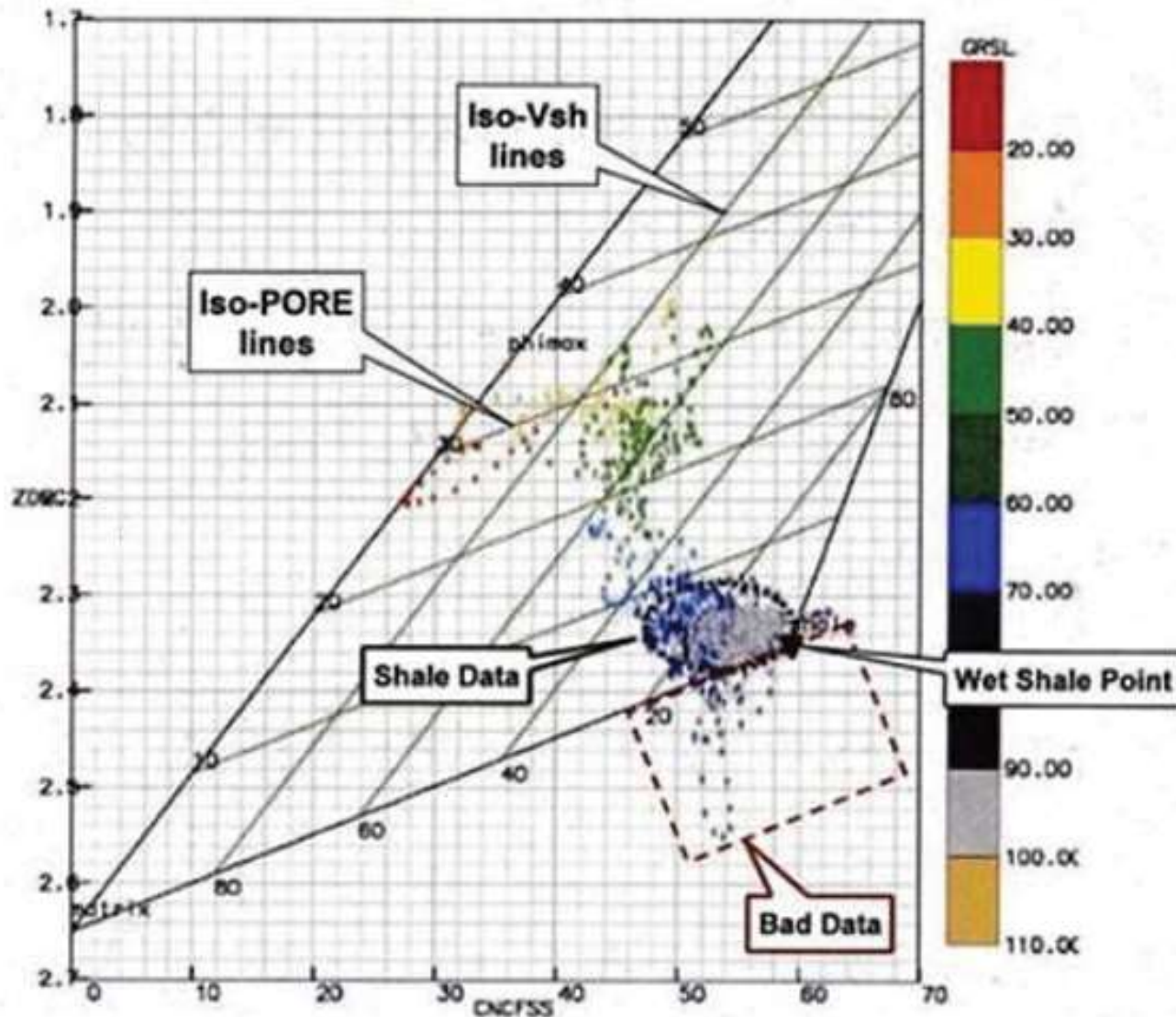
– Crossplot technique (Clean Matrix)

- Cross plots allow an estimate of mineralogy (lithology) and the determination of a consistent porosity. Measured data are plotted into the designated curve set. The position of the data point delivers the correct porosity and the mineral type or composition.
- Deviations and problems occur for the Neutron-Density-Crossplot if shale or gas is present:
 - Gas in the formation shifts the points towards low neutron porosity, because gas has a lower hydrogen concentration compared with water and in most cases the Neutron log “reads” the non-invaded zone .
 - Shale content shifts the points from the sand line towards dolomite, because apparent Neutron porosity increases (clay-bound water effect). For shaly sand a specific cross plot can be designed with a sand point, a wet-shale point and a water point. This is a component of shaly sand analysis.
- A helpful modification of the NEU-DEN Cross plot is the implementation of a third dimension: individual points are colour coded by their gamma activity. If the colour indicates low shale content (low gamma), then the point represents a clean rock (in sandstone, limestone or dolomite region). If the colour indicates high shale content and the point plots, for example, in the porous dolomite region then this is not a highly porous dolomite but probably a shaly sand.

Porosity and Mineral Composition

– Crossplot technique (Shaly Sand)

- A specific application is the interpretation of shaly sands. In a NEU-DEN-Crossplot data points of the clean sandstone are arranged along the sandstone line.



With increasing shale content the data points of the clean sandstone are shifted to higher NEU porosity (clay-bound water effect). If data points are colour coded with Gamma intensity (GR) this effect is visible as demonstrated below.

Porosity and Mineral Composition

– Crossplot technique (Shaly Sand)

- The distribution of all data points allows picking the “wet-shale” point at the highest GR point (pure shale). It characterises the shale including the clay bound water. The wet shale density could be extracted on the vertical axis and the wet shale Neutron porosity (neutron response wet shale) on the horizontal axis.
- The volumetric rock model (below) describes the porosity and density in terms of effective and shale porosity.



Shaly sand model

Porosity and Mineral Composition

– Crossplot technique (Shaly Sand)

- The equations for (measured) Neutron porosity ϕ_N and bulk density ρ_b can be solved for shale content.

$$V_{sh} = \frac{(\phi_{N,fl} - \phi_{N,ma}) \cdot (\rho_{ma} - \rho_b) - (\phi_N - \phi_{N,ma}) \cdot (\rho_{ma} - \rho_{fl})}{(\phi_{N,fl} - \phi_{N,ma}) \cdot (\rho_{ma} - \rho_{sh}) - (\phi_{N,sh} - \phi_{N,ma}) \cdot (\rho_{ma} - \rho_{fl})}$$

- Effective porosity:

$$\phi_{eff} = \frac{(\phi_N - \phi_{N,ma}) \cdot (\rho_{ma} - \rho_{sh}) - (\phi_{N,sh} - \phi_{N,ma}) \cdot (\rho_{ma} - \rho_b)}{(\phi_{N,fl} - \phi_{N,ma}) \cdot (\rho_{ma} - \rho_{sh}) - (\phi_{N,sh} - \phi_{N,ma}) \cdot (\rho_{ma} - \rho_{fl})}$$

Where

- $\phi_{N,fl}$ is the neutron response (hydrogen index) of the fluid
- $\phi_{N,ma}$ is the neutron response (hydrogen index) of the matrix
- $\phi_{N,sh}$ is the neutron response (hydrogen index) of the wet shale
- ρ_{fl} is the density of the fluid
- ρ_{ma} is the density of the matrix
- ρ_{sh} is the density of the wet shale

PGG 434 EXAMPLE A – An oil bearing (clean) sandstone

Dr. O. Olamigoke

*Source: Basic Well Logging & Formation Evaluation 1st Edition
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Goals of this exercise

- Lithological profile with description of reservoirs
- Compilation of input parameters for quantitative interpretation
- Determination of porosity
- Determination of water saturation; derivation of hydrocarbon saturation
- Determination of moveable and non-moveable oil saturation

Basic Information needed in Quantitative Log Interpretation

- **Lithology:** In quantitative analysis, there are several reasons why it is important to know the lithology of the zone. (i) Porosity logs require a lithology or matrix constant before a zone's porosity can be calculated (ii) The formation factor, used in Archie's water saturation equation vary with lithology
- **Temperature of the formation:** Formation Temperature (T_f) is also important in log analysis because the resistivities of the drilling mud (R_m) and mud filtrate (R_f), and the formation water (R_w) vary with temperature.

Data Provided

Table 1: Input data provided

Parameter	Magnitude
R_{mf} – measured at 90 °F (32 °C)	0.336 Ohm m
Formation temperature	178 °F (81 °C)
Matrix density	2.65 g/cm ³
Fluid density	1 g/cm ³
Transit time (matrix)	52 μ s/ft
Transit time (fluid)	190 μ s/ft
Archie's parameters (from core measurements)	$m = n = 2$

Definition of Terms

- The index PE (Photoelectric Effect) strongly depends on the elemental composition and is therefore a typical characteristic of various minerals.
- The maximum difference between the most negative and the most positive level is called Static Spontaneous Potential (SSP)
- Dol – Depth of Interest

Equations Needed

- Arps Equation: The temperature dependence of NaCl-based water can be described by Arps' equation (Arps, 1953)

- $F = (1/\phi^m)$

$$R_w(T_2) = R_w(T_1) \cdot \frac{T_1 + 21.5}{T_2 + 21.5}$$

- $R_o = (1/\phi^m) * R_w$

- $R_t = R_o * (S_w)^{-n}$

$$S_w = \left(\frac{R_o}{R_t} \right)^{\frac{1}{n}} = \left(\frac{R_w}{R_t} \cdot \frac{1}{\phi^m} \right)^{\frac{1}{n}}$$

$$\phi = \frac{\rho_{ma} - \rho_b}{\rho_{ma} - \rho_f}$$

$$SSP = k_{SP} \cdot \log \left(\frac{R_{mf}}{R_w} \right)$$

Log Plot

- Fig. 1 shows the plotted logs. The original limestone-calibrated Neutron porosity was transformed into sandstone-referenced Neutron porosity (NPHISS) because the reservoir is sandstone.

Log Tracks

Track		Log description	unit
First track	CAL (IN)	Caliper	Inch
	DRHO (PU)	Correction for Density	g/cm ³
Second track	PEF (BARN)	Photoelectric cross section	b/e
	GR (GAPI)	Gamma log	API
	SP (MV)	Spontaneous Potential	mV
Third track	RT (OHMM)	Formation Resistivity	Ohm-m
	RT (OHMM)	Resistivity of invaded zone	Ohm-m
	RT (OHMM)	Resistivity Micro Spherical Focusing Log	Ohm-m
Fourth track	RHOB (GC)	Bulk Density	g/cm ³
	NPHISS (DE)	Neutron Density (Sandstone converted)	fraction
Fifth track	DT (μs/FT)	Slowness	μs/ft

The original limestone-calibrated Neutron porosity was transformed into sandstone-referenced Neutron porosity (NPHSS) because the reservoir is sandstone.

Preliminary Evaluation

Define reservoir / non-reservoir

- Choose a V_{sh} cut off of 50%
- Assuming $V_{sh} = IGR$
- $V_{sh} = (GR - GR_{sa}) / (GR_{sh} - GR_{sa})$
- $0.5 = (GR - 20) / (90 - 20)$
- $GR = 55$ API

Gross Rock Thickness Calculation
based on SP: $8958 - 8793 = 165$ ft

Constant Porosity from In oil zone:
 $NPHISS = 0.2$

- $\phi = (2.65 - 2.322) / (2.65 - 1) = 0.19878$

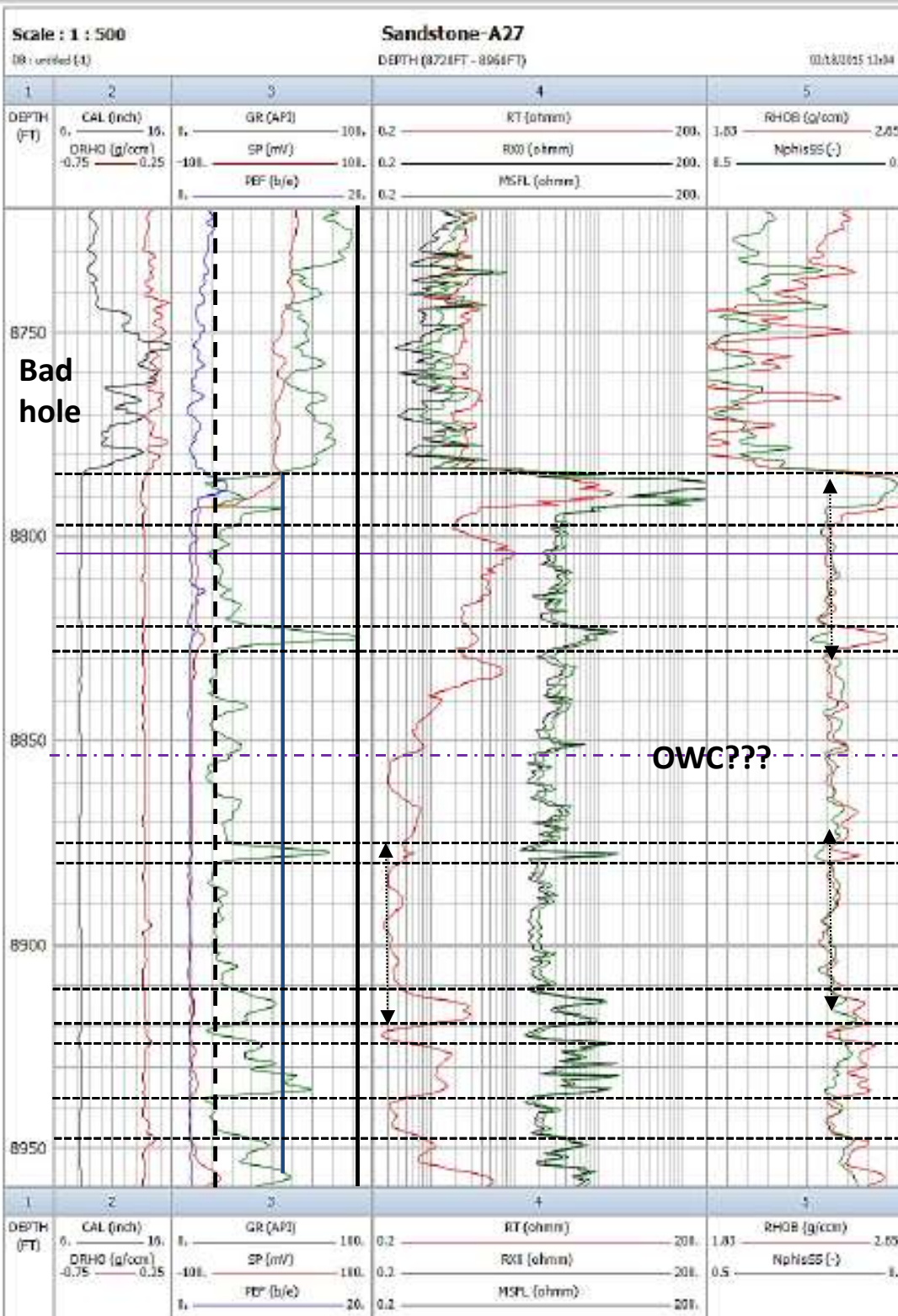
$$\phi = \frac{\rho_{ma} - \rho_b}{\rho_{ma} - \rho_f}$$

Calculation of R_w (Water Zone)

- $R_w = (0.3) * (0.2)^2 = 0.012$ Ohmm

 Sand
 Shale
 Unknown???

$$R_w = R_t * \phi^m$$



Preliminary Evaluation

Define reservoir / non-reservoir

- Choose a V_{sh} cut off of 50%
- $V_{sh} = (GR - GR_{sa}) / (GR_{sh} - GR_{sa})$
- $0.5 = (GR - 20) / (90 - 20)$
- $GR = 55$ API

24 ft Porosity Calculation

- In oil zone: $NPHISS = 0.2$

45 ft • $\phi = (2.65 - 2.322) / (2.65 - 1) = 0.19878$

$$\phi = \frac{\rho_{ma} - \rho_b}{\rho_{ma} - \rho_f}$$

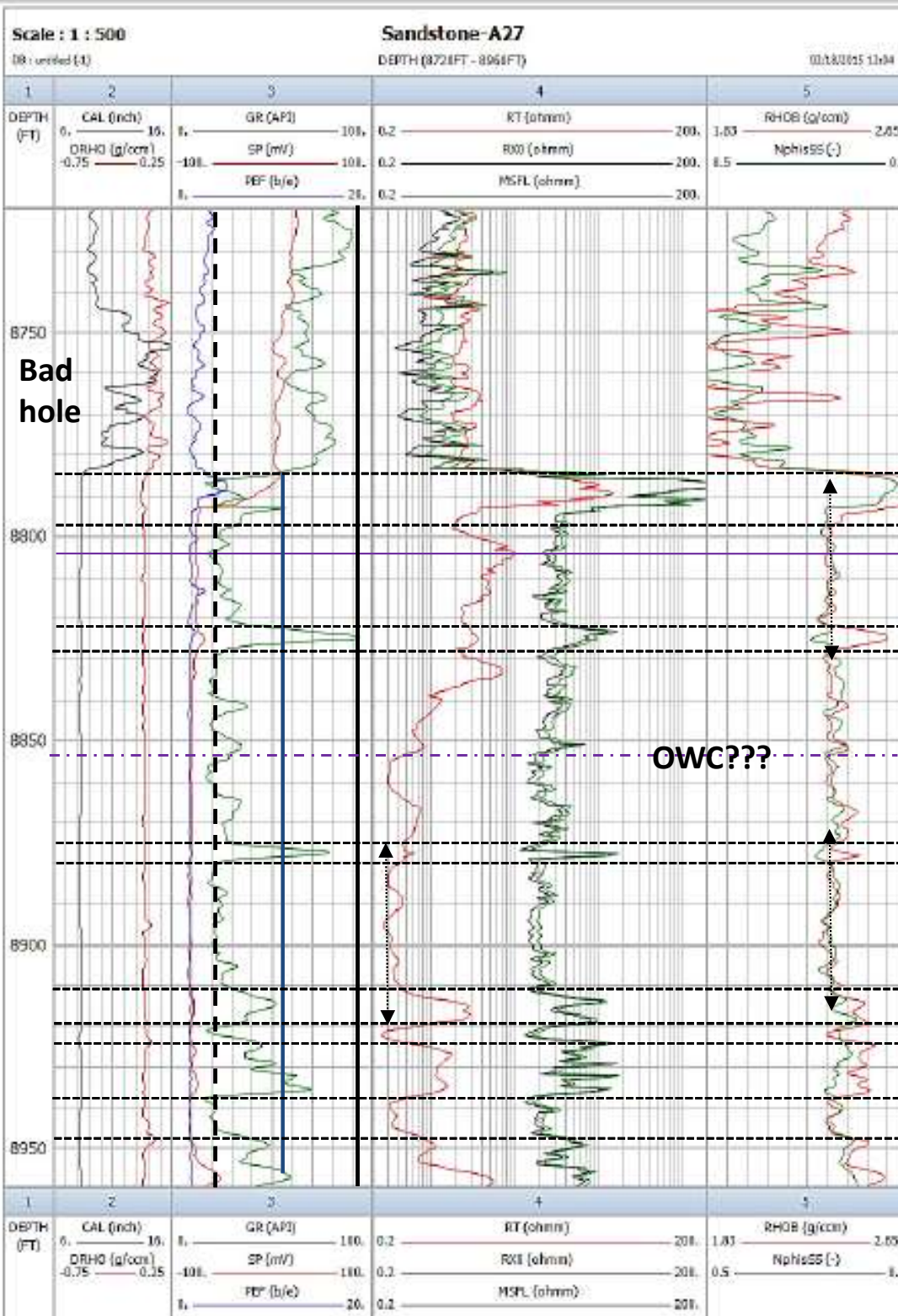
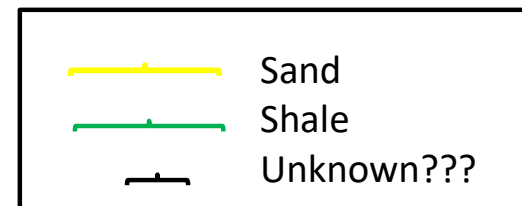
Calculation of R_w (Water Zone)

32 ft • $R_w = (0.3) * (0.2)^2 = 0.012$ Ohmm

$$R_w = R_t * \phi^m$$

6 ft

10 ft



Quick look Interpretation – QA/QC

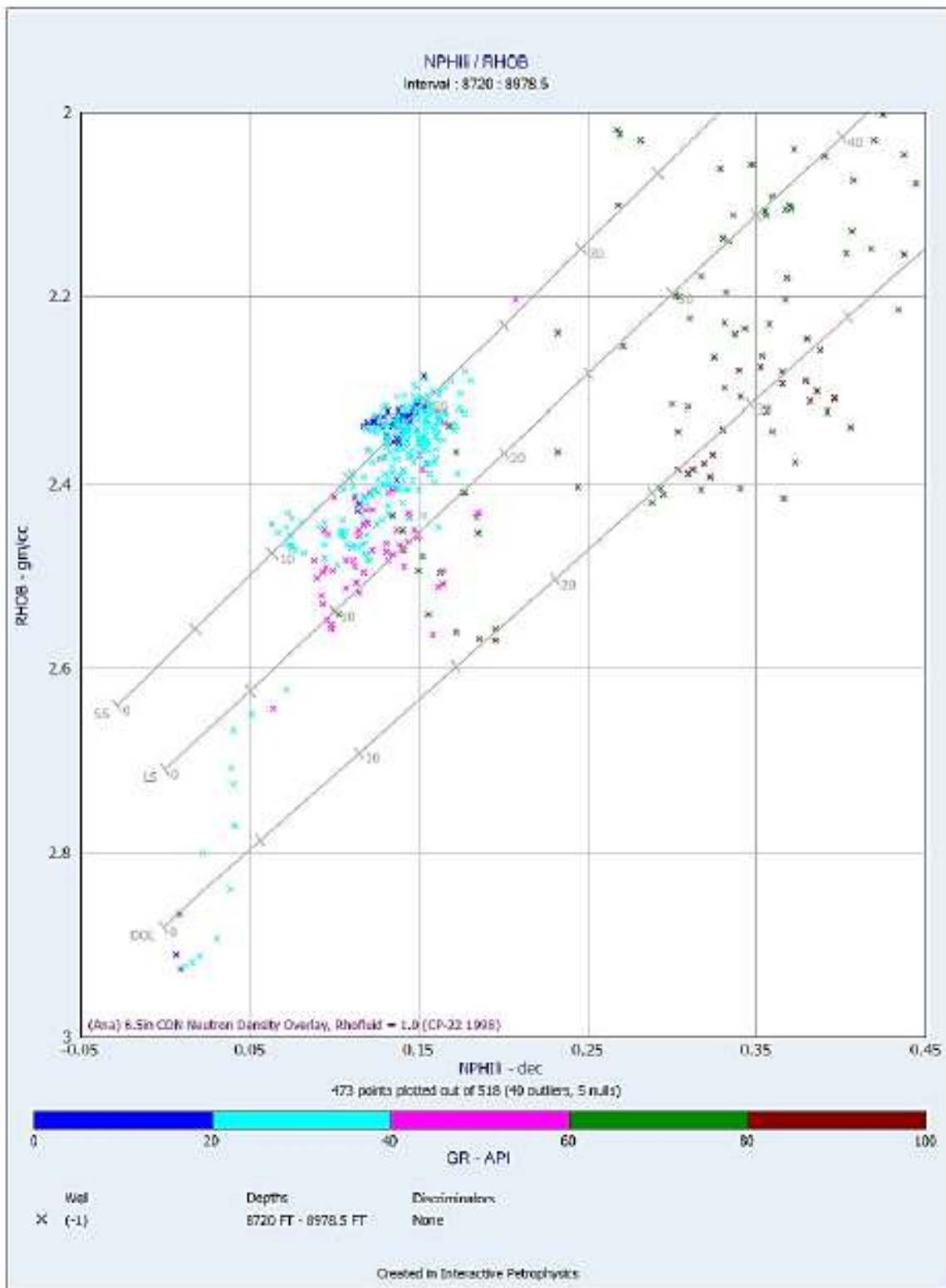
- The Caliper log (first track) shows in the uppermost part – particularly from 8745 to 8785 ft – a caliper enlargement (“bad hole”)
- A look at the Gamma log and SP log indicates this section as shale with some stability problems. The correction term for density-derived porosity DRHO is very high (up to -0.23) and does not allow any further quantitative density interpretation.

Quick look Interpretation - Lithology

The “lithology logs” GR and SP clearly separate sand and shaly zones of the section. The following profile can be derived:

- Shale zone from top to 8784 ft; bad hole.
- The zone from 8784 to 8795 ft is characterized by correct Caliper, low GR, PEF of about 5 b/e, extremely high resistivity, very low NPHI and extremely high RHOB (data in LAS-file show $> 2.9 \text{ g/cm}^3$), and low DT. All these indications – particularly the density – are typical of anhydrite. It forms a cap rock covering the reservoir.
- Sand zone (reservoir) from 8795 to 8910 ft with shaly/silty interlayers at 8822 to 8826 ft, 8876 to 8879 ft – PEF confirms a (quartz) and with 1.8 p/e. The log plot shows Neutron porosity in the porosity column. NPHISS is plotted from 0 to 0.5 and density RHOB from 2.65 g/cm^3 to 1.83 g/cm^3 . Thus, the fit of Neutron and Density Porosity is indicative of sandstone.
- The following zone is a shaly sand (increase of GR)

pretation



- The Neutron-Density cross-plot (Fig. 2) confirms lithological units.
- Most data points are close to the sandstone line (porosity 15 20%); data points in the upper right part are influenced by the shale section and caliper.
- Note that there are also some data points in the lower left corner indicating anhydrite.

Quick look Interpretation - Resistivity

- For the following quick-look interpretation of resistivity, it is important to note that the reservoir is characterized by a nearly constant porosity of about 20% - it is a homogeneous reservoir.
- The resistivity R_t below 8855 ft is very low (0.02 – 0.04 Ohm-m); this indicates a very high salinity and a strong contrast to the mud filtrate resistivity (therefore SSP is very high – about 200 mV)
- From a depth of 8800 ft upwards, the resistivity R_t increases by a factor of about 10. R_t is still below R_{xo} (R_{xo} is controlled by low saline mud filtrate). This increase of R_t must be originated by a decrease of water saturation (because porosity is constant). This part is probably the transition zone. From 8834 ft to 8795 ft, the reservoir is above transition zone.
- The nearly constant level of R_{xo} (and MSFL) confirms reservoir homogeneity and clearly indicates invasion as a result of permeability.

Compilation of input parameters for quantitative interpretation

Shale content from GR log:

- Minimum and maximum magnitude (average for the reservoir): GRmin = 16 API; GRmax = 110 API. Note: Particular GRmax cannot be taken from the shale zone of this plot because of bad hole conditions!

Fluid parameter for resistivity analysis:

- Calculation of Rmf at formation temperature 178 °F:
 - Rmf at 90 °F is 0.366 Ohm-m, conversion with Arps equation results in
 - $$R_{mf,178} = 0.336 \cdot \frac{90 + 6.77}{178 + 6.77} = 0.176 \text{ Ohmm.}$$
- Determination of Rw at formation temperature 178 °F:
 - This fundamental property for calculation of water saturation will be derived from resistivity in the water zone and from SP log analysis

Compilation of input parameters for quantitative interpretation

Calculation of R_w from water zone:

- Within the water zone, the resistivity equals R_o . For a selected part (8890 ... 8900 ft) it is on the average 0.3 Ohm-m, for this part porosity is 0.2.
- Thus, since $R_o = (1/\phi^m) * R_w$, $R_w = R_o / (1/\phi^m)$
- $R_w = (0.3) / (1/(0.2)^2) = 0.012$ Ohm-m

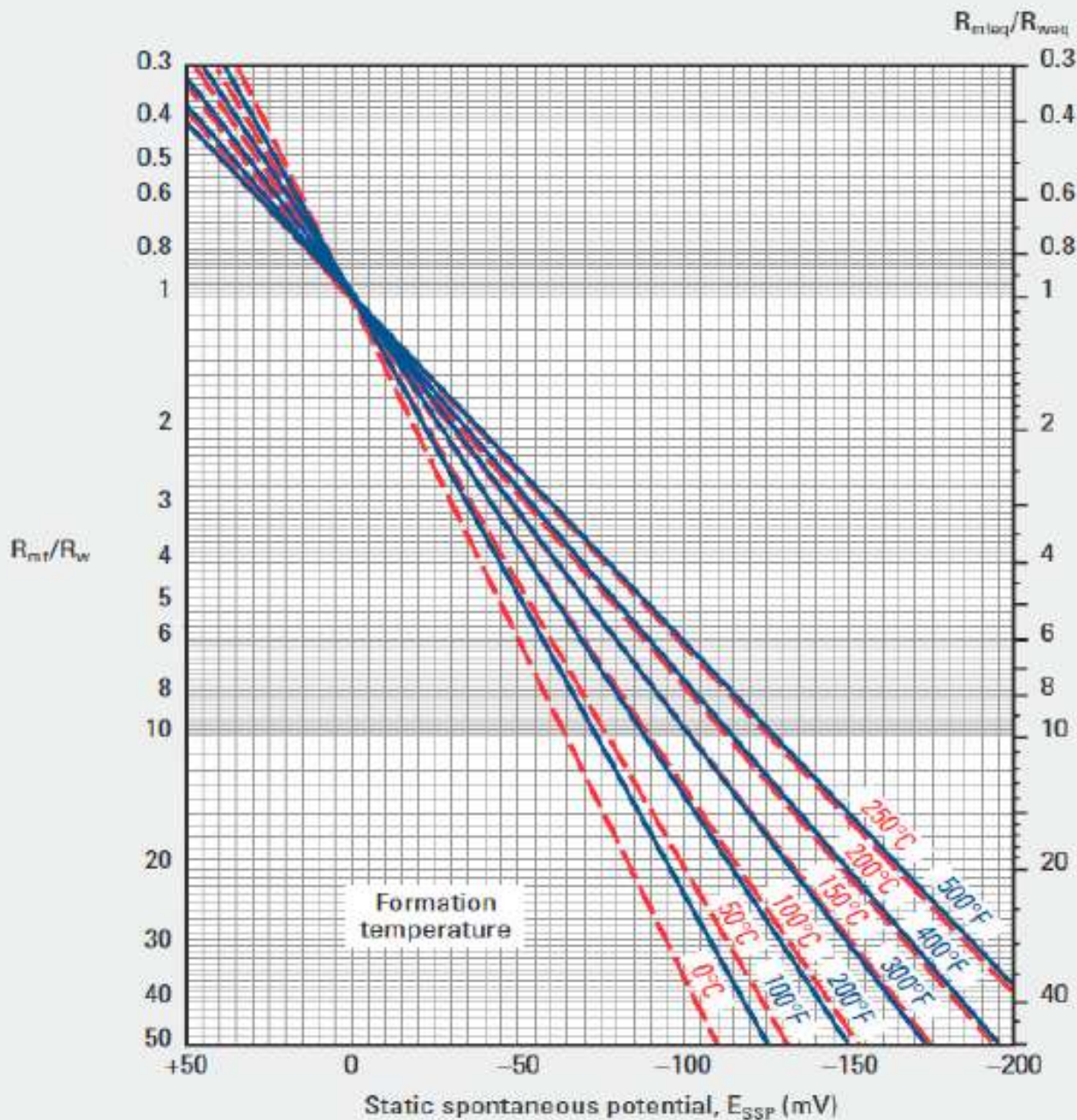
Calculation of R_w from SP:

- The Static Spontaneous Potential (SSP) is given by:

$$SSP = k_{SP} \cdot \log\left(\frac{R_{mf}}{R_w}\right)$$

- Where the factor k_{SP} is mainly controlled by temperature, this dependence is presented in the monogram in Figure 3.

Compilation of input parameters for quantitative interpretation



- Figure 3: Determination of R_w from SSP, Schlumberger Chart Book, 2000.

Compilation of input parameters for quantitative interpretation

Calculation of R_w from SP:

- The log shows a maximum SSP difference is -100 mV. At the formation temperature of 178 °F, the circle in Figure 3 results in a ratio:

$$\frac{R_{mf}}{R_w} = 15 \quad R_w = \frac{0.176}{15} = 0.012 \text{ Ohmm}$$

- This is in confidence with the value from the water zone and confirms the very high salinity of the formation water.

Quantitative interpretation for 4 reservoir zones

- Table 3 shows the measured data (from the LAS-file) for 4 depths. The calculation is demonstrated step by step for the depth of 8804 ft. For the other 3 depths results are given in Table 4.

Depth	GR	DT	RHOB	NPHI_SS	RT	RX0
8804	18.5	79.6	2.321	0.188	3.818	6.837
8820	26.7	84.9	2.291	0.226	1.242	6.025
8839	20.7	74.7	2.328	0.163	0.635	6.735
8859	16.7	76.9	2.325	0.189	0.272	5.327

Table 3: Log data from selected depths (from LAS file). The Resistivity values in the LAS file and on the log are inconsistent.

Quantitative interpretation for depth 8804ft: Shale Content

Property: Shale Content

$$V_{sh} = \frac{GR_{DOI} - GR_{MIN}}{GR_{MAX} - GR_{MIN}} = \frac{18.5 - 16}{92 - 16} = 0.041 \text{ (Shale Content is 4.1\%)}$$

- The low shale content (note that the linear equation delivers the maximum shale content) is not implemented in the following calculations. Particularly for the resistivity interpretation the very low shale content and high water conductivity allow application of Archie's equation.

Quantitative interpretation for depth 8804ft: Porosity

Property: Porosity

- Porosity from Density log. (Shale Content is 4.1%)

$$DPHI = \frac{\rho_{ma} - \rho_{DOI}}{\rho_{ma} - \rho_{fl}} = \frac{2.65 - 2.31}{2.65 - 1.00} = 0.206$$

- Porosity from Neutron log: NPHI = 0.2
- Average Porosity: $(DPHI + NPHI)/2 = 0.203$
- For comparison, Acoustic log porosity confirms the value

$$DTPHI = \frac{\Delta t_{log} - \Delta t_{ma}}{\Delta t_{fl} - \Delta t_{ma}} = \frac{79.6 - 52}{190 - 52} = 0.2$$

Quantitative interpretation for depth 8804ft: Saturation

Property: Saturation, S_w

- Formation Factor:

$$F = \frac{1}{\phi^m} = \frac{1}{(0.2)^2} = 25$$

- Formation resistivity if completely filled with water, R_o :

$$R_o = F * R_w = 25 \times 0.012 = 0.3 \text{ Ohmm}$$

- The measured resistivity is distinctly higher because non-conducting oil is present.

- $S_w = \sqrt{\frac{R_o}{R_t}} = \sqrt{\frac{0.3}{3.818}} = 0.28$ water saturation = 28%
- oil saturation = 72%

Quantitative interpretation for depth

8804ft: Non-moveable oil in the invaded zone

Property: Non-moveable oil in the invaded zone

- Saturation with respect to mud filtrate is:

$$S_{xo} = \sqrt{\frac{F * R_{mf}}{R_{xo}}} = \sqrt{\frac{25 * 0.176}{6.837}} = 0.802$$

- The invaded zone is 80% saturated with mud filtrate (it has displaced the original fluid). But there is still 20% non-displaced (non-movable) oil in the pore space
- The reservoir at 8804ft has 20% porosity and water saturation of 28%.
- The oil saturation in the non-invaded zone is 72%,
- Non-moveable oil = $1 - S_{xo} = 20\%$ of the oil are non-moveable,
- $S_{xo} - S_w = 80\% - 28\% = 52\%$ are moveable

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